

Supplementary Information

A flexible modelling framework for estimating thermal tolerance and sensitivity

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1 General Introduction

This supplement demonstrates the joint Bayesian 4-parameter logistic (4PL) framework from the main manuscript. To cite the approach and the associated R package please use:

Noble, D.W.A., Arnold, P.A., Nakagawa, S. & Pottier, P. (in preparation). A flexible modelling framework for estimating thermal tolerance and sensitivity.

The supplement has two parts. The first introduces the **bayesTLS** R package using simulated data and empirical case studies. The second reports the extended simulation results and provides the code used to reproduce the main manuscript figures.

2 Introduction to bayesTLS R Package

We use the **bayesTLS** R package included with this project. The package fits 4PL models, extracts classical TLS quantities (z , $CT_{max_{1hr}}$, and, for lethal endpoints, T_{crit}) with posterior uncertainty, and can estimate z and $CT_{max_{1hr}}$ directly so group differences and random effects are easy to interpret. It also predicts heat-injury accumulation and survival under arbitrary temperature time series, optionally with a Sharpe-Schoolfield repair kernel ([Arnold *et al.* 2025](#)).

All the data used throughout the manuscript and supplement are already preloaded into **bayesTLS**, so are easily accessible.

The **bayesTLS** package is complemented by **freqTLS**, which fits the same models in a frequentist likelihood framework. **freqTLS** is faster and useful for large datasets, while **bayesTLS** is useful when posterior draws are needed for uncertainty propagation. This tutorial focuses on **bayesTLS**.

To access the **bayesTLS** R package, install it once from GitHub. Installing **bayesTLS** pulls in its modelling dependencies (**brms**, **dplyr**, **ggplot2**, **posterior**, **tibble**, **patchwork**) automatically; the tutorial also uses a few small helper packages:

```
# Install bayesTLS from GitHub.
# install.packages("remotes") # if needed
remotes::install_github("danielnoble/bayesTLS")

# Install helper packages used in the tutorial.
install.packages(c("here", "purrr", "tidyr"))
```

Every fit below passes `file = file.path(models_dir, "...")`, so re-running a chunk reloads the saved model instead of refitting it. For this tutorial, the folder models are stored in `output/models`. If you don't already have this folder it will create it for you. Run this setup chunk once, before the analyses that follow:

```
# Load bayesTLS and helper packages.
# Install pacman first if needed.
if (!requireNamespace("pacman", quietly = TRUE)) install.packages("pacman")
pacman::p_load(bayesTLS, brms, dplyr, tidyr, purrr, ggplot2, here)

set.seed(123)

# Cache fitted brms models in output/models.
models_dir <- here::here("output", "models")
if (!dir.exists(models_dir)) dir.create(models_dir, recursive = TRUE)
```

The functions in **bayesTLS** chain together as a single pipeline with the following logic:

1. **Standardise** raw experimental data to a common column schema (`temp`, `duration`, `n_total`, `n_surv`, ...). This step is important because it also stores meta-data about the centering temperatures used which ensures calculations are made correctly.

Table S1: Core **bayesTLS** functions for a standard workflow, following the five pipeline steps. **tls()** is the general entry point for deriving TDT quantities from any fitted 4PL, including hand-coded **brms** models. See Table S2 for the full list of exported functions.

Step	Function	Purpose
Standardise	<code>standardize_data()</code>	Reshape raw experimental data to the package schema (temp, duration)
Fit	<code>fit_4pl()</code>	Fit the joint Bayesian 4PL in one call (builds priors and the brms for
Extract	<code>extract_tdt()</code>	Derive z , CT_{max} and (for lethal endpoints) T_{crit} from a 'fit_4pl()' fit
Extract	<code>tls()</code>	Derive the same quantities per moderator group from ANY fitted 4PL
Predict	<code>predict_survival_curves()</code>	Posterior survival across a temperature x duration grid, with credible
Predict	<code>predict_heat_injury()</code>	Heat-injury accumulation and predicted survival under a fluctuating
Plot	<code>plot_tdt_curve()</code>	Visualise the TDT (LT_x) curve on a log-time axis; companion 'plot'

2. **Fit** the joint Bayesian 4PL with the built-in `beta_binomial(link = "identity")` family and asymptote-bounded reparameterisation. We also have other families built in depending on the response variable distribution.
3. **Extract** z , CT_{max} (at a chosen reference exposure) and, for lethal endpoints, T_{crit} — all as transforms of the fitted posterior, with full uncertainty. **tls()** is the general extractor: it works on any fitted 4PL — the `fit_4pl()` workflow or a hand-coded **brms** model — and returns results **per moderator group** (by =, e.g. species, sex or oxygen treatment) as tidy-long summaries plus the underlying posterior draws. **tls()** and the workflow-specific `extract_tdt()` share one `target_surv` argument that sets the survival threshold the curve is read at — the relative midpoint (default), the absolute LT_{50} , or any LT_x — and return the same quantities for a `fit_4pl()` fit.
4. **Predict** survival across a temperature $\times$ duration grid, and heat-injury accumulation with predicted survival under fluctuating field temperature traces — both group-aware (by =) and with optional damage repair.
5. **Plot** each step — survival curves, tolerance landscapes and heat-injury trajectories — with a shared project theme so figures across the workflow read consistently.

Every exported function is documented (`?fit_4pl`, etc.) and tested against known simulation truths. The supplement first demonstrates the pipeline in controlled simulations and then applies it to the snow gum leaf PSII and *Drosophila* datasets.

2.1 Description of Functions

2.1.0.1 Core functions

Table S1 lists the key functions that are part of the **bayesTLS** package. Generally, these are the ones users will use most. These functions are used for standardising data, modelling, deriving the TLS quantities of interest, and predicting heat injury and survival.

In addition, if **brms** 4PL models are fit manually to data we also provide a **tls()** function that works more generally across fitted models to derive relevant TLS quantities.

2.1.0.2 Full package functions

There are many helper functions in the package that maybe useful for various users. Table S2 lists every exported function in the library, grouped by pipeline stage with a description of what they are designed to do.

2.2 Empirical model specifications used in the manuscript

The manuscript case studies all use the same 4PL response surface, but we fit it in the parameterisation that makes each biological comparison easiest to read. In the default **midpoint** parameterisation, temperature enters the midpoint directly,

$$\text{mid}_{ij} = \beta_0 + \beta_1(T_{ij} - \bar{T}) + \mathbf{x}_{ij}^\top \gamma + b_j, \quad (\text{S1})$$

so relative-threshold z and CT_{max} are derived from the fitted midpoint line. In the **direct** parameterisation, the same midpoint is written in terms of CT_{max} and z :

$$\begin{aligned} \text{mid}_{ij}(T) &= \log_{10} t_{\text{ref}} + \frac{CT_{max,ij}(t_{\text{ref}}) - T_{ij}}{z_{ij}}, \\ CT_{max,ij}(t_{\text{ref}}) &= \alpha_0 + \mathbf{x}_{ij}^\top \alpha + b_j, \\ \log z_{ij} &= \zeta_0 + \mathbf{x}_{ij}^\top \zeta + u_j. \end{aligned} \quad (\text{S2})$$

Here $\mathbf{x}_{ij}$ contains fixed covariates such as species, oxygen treatment, recovery condition or sex, while b_j and u_j denote optional group-level effects. Direct fits place random effects inside the relevant `ctmax =` or `z =` formula; midpoint fits place them on `mid` or through an explicit `mid =` formula. In both parameterisations, **bayesTLS** constrains the lower and upper asymptotes to disjoint intervals and estimates the steepness parameter on a positive scale. Temperature effects can also be included on the asymptotes and steepness where the data support them.

We fit the empirical examples below with the Bayesian **bayesTLS** implementation through **brms** and Stan because posterior draws make uncertainty propagation through `tls()` and `predict_heat_injury()` straightforward. The same likelihoods can also be fit in a frequentist likelihood framework with **freqTLS**; for the same data and parameterisation these fits should give similar point estimates, with uncertainty propagated by the Delta method or parametric bootstrap.

The manuscript examples use the following model specifications:

- **Cereal aphids:** lethal survival counts were fit with a beta-binomial 4PL, placing species directly on $CT_{max_{1hr}}$ and z (`ctmax = ~ 0 + species`, `z = ~ 0 + species`). No random effects were included. The fitted species-specific posterior was then projected under the unmodified Wuhan May-June 2016 air-temperature trace with `predict_heat_injury(by = "species")`.
- **Zebrafish oxygen experiment:** lethal survival counts were fit with a beta-binomial 4PL, placing oxygen treatment directly on $CT_{max_{1hr}}$ and z (`ctmax = ~ 0 + oxygen`, `z = ~ 0 + oxygen`). No random effects were included.

Table S2: Functions exported by the TDT analysis library, grouped by pipeline stage. Each function is documented with roxygen2 in its R file; runnable examples accompany every function.

Stage	Function	Purpose
Data	<code>standardize_data()</code>	Rename raw columns to project schema; attach metadata
Model spec	<code>make_4pl_priors()</code>	Weakly informative priors for the disjoint-bounds 4PL
Model spec	<code>make_4pl_formula()</code>	brms non-linear formula; identity link, all four sub-parameters
Fit	<code>fit_4pl()</code>	Joint Bayesian 4PL fit; returns a ‘bayes_tls’ object (using <code>brmsfit</code>)
Inspection	<code>print.bayes_tls()</code>	Compact one-screen header: data shape, T_bar, asymptote
Inspection	<code>summary.bayes_tls()</code>	Delegates to ‘summary()’ on the underlying ‘brmsfit’; returns a list
Inspection	<code>plot.bayes_tls()</code>	Delegates to ‘plot()’ on the underlying ‘brmsfit’; returns a plot
Inspection	<code>has_fit()</code>	Predicate: is a ‘bayes_tls’ workflow fitted (TRUE) or saved (FALSE)
Inspection	<code>extract_4pl_pars()</code>	Posterior draws of the four natural-scale 4PL parameters
Inspection	<code>get_brmsfit()</code>	Return the underlying ‘brmsfit’ object held inside a ‘bayes_tls’
Inspection	<code>print.tls()</code>	Compact print method for a ‘tls’ object: shows the per-temperature
Accessors	<code>get_z_draws()</code>	Per-draw posterior of z from an ‘extract_tdt()’ output
Accessors	<code>get_ctmax_draws()</code>	Per-draw posterior of CTmax_1hr from an ‘extract_tdt()’ output
Accessors	<code>get_tcrit_draws()</code>	Per-draw posterior of T_crit from an ‘extract_tdt(lethal)’ output
Accessors	<code>get_z_summary()</code>	Posterior summary (median + 95% CrI) of z from an ‘extract_tdt()’
Accessors	<code>get_ctmax_summary()</code>	Posterior summary (median + 95% CrI) of CTmax from an ‘extract_tdt()’
Accessors	<code>get_tcrit_summary()</code>	Posterior summary (median + 95% CrI) of T_crit from an ‘extract_tdt(lethal)’
Accessors	<code>get_tls_draws()</code>	Per-draw z, CTmax and (lethal) T_crit merged on ‘draws’
Accessors	<code>get_tls_summary()</code>	Tidy combined posterior summary (the summary component)
Accessors	<code>get_surv_draws()</code>	Per-draw posterior of survival probability from ‘predict_survival_curves()’
Accessors	<code>get_hi_draws()</code>	Per-draw posterior of heat-injury integrals from ‘predict_survival_curves()’
Diagnostics	<code>diagnose_tdt_fit()</code>	Rhat / ESS / divergences / treedepth, with pass-fail flags
Diagnostics	<code>tdt_parameter_table()</code>	Posterior summary on the natural scale (low, up, k, mid, high)
Diagnostics	<code>bayes_R2_tls()</code>	Bayesian R ² for a fitted ‘bayes_tls’ workflow (posterior predictive)
Predictions	<code>predict_survival_curves()</code>	Survival curves on a temp x duration grid with credible intervals
Predictions	<code>derive_tdt_landscape()</code>	Dense 2-D survival surface for a heatmap.
Predictions	<code>summarise_observed_survival()</code>	Observed survival mean and SE per (temp, duration) combination
TDT quantities	<code>derive_tdt_curve()</code>	Time to the threshold survival across temperature. Derivative
TDT quantities	<code>derive_temperature_for_duration()</code>	Temperature at threshold survival for a fixed exposure duration
TDT quantities	<code>derive_z()</code>	Thermal sensitivity z read directly from the posterior (not CTmax)
TDT quantities	<code>extract_tdt()</code>	Bundled wrapper: z and CTmax always; T_crit optional
TDT quantities	<code>tls()</code>	One call returning z, CTmax and (when ‘lethal = TRUE’) T_crit
TDT quantities	<code>tls_z()</code>	Convenience wrapper for ‘tls(params = "z")’ — z only
TDT quantities	<code>tls_ctmax()</code>	Convenience wrapper for ‘tls(params = "ctmax")’ — CTmax only
TDT quantities	<code>tls_tcrit()</code>	Convenience wrapper for ‘tls(params = "tcrit", lethal = TRUE)’
Two-stage	<code>ts_stage1()</code>	Classical Stage 1: per-temperature dose-response (binomial)
Two-stage	<code>ts_stage2()</code>	Classical Stage 2: OLS of log10(LT50) on temperature

- **Snow gum PSII:** retained photosystem-II function was fit as a continuous proportion with a Beta likelihood. Recovery treatment was placed directly on $CT_{max_{1hr}}$ and z , with a plant-level random intercept on CT_{max} ($ctmax = \sim 0 + \text{recovery} + (1 \mid \text{plant})$, $z = \sim 0 + \text{recovery}$). Because this endpoint is sublethal, we report z and CT_{max} but not T_{crit} .
- **Vinegar fly mortality:** mortality counts were fit with beta-binomial 4PLs. We fit separate sex-specific models for comparison with Ørsted *et al.* (2024) and a joint midpoint model with sex and its temperature interaction on mid ($mid = \sim \text{sex} * \text{temp_c}$) to test sex differences. The main mortality summaries use the absolute LT50 threshold and a 4-hour reference exposure; the heat-injury forecasts use the relative threshold from the sex-specific lethal fits and the shaded Rennes microclimate trace from the Ørsted *et al.* NicheMapR workflow.

The exact formulas, priors, sampler settings and diagnostics are shown in the code chunks for each case study. The direct zebrafish and aphid examples use the package’s default four-chain sampler settings; the snow gum, vinegar-fly and sublethal-extension models use the longer or more conservative settings annotated beside each fit.

3 A Tutorial with Simulations

This tutorial uses simulated data to show how the Bayesian hierarchical 4PL recovers thermal sensitivity (z) and $CT_{max_{1hr}}$ in both binomial and beta-binomial (overdispersed) settings. The joint model propagates uncertainty through downstream quantities and accommodates extensions that are awkward in a two-stage analysis. The subsequent case studies apply the same workflow to data presented in the main manuscript.

3.0.1 Simulating Data

3.0.1.1 Step 1 — Setting up parameters

We choose values that might be typical of a thermal mortality assay. The four parameters are the lower and upper asymptotes (ℓ , up), the slope on the $\log_{10}$ -time axis (k), and a midpoint that varies linearly with temperature: $mid(T) = \beta_0 + \beta_1(T - \bar{T})$. Each value has a clear biological meaning (detailed in the code below).

```
# Define the generating parameters for the tutorial simulation.
true_params <- list(
  ell      = 0.03,    # 3% of individuals survive even at the longest exposures
  up       = 0.97,    # 97% are alive at very short exposures
  k        = 8,       # steepness of the survival drop on the log10(t) axis
  m_beta0  = 1.5,     # midpoint at the grand-mean temperature: ~31.6 min
  m_beta1  = -0.15,   # midpoint drops 0.15 log10-min per °C; z = -1/beta1  6.7 °C
  T_bar    = 34       # grand-mean temperature
)
```

The implied thermal sensitivity is $z = -1/\beta_1 \approx 6.7$ °C: a temperature change of this magnitude shifts LT50 by a factor of 10. Saavy readers will note that this is technically relative z , but because our asymptotes are 0.03 and 0.97, despite these shifts from 0 and 1, we are still at a

midpoint of 50% survival. So, in this case, they are the same. The same parameters determine the uncentred TDT intercept $\alpha = \beta_0 + \frac{1}{k} \log \frac{up-0.5}{0.5-\ell} - \beta_1 \bar{T}$ and the 1-hour critical thermal limit $CT_{max_{1hr}} = (\log_{10} 60 - \alpha) / \beta_1$. We use these known values to assess how well the models are performing, but do a proper simulation in Section 8.

We first calculate α , z , and $CT_{max_{1hr}}$ from the generating parameters to provide reference values for the fitted models.

```
# Compute the true TDT summaries implied by the simulation parameters.
true_z      <- -1 / true_params$m_beta1

# Convert the centred midpoint model to the original temperature scale.
true_alpha  <- true_params$m_beta0 +
  (1 / true_params$k) *
  log((true_params$up - 0.5) / (0.5 - true_params$ell)) -
  true_params$m_beta1 * true_params$T_bar
true_CTmax  <- (log10(60) - true_alpha) / true_params$m_beta1

# Tcrit uses the geometric midpoint of the default heat-injury rate range.
true_T_crit <- true_CTmax - 2.5 * true_z
```

3.0.1.2 Step 2 — Simulated study design

We mimic a well-replicated TDT experiment: five assay temperatures crossed with six log-spaced exposure durations, with 30 replicates of 30 individuals per cell. The resulting 900 trials and 27,000 individuals make parameter recovery clear but the case studies below and simulations illustrate less information-rich designs.

```
# Build the temperature-by-duration sampling design for the simulation.
temps      <- c(30, 32, 34, 36, 38)          # 5 assay temperatures (°C)
durations  <- c(1, 5, 15, 45, 135, 405)      # 6 durations (minutes, log-spaced)
n_rep      <- 30                             # 30 replicates per cell
n_per_rep  <- 30                             # 30 individuals per replicate

design <- tidyr::expand_grid(
  T    = temps,
  t    = durations,
  rep  = seq_len(n_rep)
) |>
  dplyr::mutate(
    log10_t = log10(t),
    T_c     = T - true_params$T_bar,          # mean-centred T
    mid     = true_params$m_beta0 + true_params$m_beta1 * T_c, # mid(T)
    p_true  = true_params$ell +
      (true_params$up - true_params$ell) /
      (1 + exp(true_params$k * (log10_t - mid))),
    n       = n_per_rep
  )
```


3.0.1.3 Step 3 — Look at the truth before simulating

Before generating noisy data, let's plot the *true* survival surface (Figure [S1](#)) so we know what the model is being asked to recover. Each line is the 4PL at one assay temperature, plotted against $\log_{10}$ exposure time.

```
# Plot the true survival surface before sampling noise is added.
truth_grid <- tidyr::expand_grid(
  T = temps,
  t = 10 ^ seq(log10(0.5), log10(1000), length.out = 200)
) |>
dplyr::mutate(
  log10_t = log10(t),
  T_c      = T - true_params$T_bar,
  mid      = true_params$m_beta0 + true_params$m_beta1 * T_c,
  p_true   = true_params$ell +
    (true_params$up - true_params$ell) /
    (1 + exp(true_params$k * (log10_t - mid)))
)

ggplot2::ggplot(truth_grid,
  ggplot2::aes(log10_t, p_true, colour = factor(T))) +
ggplot2::geom_hline(yintercept = 0.5, linetype = "dashed", colour = "grey50") +
ggplot2::geom_line(linewidth = 0.8) +
ggplot2::labs(
  x = expression(log[10]~exposure~time~(min)),
  y = "True survival probability",
  colour = "Temperature (°C)"
) +
ggplot2::theme_minimal()
```

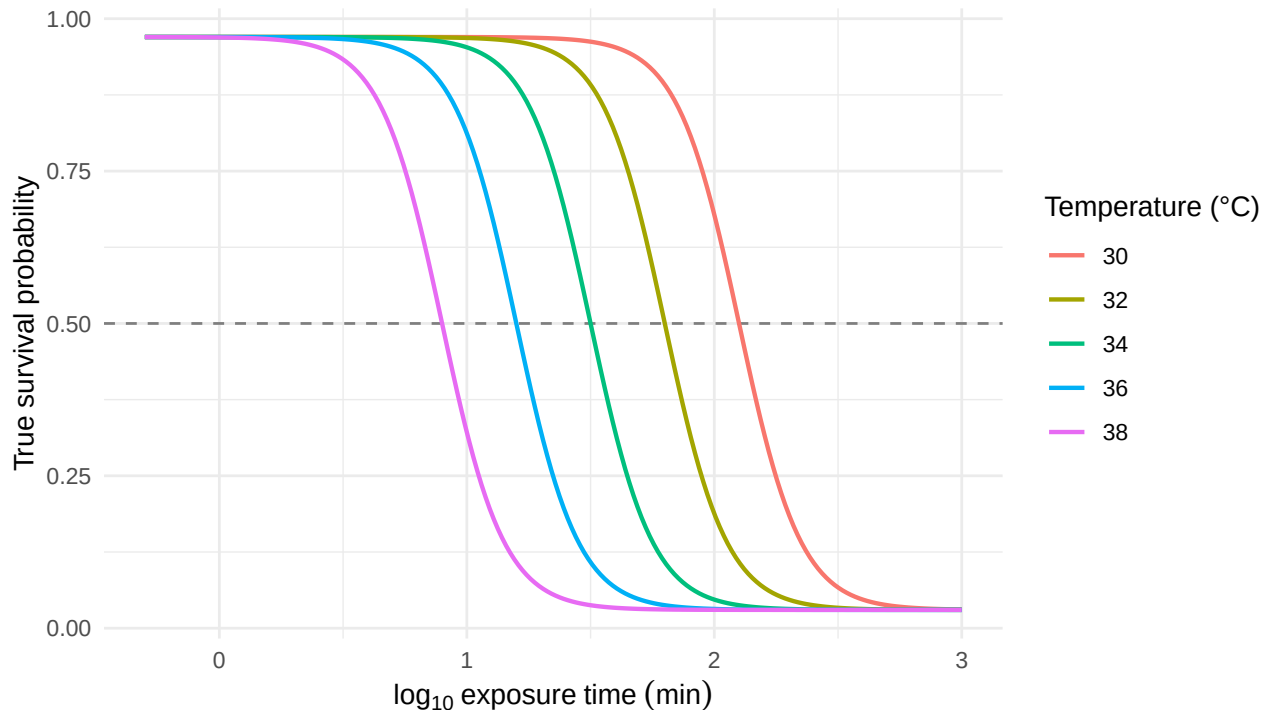


Figure S1: The known 4PL dose-response surface used to generate the simulated data, evaluated on a dense time grid at each of the five assay temperatures. The horizontal line marks 50% survival; where each curve crosses it gives the true LT50 at that temperature.

Notice the curves shift left as temperature rises: hotter temperatures kill faster, so the LT50 is reached at shorter durations.

3.0.1.4 Step 4 — Generate noisy data: binomial and beta-binomial

Survival counts may exhibit two forms of variation. Under **binomial** sampling, each replicate's survival count is drawn from $\text{Binomial}(n, p)$, where p is given by the 4PL surface. **Beta-binomial** sampling adds *overdispersion*: replicate-to-replicate variation in p , for example from cohort effects or unmodelled heterogeneity. When TDT data show this extra variation, the beta-binomial likelihood is more appropriate (note that our 4PL fit function defaults to beta-binomial to protect against mis-fits - over- or under-dispersion).

```
# Simulate binomial and beta-binomial counts from the true 4PL surface.
# (a) Binomial: y ~ Binomial(n, p_true)
sim_bin <- design |>
  dplyr::mutate(y = rbinom(dplyr::n(), size = n, prob = p_true))

# (b) Beta-binomial: replicate-level p drawn from Beta(p*phi, (1-p)*phi)
phi <- 5 # smaller phi = more overdispersion
sim_bb <- design |>
  dplyr::mutate(
    p_draw = rbeta(dplyr::n(), p_true * phi, (1 - p_true) * phi),
    y       = rbinom(dplyr::n(), size = n, prob = p_draw)
  )
```

3.0.2 The classical two-stage TDT pipeline

We first run the simulated data through the **classical two-stage TDT pipeline** (Ørsted *et al.* 2022; Rezende *et al.* 2014), which is dominant in the thermal-tolerance literature.

1. **Stage 1.** At each assay temperature, fit a logistic dose-response curve to the count data and read off a point estimate of $\log_{10} \text{LT50}_T$.
2. **Stage 2.** Regress those Stage-1 point estimates on assay temperature with ordinary least squares, then derive $z = -1/\beta_1$ and $CT_{max_{1hr}} = (\log_{10} 60 - \alpha)/\beta_1$ from the fitted line.

3.0.2.1 Stage 1 — Dose-response curves at each temperature

We fit a binomial GLM with a logit link separately at each assay temperature. This is a common Stage-1 specification in the TDT literature: a two-parameter logistic with asymptotes constrained to 0% and 100% (e.g., Ørsted *et al.* 2024). The midpoint $\log_{10} \text{LT50}_T = -\beta_0/\beta_1$ falls out of the GLM's two coefficients.

```
# Fit the per-temperature dose-response curves for the classical Stage-1 analysis.
# Stage 1 uses the package function ts_stage1(): a per-temperature binomial GLM
# (the field-default Stage 1). Here we assume stage 1 fit likelihoods are correctly specified.
stage1_bin <- ts_stage1(sim_bin, temp = "T", duration = "t",
  n_surv = "y", n_total = "n", family = "binomial")
```

Table S3: Per-temperature Stage-1 estimates of $\log_{10}$ LT50 from a binomial GLM with logit link, fit separately at each of the five assay temperatures. Truth values are the simulated midpoints (which coincide with $\log_{10}$ LT50 here because the simulation asymptotes are near 0 and 1 and change equally).

T (°C)	Truth $\log_{10}$ (LT50)	Binomial $\log_{10}$ (LT50)	Beta-binomial $\log_{10}$ (LT50)
30	2.1	2.07	2.023
32	1.8	1.78	1.77
34	1.5	1.49	1.529
36	1.2	1.21	1.211
38	0.9	0.92	0.932

```
stage1_bb <- ts_stage1(sim_bb, temp = "T", duration = "t",
                      n_surv = "y", n_total = "n", family = "betabinomial")

# Format the Stage-1 LT50 estimates and simulation truth.
stage1_table <- dplyr::full_join(
  stage1_bin |> dplyr::select(temp, bin_log10_lt50 = log10_lt50),
  stage1_bb |> dplyr::select(temp, bb_log10_lt50 = log10_lt50),
  by = "temp"
) |>
dplyr::mutate(
  truth_log10_lt50 = true_params$m_beta0 +
    true_params$m_beta1 * (temp - true_params$T_bar)
) |>
dplyr::select(temp, truth_log10_lt50, bin_log10_lt50, bb_log10_lt50) |>
dplyr::rename(
  `T (°C)` = temp,
  `Truth log10(LT50)` = truth_log10_lt50,
  `Binomial log10(LT50)` = bin_log10_lt50,
  `Beta-binomial log10(LT50)` = bb_log10_lt50
)

tinytable::tt(stage1_table, digits = 3, escape = TRUE)
```

With the simulation's true asymptotes near 0 and 1, the two-parameter dose-response function forced-asymptote constraint introduces negligible bias here, so each Stage-1 estimate sits close to the simulation truth.

3.0.2.2 Stage 2 — log-linear TDT regression

Stage 2 regresses Stage-1 $\log_{10}$ LT50 on assay temperature with ordinary least squares. The classical TDT quantities — $z = -1/\beta_1$ and $CT_{max_{1hr}} = (\log_{10} 60 - \alpha)/\beta_1$ — fall out of the fitted line.

```
# Regress log10(LT50) on temperature and derive z, CTmax and Tcrit.
stage2_bin <- ts_stage2(stage1_bin, t_ref = 60, time_multiplier = 1)
stage2_bb  <- ts_stage2(stage1_bb,  t_ref = 60, time_multiplier = 1)
```

Figure S2 visualises the Stage-2 fit. Each point is one Stage-1 $\log_{10}$ LT50 estimate from Table S3; the solid line is the OLS regression through them. The slope is β_1 , from which $z = -1/\beta_1$ — i.e., the temperature increase that causes LT50 to change by a factor of 10. The line's height at $\log_{10} 60 \approx 1.78$ (1 hour) is at the temperature equal to $CT_{max_{1hr}}$ — the vertical dotted line drops from this intersection straight to the temperature axis to mark it. The dashed grey line is the simulation truth.

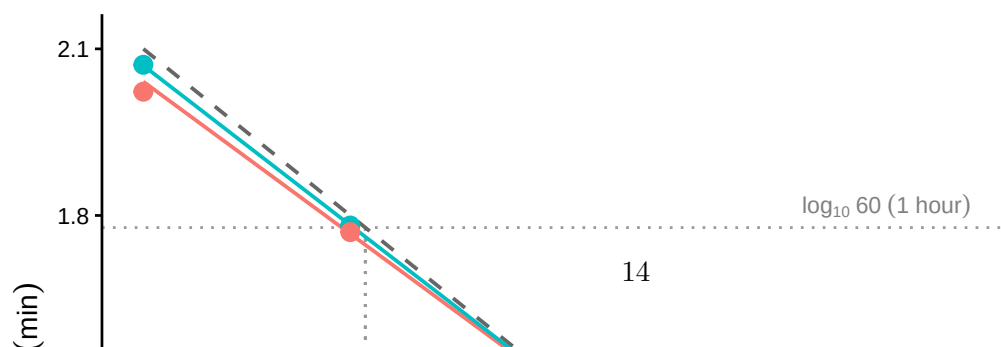
```

# Plot the Stage-2 regression and CTmax reference point.
stage1_combined <- dplyr::bind_rows(
  stage1_bin |> dplyr::mutate(dataset = "Binomial"),
  stage1_bb  |> dplyr::mutate(dataset = "Beta-binomial")
)

truth_line <- tibble::tibble(
  T = seq(min(temps), max(temps), length.out = 100)
) |>
dplyr::mutate(
  log10_lt50 = true_params$m_beta0 +
    true_params$m_beta1 * (T - true_params$T_bar)
)

ggplot2::ggplot(stage1_combined,
  ggplot2::aes(x = temp, y = log10_lt50, colour = dataset)) +
  ggplot2::geom_hline(yintercept = log10(60),
    linetype = "dotted", colour = "grey60") +
  ggplot2::geom_segment(
    ggplot2::aes(x = true_CTmax, xend = true_CTmax,
      y = log10(60), yend = -Inf),
    inherit.aes = FALSE,
    linetype = "dotted", colour = "grey60"
  ) +
  ggplot2::geom_line(data = truth_line,
    ggplot2::aes(x = T, y = log10_lt50),
    inherit.aes = FALSE,
    linetype = "dashed", colour = "grey40",
    linewidth = 0.7) +
  ggplot2::geom_smooth(method = "lm", se = FALSE, linewidth = 0.7) +
  ggplot2::geom_point(size = 3) +
  ggplot2::annotate("text", x = max(temps), y = log10(60),
    label = "log[10]~60~(1*' hour')", parse = TRUE,
    hjust = 1, vjust = -0.4, colour = "grey50", size = 3) +
  ggplot2::annotate("text", x = true_CTmax, y = -Inf,
    label = "CT[max[1*hr]]", parse = TRUE,
    hjust = -0.15, vjust = -0.7, colour = "grey50", size = 3) +
  ggplot2::labs(
    x = "Assay temperature (°C)",
    y = expression(log[10]~LT[50]~(min)),
    colour = "Stage-1 dataset"
  ) +
  ggplot2::theme_classic()

```



Point estimates of z and $CT_{max_{1hr}}$ are then used as is often without uncertainty associated with these estimates. We compare these to the joint Bayesian results to show the 4PL gives the same values under these ideal conditions.

! Important

Classical two-stage analyses often treat Stage-1 LT_{50} estimates as known when fitting Stage-2 models. In other words, uncertainty in LT_{50} values are never accounted for in the Stage-2 regression. This can have consequences when estimating slopes and intercepts in Stage-2. Despite being a potential issue, it is also challenging to correctly propagate error to z and $CT_{max_{1hr}}$. Incorrectly doing so in Stage-2 may have bigger impacts. Without propagating uncertainty properly these will systematically bias sampling uncertainty in z and $CT_{max_{1hr}}$ compromising inferences and leading to optimistic predictions. The Delta-method or bootstrap procedures can be used to propagate uncertainty, though, these are rarely used in practice. In addition, even if uncertainty was correctly propagated, a likelihood approach makes it harder to propagate uncertainty through downstream predictions (e.g., heat injury).

3.0.3 A joint Bayesian 4PL

Instead of fitting separate logistic curves and then regressing their LT_{50} estimates, the joint fit estimates ℓ , up , k , β_0 , and β_1 simultaneously from the raw counts. We use the same simulated data to derive z and $CT_{max_{1hr}}$ directly from this model from the posteriors of these parameters.

3.0.3.1 Specifying the 4PL in brms

The model has two pieces. The first is the 4PL itself — probability as a function of $\log_{10}$ exposure time:

$$p = \ell + \frac{up - \ell}{1 + \exp(k(\log_{10} t - \text{mid}))}, \quad (\text{S3})$$

where ℓ and up are the lower and upper asymptotes, k is the slope on the $\log_{10} t$ axis, and mid is the value of $\log_{10} t$ at which the curve crosses the midpoint between ℓ and up . The second piece lets the midpoint vary linearly with mean-centred temperature:

$$\text{mid} = \beta_0 + \beta_1 (T - \bar{T}), \quad (\text{S4})$$

so that β_1 encodes how fast the dose-response curve shifts on the time axis as assay temperature changes; $\bar{T}$ is the grand-mean temperature, included so that β_0 is interpretable as the midpoint at the average assay temperature.

The function library exposes this model as a single call: `fit_4pl()` constructs the brms `bf()` formula, the default weakly-informative priors, and the sampling controls, and returns a workflow object containing the fit. Internally the formula uses the disjoint-bounds reparameterisation described in §[Loading the function library] — `low = 0.001 + \text{inv\_logit}(\text{lowraw}) \cdot 0.498` and `up = 0.501 + \text{inv\_logit}(\text{upraw}) \cdot 0.498` — so the

4PL output is guaranteed to lie in (0,1) regardless of where MCMC takes the unconstrained `lowraw`, `upraw`, and `logk` parameters. The default also puts a `temp_c` slope on every 4PL sub-parameter; when temperature has no real effect on a given parameter, the slope shrinks toward zero under the prior. We inspect the brms formula `fit_4pl()` builds with:

```
# Show the 4PL brms formula fit_4pl() builds.
f <- make_4pl_formula()
cat("4PL dose-response (disjoint-bounds reparameterisation; survival stays in (0, 1)):\n\n")
```

4PL dose-response (disjoint-bounds reparameterisation; survival stays in (0, 1)):

```
cat("  ", paste(deparse(f$formula[[2]]), collapse = ""),
    " ~ low + (up - low) / (1 + exp(k * (logd - mid)))\n", sep = "")
```

```
n_surv | trials(n_total) ~ low + (up - low) / (1 + exp(k * (logd - mid)))
```

```
cat("      low = 0.001 + inv_logit(lowraw) * 0.498\n")
```

```
low = 0.001 + inv_logit(lowraw) * 0.498
```

```
cat("      up  = 0.501 + inv_logit(upraw)  * 0.498\n")
```

```
up  = 0.501 + inv_logit(upraw)  * 0.498
```

```
cat("      k   = exp(logk)\n\n")
```

```
k   = exp(logk)
```

```
cat("  Sub-parameter linear predictors (default: a temp_c slope on each):\n")
```

Sub-parameter linear predictors (default: a `temp_c` slope on each):

```
for (nm in c("lowraw", "upraw", "logk", "mid"))
  cat(sprintf("      %-6s ~ %s\n", nm, paste(deparse(f$pforms[[nm]][[3]]), collapse = "")))
```

```
lowraw ~ temp_c
upraw  ~ temp_c
logk   ~ temp_c
mid    ~ temp_c
```


! Important

The default model puts a `temp_c` slope on all four 4PL sub-parameters and includes no random effects. How random effects are added depends on the parameterisation: in the **midpoint** parameterisation (used here) pass `random_effects = c("Date", "Tank", ...)` to `fit_4pl()` to place random intercepts on `mid`; in the **direct** CTmax/z parameterisation they instead go *inside* the `ctmax/z` formulas, e.g. `ctmax = ~ 0 + group + (1 | batch)`. The simulation here has no batch structure so we leave them off; the snow gum leaf case study below uses the direct form with a per-plant random intercept (`ctmax = ~ 0 + recovery + (1 | plant)`).

3.0.3.2 Standardising and fitting

The first step is to pass the simulated data through `standardize_data()` so the columns match the schema the rest of the library expects (`temp`, `duration`, `logd`, `temp_c`, `n_total`, `n_surv`). This function also keeps track of your reference temperature, time units and centering so it's important to use for calculations to be done correctly.

```
# Convert the simulated counts into the bayesTLS modelling schema.
std_bin <- standardize_data(sim_bin,
  temp = "T", duration = "t",
  n_total = "n", n_surv = "y",
  duration_unit = "minutes")
std_bb <- standardize_data(sim_bb,
  temp = "T", duration = "t",
  n_total = "n", n_surv = "y",
  duration_unit = "minutes")
```

We now call `fit_4pl()` twice — once with the built-in `binomial(link = "identity")` family for the non-overdispersed simulation, and once with the default `beta_binomial(link = "identity")` for the overdispersed case. We store the model fits using `brms's file = ... mechanism`, so subsequent code calls just reload the cached fits unless the formula or data changes.

```
# Fit the tutorial 4PL models with and without overdispersion.
# Reuse cached fits; delete the .rds files to force refitting.
wf_bin <- fit_4pl(std_bin,
  family = binomial(link = "identity"),
  chains = 4, iter = 2000, cores = 4, seed = 123,
  file = file.path(models_dir, "sim_4pl_binomial_v2"),
  file_refit = "never")

wf_bb <- fit_4pl(std_bb,
  chains = 4, iter = 2000, cores = 4, seed = 123,
  file = file.path(models_dir, "sim_4pl_betabinomial_v2"),
  file_refit = "never")
```

```
# Show brms summaries and diagnostics for each fit.
summary(wf_bin)
```

```
Family: binomial
Links: mu = identity
Formula: n_surv | trials(n_total) ~ (0.001 + inv_logit(lowraw) * 0.498) + ((0.501 + inv_logit(
  lowraw ~ temp_c
  upraw ~ temp_c
  logk ~ temp_c
  mid ~ temp_c
Data: data (Number of observations: 900)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Regression Coefficients:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
lowraw_Intercept	-2.82	0.09	-3.01	-2.66	1.00	2519	2773
lowraw_temp_c	-0.01	0.03	-0.07	0.05	1.00	2787	2728
upraw_Intercept	2.68	0.07	2.56	2.82	1.00	3357	2710
upraw_temp_c	-0.02	0.02	-0.07	0.03	1.00	3350	2790
logk_Intercept	2.12	0.03	2.07	2.17	1.00	2971	2979
logk_temp_c	0.00	0.01	-0.02	0.02	1.00	3151	2954
mid_Intercept	1.50	0.00	1.49	1.51	1.00	3931	2847
mid_temp_c	-0.15	0.00	-0.15	-0.15	1.00	5742	2551

Draws were sampled using `sample(hmc)`. For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
summary(wf_bb)
```

```
Family: beta_binomial
Links: mu = identity
Formula: n_surv | trials(n_total) ~ (0.001 + inv_logit(lowraw) * 0.498) + ((0.501 + inv_logit(
  lowraw ~ temp_c
  upraw ~ temp_c
  logk ~ temp_c
  mid ~ temp_c
Data: data (Number of observations: 900)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Regression Coefficients:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
lowraw_Intercept	-3.05	0.25	-3.61	-2.63	1.00	2222	2317
lowraw_temp_c	0.10	0.09	-0.06	0.28	1.00	2086	2207

upraw_Intercept	2.79	0.14	2.52	3.08	1.00	3021	2743
upraw_temp_c	0.03	0.05	-0.06	0.13	1.00	3072	2656
logk_Intercept	1.99	0.06	1.89	2.10	1.00	2980	2596
logk_temp_c	-0.01	0.02	-0.06	0.03	1.00	2681	2243
mid_Intercept	1.52	0.01	1.50	1.54	1.00	3323	2694
mid_temp_c	-0.15	0.00	-0.16	-0.14	1.00	5046	3113

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
phi	4.88	0.43	4.08	5.74	1.00	3502	3194

Draws were sampled using `sample(hmc)`. For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
# Extract the underlying brmsfit objects for brms helpers.
fit_bin <- get_brmsfit(wf_bin)
fit_bb  <- get_brmsfit(wf_bb)
```

We also summarise the proportion of outcome variation explained by each fitted model using Bayesian R^2 . For any fitted workflow this is a single `bayes_R2_tls()` call. We define a small helper that maps it over a named list of fits and formats the result into a median [95% CrI] table, then reuse it in every case study below.

```
# bayes_R2_tls() pulls the brmsfit and returns a tidy one-row tibble
# (estimate / est_error / lower / upper) -- one call per fit:
bayes_R2_tls(wf_bin)
```

```
# A tibble: 1 x 4
  estimate est_error lower upper
  <dbl>     <dbl> <dbl> <dbl>
1   0.988   0.000130 0.987 0.988
```

```
# For the tables we map that single call over a named list of fits and format it
# as "median [95% CrI]"; this thin formatter is reused in every case study below.
bayes_r2_table <- function(fits, digits = 3) {
  purrr::imap_dfr(fits, function(wf, nm) {
    r <- bayes_R2_tls(wf)
    tibble::tibble(
      Model      = nm,
      `Bayesian R²` = format_interval(r$estimate, r$lower, r$upper, digits = digits)
    )
  })
}
```

Table S4: Bayesian R^2 for the two tutorial joint 4PL fits (binomial and beta-binomial), via `bayes_R2_tls()` (a tidy wrapper around `brms::bayes_R2()`). Cells are the posterior median with the 95% credible interval.

Model	Bayesian R^2
Binomial 4PL	0.988 [0.987, 0.988]
Beta-binomial 4PL	0.926 [0.924, 0.928]

```
# Format Bayesian R2 for the tutorial fits.
bayes_r2_table(list(
  "Binomial 4PL"      = wf_bin,
  "Beta-binomial 4PL" = wf_bb
)) |> tinytable::tt(escape = TRUE)
```

3.0.3.3 Checking parameter recovery

We assess recovery of true values by comparing posterior medians and 95% credible intervals with the known generating values (Table S5). Given the large simulated sample, the posterior should concentrate near those values.

```
# tdt_parameter_table() returns the per-parameter posterior summary on the
# natural 4PL scale (median + 95% CrI) directly from a fit, so we don't roll our
# own. We compare the four 4PL parameters and the temperature slope to the known
# truth; the z row it also returns is the derived quantity tabulated in
# @sec-joint-derive. format_interval() renders each cell as "median [2.5%, 97.5%]".
truth <- tibble::tibble(
  Parameter = c("low (lower asymptote)", "up (upper asymptote)", "k (slope)",
    "mid intercept (at T_bar)", "mid temp_c slope"),
  Truth      = round(c(true_params$ell, true_params$up, true_params$k,
    true_params$m_beta0, true_params$m_beta1), 3)
)
recovery_cell <- function(wf) tdt_parameter_table(wf) |>
  dplyr::filter(parameter != "z (°C)") |>
  dplyr::transmute(Parameter = parameter,
    cell = format_interval(median, lower, upper, digits = 3))
recovery <- truth |>
  dplyr::left_join(recovery_cell(wf_bin) |> dplyr::rename(Binomial = cell),
    by = "Parameter") |>
  dplyr::left_join(recovery_cell(wf_bb) |> dplyr::rename(`Beta-binomial` = cell),
    by = "Parameter")

tinytable::tt(recovery, digits = 3, escape = TRUE)
```

We can also overlay the posterior fitted curves on the simulated data (Figure S3). If the model has recovered the true surface, the fitted lines should track the simulated proportions at every assay

Table S5: Posterior medians and 95% credible intervals for the four 4PL parameters and the temperature slope, compared to the known truth used to generate the simulated data. Recovery is judged successful when the credible interval covers the truth.

Parameter	Truth	Binomial	Beta-binomial
low (lower asymptote)	0.03	0.029 [0.024, 0.034]	0.024 [0.014, 0.035]
up (upper asymptote)	0.97	0.967 [0.963, 0.971]	0.97 [0.962, 0.977]
k (slope)	8	8.316 [7.893, 8.771]	7.327 [6.589, 8.176]
mid intercept (at T_{bar})	1.5	1.501 [1.493, 1.51]	1.518 [1.497, 1.538]
mid temp_c slope	-0.15	-0.15 [-0.152, -0.146]	-0.15 [-0.157, -0.142]

temperature.

```
# Plot fitted survival curves against the simulated observations.
# Use the library's predict_survival_curves() and plot_survival_curves().
psc_bin <- predict_survival_curves(
  wf_bin, temps = temps,
  durations = 10 ^ seq(log10(0.5), log10(800), length.out = 100),
  ndraws = 200
)

plot_survival_curves(psc_bin, observed = std_bin, log_time = TRUE) +
  ggplot2::scale_x_log10() +
  ggplot2::labs(x = expression(exposure~time~(min)))
```

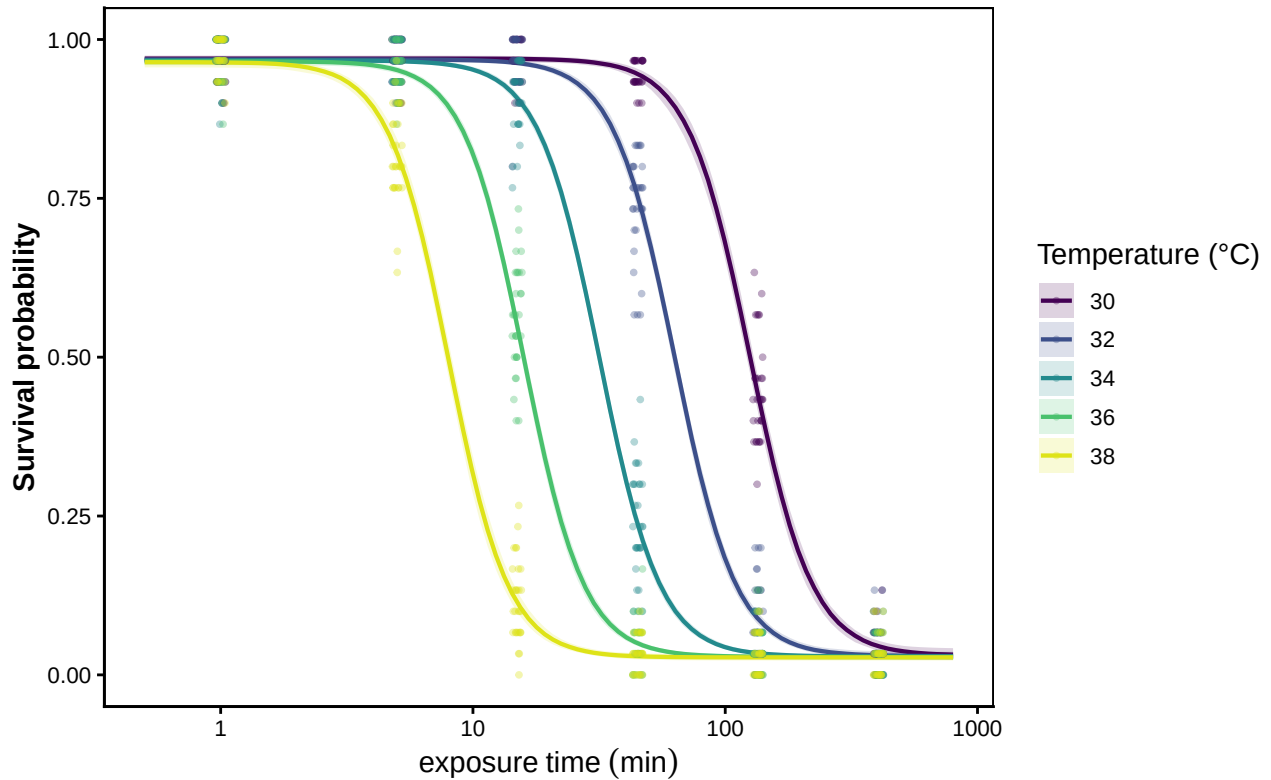


Figure S3: Posterior fitted 4PL curves (solid lines, with shaded 95% credible bands) overlaid on the simulated binomial data (points). One panel-coloured line per assay temperature. Where the fitted curves track the simulated proportions across all five temperatures, the model has recovered the underlying dose-response surface.

3.0.4 Deriving z , $CT_{max_{1hr}}$, and T_{crit} from the posterior

The classical TLS quantities are transformations of the fitted joint posterior. The general extractor `tls()` (used in the tutorial below) and the workflow-specific `extract_tdt()` both return the first two quantities below, and the third when `lethal = TRUE`; they share the same `target_surv`, `t_ref` and `lethal` arguments and agree for a `fit_4pl()` fit:

- z — thermal sensitivity is read directly from the posterior as $z = -1/(d \log_{10} LT_{\text{threshold}}/dT)$

per draw (Equation 7 of the manuscript). Under the relative threshold, $\log_{10} LT(T) = \text{mid}(T)$ and the derivative is simply the linear slope, so $z = -1/\beta_1$ exactly.

- $CT_{max_{1hr}}$ — temperature at which the threshold is reached after 1 h of exposure, obtained from the fitted 4PL at the reference time, per posterior draw.
- T_{crit} (only when `lethal = TRUE`) — the temperature below which damage accumulates more slowly than a chosen rate floor. The damage rate at temperature T in the TDT framework is $r(T) = 100 \cdot 10^{(T-CT_{max,1hr})/z}$ % LT50-dose per hour, so inverting for a chosen rate floor r^* gives

$$T_{crit}(r^*) = CT_{max,1hr} + z \cdot \log_{10}(r^*/100). \quad (\text{S5})$$

The threshold one wishes to use is set by the `target_surv` argument (shared by `tls()` and `extract_tdt()`). The default, `"relative"`, evaluates at $(\text{low} + \text{up})/2$ per draw, which equals classical absolute 50% survival when asymptotes are 0 and 1 but accommodates drifting upper asymptotes (for example, background mortality unrelated to heat stress). Passing `target_surv = "absolute"` or any numeric value in $(0, 1)$ instead uses an absolute survival threshold. Because the simulated asymptotes are near 0 and 1, both definitions give essentially the same results here.

How do we estimate T_{crit} ? Different r^* values imply different T_{crit} thresholds, and the literature does not converge on one value: breakpoint analyses of $\log_{10}(\text{time-to-thermal-failure})$ against temperature place the critical temperature anywhere from roughly 27 to 40 °C depending on the assay and taxon (including *Drosophila suzukii* and *Lemna gibba*) (Faber *et al.* 2026). Rather than fix one r^* , we adopt a plausible operational range — $r^* \in [0.1, 1]$ % HI per hour, the heat-injury rate at which net damage begins to accrue — corresponding to approximately 100 to 1000 hours (about 4 to 42 days) of continuous exposure to reach LT_{50} . We integrate over a uniform prior on $\log_{10} r^*$ across this range. The resulting posterior for T_{crit} includes parameter uncertainty in $CT_{max,1hr}$ and z as well as operational uncertainty in r^* . Its median is $CT_{max,1hr} - 2.5 \cdot z$, corresponding to $r^* \approx 0.3$ % HI per hour.

⚠ T_{crit} is meaningful only for lethal-endpoint data

The rate-multiplier T_{crit} in Equation Equation S5 interprets z as the temperature sensitivity of *damage accumulation*, which is appropriate for lethal-endpoint dose-response data and can be used to make a prediction about T_{crit} because T_{crit} should be the temperature where damage begins to accumulate (0.1-1% is what we assume). Setting `lethal = TRUE` (in `tls()` or `extract_tdt()`) is therefore able to predict T_{crit} with lethal data; with `lethal = FALSE` the extractors return only z and $CT_{max_{1hr}}$. Users with lethal-endpoint data can set `lethal = TRUE` to additionally derive an *estimate* of T_{crit} . Note that this is only a prediction and does not negate the need to follow up with studies to validate (see Faber *et al.* 2026).

For **sublethal** endpoints fitted z instead describes performance loss and can be very different from z derived from lethal data. Substituting that value into Equation Equation S5 can shift T_{crit} substantially and produce non-sensical T_{crit} estimates. For example, a smaller sublethal z can produce a T_{crit} several °C higher than the lethal estimate despite a similar $CT_{max_{1hr}}$. Interpreting T_{crit} as an intersection between thermal-performance and thermal-death-time curves requires both lethal and sublethal data (Faber *et al.* 2026; Jørgensen *et al.* 2021; Ørsted *et al.* 2022).

Returned quantities (e.g., $CT_{max,1hr}$ and z) inherit the same posterior, so every credible interval reflects joint uncertainty in $(\ell, \text{up}, k, \beta_0, \beta_1)$. The opt-in T_{crit} pools that same posterior with the additional uniform prior on $\log_{10} r^*$.

i Note

The default `fit_4pl()` puts a `temp_c` slope on every 4PL sub-parameter, so ℓ , `up` and k can each carry a linear effect of temperature in addition to the linear-mid term — when those slopes shrink to zero the model reduces to the constant-shape case. However, when the data carry a real temperature signal on any of the shape parameters, the asymmetry correction $\frac{1}{k(T)} \log \frac{\text{up}(T)-0.5}{0.5-\ell(T)}$ varies with T and the resulting $\log_{10} \text{LT50}(T)$ curve is bent rather than straight:

$$\log_{10} \text{LT50}(T) = \beta_0 + \beta_1(T - \bar{T}) + \frac{1}{k(T)} \log \frac{\text{up}(T)-0.5}{0.5-\ell(T)}. \quad (\text{S6})$$

The correction factor is needed when calculating an ‘absolute’ threshold (e.g., 50% mortality - `target_surv = "absolute"`). Under an *absolute* threshold the asymmetry-correction term $\frac{1}{k(T)} \log \frac{\text{up}(T)-0.5}{0.5-\ell(T)}$ bends the curve when `up`, ℓ or k carry T effects; then a per-draw **local** $z(T) = -1/(\text{d} \log_{10} \text{LT}_{0.5}/\text{d}T)$ is computed at each assay temperature. The reported z is the per-draw mean of those local values over the assay range — with the full per-temperature $z(T)$ available via `z_local = TRUE`.

If using a ‘relative threshold’ (`target_surv = "relative"`) then the calculations do not require the correction factor. $CT_{max,1hr}$ is the temperature at which $\log_{10} \text{LT}(T)$ crosses t_{ref} , per draw. For z , under the relative threshold $\log_{10} \text{LT}(T) = \text{mid}(T)$ is exactly linear, so $z = -1/\beta_1$ per draw. The worked example in Section 3.0.6 demonstrates this with a simulation where `up` and k carry strong T effects.

Under `target_surv = "absolute"`, `extract_tdt()` includes the asymmetry-correction term automatically, evaluating it per draw from the same posterior of ℓ , `up` and k so that any temperature-dependence of those parameters is propagated. The draws used to compute z and $CT_{max,1hr}$ are subsampled once across all parameters, so the two quantities share draws — their correlation, and the joint pairing used for T_{crit} , is preserved.

```
# tls() is the general one-call extractor: from any fitted 4PL it returns z,
# CTmax and -- when lethal = TRUE -- T_crit, as a tidy-long summary ($summary)
# plus the underlying posterior draws ($draws). One call per fit; t_ref = 60 min
# sets the 1-hour reference and lethal = TRUE adds T_crit (the endpoint is lethal).
tl_bin <- tls(wf_bin, target_surv = "relative", t_ref = 60, lethal = TRUE, ndraws = 1000)
tl_bb <- tls(wf_bb, target_surv = "relative", t_ref = 60, lethal = TRUE, ndraws = 1000)

# One row per quantity (z, CTmax, Tcrit): posterior median and 95% interval.
tl_bin$summary
```

```
# A tibble: 3 x 4
  quantity median lower upper
  <chr>      <dbl> <dbl> <dbl>
1 z          6.69  6.55  6.83
2 CTmax      32.1  32.1  32.2
```


Table S6: Posterior medians and 95% credible intervals for thermal sensitivity (z), the critical thermal limit at a 1-hour reference duration ($CT_{max_{1hr}}$), and the rate-multiplier critical temperature (T_{crit} , integrated over $r^* \in [0.1, 1]$ % HI per hour). All three derived via `tls()` from a single posterior. Each row's truth value is computed analytically from `true_params`.

Quantity	Truth	Binomial	Beta-binomial
z (°C)	6.67	6.69 [6.55, 6.83]	6.67 [6.37, 7.03]
CTmax at 1 hr (°C)	32.15	32.15 [32.08, 32.21]	32.26 [32.1, 32.41]
T_{crit} (°C)	15.48	15.48 [12.19, 18.54]	15.57 [12.08, 18.89]

3 Tcrit 15.5 12.2 18.5

Table S6 reads each quantity's median and interval straight off these `tls()` summaries, and Figure S4 plots the matching `$draws`.

```
# tls()$summary returns one row per quantity (z, CTmax, Tcrit) with median /
# lower / upper, in that order. format_interval() (vectorised) renders each cell
# as "median [2.5%, 97.5%]".
tls_recovery <- tibble::tibble(
  Quantity      = c("z (°C)", "CTmax at 1 hr (°C)", "T_crit (°C)"),
  Truth         = round(c(true_z, true_CTmax, true_T_crit), 2),
  Binomial      = format_interval(tl_bin$summary$median, tl_bin$summary$lower, tl_bin$summary$upper),
  `Beta-binomial` = format_interval(tl_bb$summary$median, tl_bb$summary$lower, tl_bb$summary$upper)
)

tinytable::tt(tls_recovery, digits = 3, escape = TRUE)
```

```

# Plot posterior distributions of the recovered TLS quantities. tls()$draws is
# already tidy-long (quantity, .draw, value), so no reshaping is needed -- bind
# the two fits and relabel the quantities for the facets.
post_long <- dplyr::bind_rows(
  tl_bin$draws |> dplyr::mutate(model = "Binomial"),
  tl_bb$draws |> dplyr::mutate(model = "Beta-binomial")) |>
dplyr::mutate(quantity = factor(quantity,
  levels = c("z", "CTmax", "Tcrit"),
  labels = c("z (°C)",
    "CTmax_1hr (°C)",
    "T_crit (°C)")))

truth_long <- tibble::tibble(
  quantity = factor(c("z", "CTmax", "Tcrit"),
    levels = c("z", "CTmax", "Tcrit"),
    labels = levels(post_long$quantity)),
  truth = c(true_z, true_CTmax, true_T_crit)
)

ggplot2::ggplot(post_long, ggplot2::aes(value, fill = model)) +
  ggplot2::geom_density(alpha = 0.5, colour = NA) +
  ggplot2::geom_vline(data = truth_long,
    ggplot2::aes(xintercept = truth),
    linetype = "dashed", colour = "black") +
  ggplot2::facet_wrap(~ quantity, scales = "free") +
  ggplot2::labs(x = NULL, y = "Posterior density", fill = "Model") +
  ggplot2::theme_classic()

```

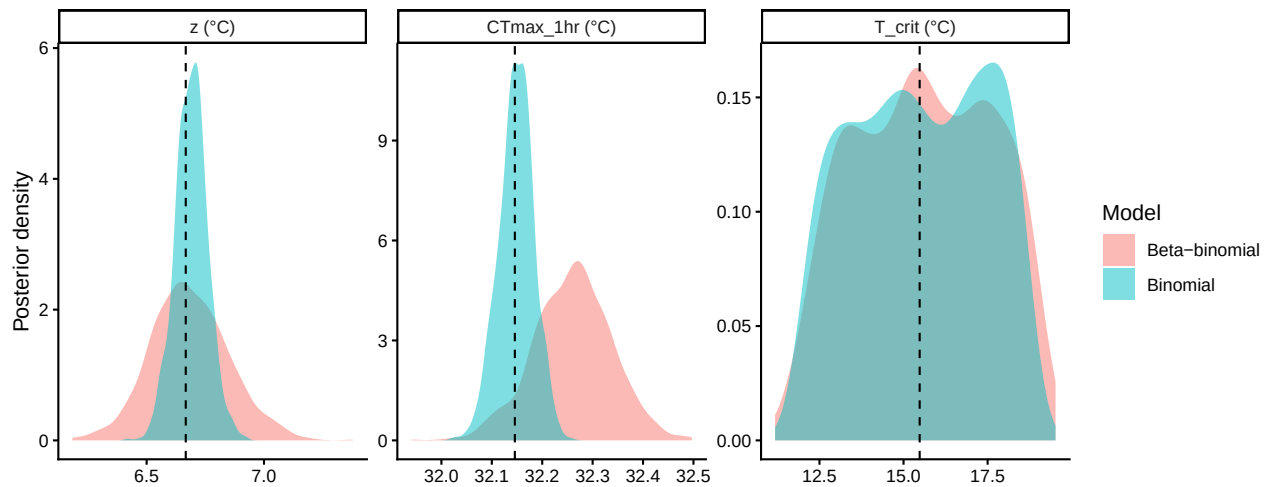


Figure S4: Posterior densities of z , $CT_{max_{1hr}}$, and T_{crit} from the binomial (blue) and beta-binomial (orange) joint fits. Vertical dashed lines mark the simulation truth. All three posteriors concentrate around their respective truths, illustrating that a single call to `tls()` produces each classical TLS quantity with posterior uncertainty and no separate Stage-2 regression.

Both fits recover all three simulation truths (Table S6, Figure S4), with credible intervals reflecting joint uncertainty in the 4PL parameters. In the joint approach, every classical TLS quantity is derived from the same fitted posterior.

Posterior summary statistics — mean, SD, median, and 95% credible interval — follow directly from the `$draws` slot of the same `tls()` output:

```
# Posterior summaries of z, CTmax and T_crit per likelihood. tls()$draws is
# tidy-long already (quantity, .draw, value), so we summarise each quantity x fit
# directly. The inline numbers below read off this table.
uncert_summary <- dplyr::bind_rows(
  tl_bin$draws |> dplyr::mutate(model = "Binomial"),
  tl_bb$draws  |> dplyr::mutate(model = "Beta-binomial")) |>
dplyr::group_by(quantity, likelihood = model) |>
dplyr::summarise(mean = mean(value), sd = stats::sd(value),
  median = stats::median(value),
  lower = stats::quantile(value, 0.025, names = FALSE),
  upper = stats::quantile(value, 0.975, names = FALSE),
  .groups = "drop") |>
dplyr::mutate(
  quantity = factor(quantity, c("z", "CTmax", "Tcrit"),
    c("z (°C)", "CTmax at 1 hr (°C)", "T_crit (°C)")),
  likelihood = factor(likelihood, c("Binomial", "Beta-binomial"))
) |>
dplyr::arrange(likelihood, quantity)

# Scalars for inline reporting, read straight off uncert_summary (single source).
g <- function(q, lik, col)
  uncert_summary[[col]][uncert_summary$quantity == q & uncert_summary$likelihood == lik]
sd_z_bin <- round(g("z (°C)", "Binomial", "sd"), 2)
sd_z_bb <- round(g("z (°C)", "Beta-binomial", "sd"), 2)
sd_ct_bin <- round(g("CTmax at 1 hr (°C)", "Binomial", "sd"), 2)
sd_ct_bb <- round(g("CTmax at 1 hr (°C)", "Beta-binomial", "sd"), 2)
ci_z_bin <- round(g("z (°C)", "Binomial", "upper") - g("z (°C)", "Beta-binomial", "upper"), 2)
ci_ct_bin <- round(g("CTmax at 1 hr (°C)", "Binomial", "upper") - g("CTmax at 1 hr (°C)", "Beta-binomial", "upper"), 2)

uncert_summary |>
dplyr::rename(Quantity = quantity, Likelihood = likelihood, Mean = mean,
  SD = sd, Median = median, Lower = lower, Upper = upper) |>
tinytable::tt(digits = 3, escape = TRUE)
```

Both z and $CT_{max_{1hr}}$ are estimated precisely, and their credible intervals come directly from the joint posterior. A frequentist joint fit could similarly quantify transformation uncertainty using the Delta method or using bootstrap approaches.

Under the binomial likelihood, the posterior SD for z is 0.07 °C and its 95% credible interval is 0.27 °C wide; $CT_{max_{1hr}}$ has SD 0.03 °C and interval width 0.14 °C.

Table S7: Posterior summaries of z , $CT_{max_{1hr}}$, and T_{crit} from the joint Bayesian 4PL: posterior mean, standard deviation, median, and 95% credible interval, reported separately for the binomial and beta-binomial fits.

Quantity	Likelihood	Mean	SD	Median	Lower	Upper
z (°C)	Binomial	6.69	0.0691	6.69	6.55	6.83
CTmax at 1 hr (°C)	Binomial	32.15	0.0347	32.15	32.08	32.21
T_{crit} (°C)	Binomial	15.48	1.9572	15.48	12.19	18.54
z (°C)	Beta-binomial	6.68	0.1663	6.67	6.37	7.03
CTmax at 1 hr (°C)	Beta-binomial	32.26	0.0776	32.26	32.1	32.41
T_{crit} (°C)	Beta-binomial	15.6	2.0115	15.57	12.08	18.89

The beta-binomial fit is slightly wider for both quantities (SD for z : 0.17 °C; for $CT_{max_{1hr}}$: 0.08 °C) because ϕ captures extra-binomial variation and propagates it into the derived quantities. A two-stage pipeline requires an additional bootstrap or delta-method calculation to achieve comparable propagation.

For per-draw functions of *several* quantities at once, the wide-paired form — one row per draw, with the quantities as columns — is more convenient than the long `$draws` above. `extract_tdt()` returns the bundled posterior object (per-quantity slots, the LT50 curve, plus the `get_*`() accessors), and `get_tls_draws()` pulls z , $CT_{max_{1hr}}$ and T_{crit} aligned on `.draw` into one tibble (an inner join, so the quantities keep their joint pairing — and join within group for grouped fits). That pairing is what lets a between-group contrast, a ratio, or a derived index be computed per draw with correctly-correlated uncertainty:

```
# extract_tdt() -> bundled object; get_tls_draws() -> one row per draw with
# z / CTmax / T_crit aligned on `.draw`. The merge is an inner join on `.draw`,
# so the quantities stay jointly paired (a column-bind could not).
et_bin <- extract_tdt(wf_bin, target_surv = "relative", t_ref = 60, lethal = TRUE, ndraws = 100)
d_bin <- get_tls_draws(et_bin)
head(d_bin)
```

```
# A tibble: 6 x 4
  .draw      z CTmax T_crit
  <int> <dbl> <dbl> <dbl>
1     1  6.71  32.2  14.3
2     2  6.80  32.2  17.7
3     3  6.62  32.2  14.5
4     4  6.71  32.1  16.7
5     5  6.75  32.2  13.9
6     6  6.61  32.2  13.2
```

```
# get_tls_summary() is the summary companion -- it returns the same tidy table as
# tls()$summary above (same quantities, same threshold), confirming the bundled
```

```
# extract_tdt() object and the general tls() extractor agree for a fit_4pl() fit:
get_tls_summary(et_bin)
```

```
# A tibble: 3 x 4
  quantity median lower upper
  <chr>      <dbl> <dbl> <dbl>
1 z          6.69  6.56  6.83
2 CTmax       32.2  32.1  32.2
3 Tcrit       15.5  12.2  18.6
```

```
# z and CTmax co-vary in the posterior -- a relationship only visible on PAIRED
# draws (summarising the two separately would erase it):
round(stats::cor(d_bin$z, d_bin$CTmax), 2)
```

```
[1] -0.51
```

```
# Because they share a draw, derived per-draw quantities carry the joint
# uncertainty. CTmax_1hr - 2.5*z is T_crit at the central rate (r* ~ 0.3% HI/h);
# the full T_crit column is wider because it ALSO integrates the r* prior:
qs <- c(0.025, 0.5, 0.975)
stats::quantile(d_bin$CTmax - 2.5 * d_bin$z, qs) # paired z & CTmax, central rate
```

```
      2.5%      50%      97.5%
15.02896 15.44091 15.79009
```

```
stats::quantile(d_bin$T_crit, qs) # + rate uncertainty -> wider
```

```
      2.5%      50%      97.5%
12.24841 15.46710 18.61087
```

! Important

For real data, it is important to compare binomial and beta-binomial fits rather than assuming either likelihood is adequate. When extra dispersion is present, the choice can affect uncertainty and inference.

3.0.5 Comparing the two pipelines

3.0.5.1 Point-estimate agreement on the same data

Table S8 puts the two-stage point estimates next to the joint Bayesian posterior medians for both simulated datasets.

Table S8: Point estimates (two-stage) and posterior medians (joint Bayesian) for thermal sensitivity (z) and $CT_{max_{1hr}}$, recovered by each of four estimator-dataset combinations from the same simulated datasets. Truth values are the simulation parameters.

Quantity	Truth	Two-stage (binomial)	Joint Bayesian (binomial)	Two-stage (beta-binomial)
z ($^{\circ}\text{C}$)	6.67	6.96	6.69	7.3
CTmax at 1 hr ($^{\circ}\text{C}$)	32.15	32.03	32.15	31.9

```
# Compare tutorial estimates from the joint and two-stage workflows. The joint
# medians come straight off the tls() summaries derived above; match() pulls the
# z and CTmax rows by name (so the result is independent of row order).
compare_table <- tibble::tibble(
  quantity = c("z ( $^{\circ}\text{C}$ )", "CTmax at 1 hr ( $^{\circ}\text{C}$ )"),
  truth     = c(true_z, true_CTmax),
  ts_bin    = c(stage2_bin$summary$z, stage2_bin$summary$CTmax_1hr),
  joint_bin = t1_bin$summary$median[match(c("z", "CTmax"), t1_bin$summary$quantity)],
  ts_bb     = c(stage2_bb$summary$z, stage2_bb$summary$CTmax_1hr),
  joint_bb  = t1_bb$summary$median[match(c("z", "CTmax"), t1_bb$summary$quantity)]
)

compare_table |>
  dplyr::rename(
    Quantity      = quantity,
    Truth         = truth,
    `Two-stage (binomial)` = ts_bin,
    `Joint Bayesian (binomial)` = joint_bin,
    `Two-stage (beta-binomial)` = ts_bb,
    `Joint Bayesian (beta-binomial)` = joint_bb
  ) |>
  tynytble::tt(digits = 3, escape = TRUE)
```

All four estimators give similar point estimates for z and $CT_{max_{1hr}}$, close to the simulation truth. Under this simple generating scenario, the joint Bayesian 4PL reproduces the classical pipeline's central estimates while retaining a coherent posterior for downstream inference.

3.0.5.2 One model, two parameterisations

`fit_4pl()` defaults to the **midpoint** parameterisation: it estimates a linear midpoint $\text{mid}(T)$ and derives z and CT_{max} from its slope and intercept (Equation 7 of the manuscript). The **direct** parameterisation instead places **brms** non-linear parameters on z and the CT_{max} deviation themselves, requested by passing `ctmax = ~ 1` and `z = ~ 1`. The two are reparameterisations of one likelihood, so fitting the *same* simulated data both ways and extracting with `tls()` should return the same posterior for z and $CT_{max_{1hr}}$.

```
# Refit the binomial tutorial data with the DIRECT (z, CTmax) parameterisation.
# Same data, same family -- only the way the midpoint is written changes.
# Cached like the fits above; the .rds is created on first render.
wf_direct <- fit_4pl(std_bin,
  ctmax = ~ 1, z = ~ 1,
  family = binomial(link = "identity"),
  chains = 4, iter = 2000, cores = 4, seed = 123,
  file      = file.path(models_dir, "sim_4pl_direct"),
  file_refit = "never")
```

A single `tls()` call turns either fit into a tidy table of the derived quantities — one row per quantity, with the posterior median and 95% interval. It is the one-call extractor: pass it *any* fitted 4PL and it returns z , $CT_{max_{1hr}}$ and (for a lethal endpoint) T_{crit} , whatever the parameterisation.

```
# One call per fit -- no separate extract step. target_surv picks the threshold,
# t_ref sets the reference duration (minutes), lethal = TRUE adds T_crit.
tls_mid <- tls(wf_bin, target_surv = "relative", t_ref = 60, lethal = TRUE)
tls_dir <- tls(wf_direct, target_surv = "relative", t_ref = 60, lethal = TRUE)

# What a single tls() call returns for the direct fit:
tls_dir$summary
```

```
# A tibble: 3 x 4
  quantity median lower upper
  <chr>      <dbl> <dbl> <dbl>
1 z          6.64  6.51  6.77
2 CTmax      32.1  32.1  32.2
3 Tcrit      15.5  12.3  18.7
```

Table [S9](#) lays the two parameterisations side by side, reading z and $CT_{max_{1hr}}$ straight off each `tls()` summary.

```
# Pull the z and CTmax rows from each tls() summary (rows come back in the order
# z, CTmax, Tcrit) and lay them side by side. format_interval() is vectorised.
mid <- dplyr::filter(tls_mid$summary, quantity %in% c("z", "CTmax"))
dir <- dplyr::filter(tls_dir$summary, quantity %in% c("z", "CTmax"))

param_equiv <- tibble::tibble(
  Quantity      = c("z (°C)", "CTmax at 1 hr (°C)"),
  Truth         = round(c(true_z, true_CTmax), 2),
  `Midpoint (default)` = format_interval(mid$median, mid$lower, mid$upper, digits = 2),
  `Direct (z, CTmax)`  = format_interval(dir$median, dir$lower, dir$upper, digits = 2)
)

tinytable::tt(param_equiv, digits = 3, escape = TRUE)
```

Table S9: Posterior medians and 95% credible intervals for z and $CT_{max_{1hr}}$ recovered from the same simulated (binomial) data fit two ways — the default midpoint parameterisation and the direct (z, CT_{max}) parameterisation — both extracted with the single-call `tls()`. Truth values are the simulation parameters.

Quantity	Truth	Midpoint (default)	Direct (z, CT_{max})
z (°C)	6.67	6.69 [6.56, 6.83]	6.64 [6.51, 6.77]
CT_{max} at 1 hr (°C)	32.15	32.15 [32.08, 32.21]	32.13 [32.06, 32.19]

The midpoint and direct fits return the same z and $CT_{max_{1hr}}$ to within Monte Carlo error, confirming they are two ways of writing one model. The midpoint form is the default because its linear $\text{mid}(T)$ makes z a closed-form transform of a single slope (Equation 7 of the manuscript); the direct form is available when a prior or hypothesis is most naturally expressed on z or CT_{max} directly.

3.0.6 A worked example: temperature effects on every shape parameter

The default `fit_4pl()` places a `temp_c` slope on ℓ , up , and k , estimating temperature effects on curve shape alongside the linear midpoint. Strictly speaking, this isn't necessary if you are interested in *relative* measures so long as they are on the midpoint. However, these are needed for *absolute* measures because of the need for the correction factor.

Here, we test recovery when the constant-shape assumption fails, we re-simulate beta-binomial data on the design in Section 3.0.1.2 with up declining and k increasing sharply with T . We then apply the same `fit_4pl()` and `extract_tdt()` calls used above.

```
# Prepare the extended simulation dataset for the package workflow example.
# Same factorial design as @sec-sim-step2; re-declared here so this chunk is
# self-contained.
temps      <- c(30, 32, 34, 36, 38)
durations  <- c(1, 5, 15, 45, 135, 405)
n_rep      <- 30
n_per_rep  <- 30
phi        <- 5

# Set truth on the raw parameter scale used by fit_4pl().
true_params_ext <- list(
  ell_raw  = qlogis(0.05 / 0.49),           # ell = 0.05
  up_raw_0 = qlogis((0.92 - 0.51) / 0.49), # up = 0.92 at T_bar
  up_raw_T = -0.60,                        # strong decline in up with T
  log_k_0   = log(8),                      # k = 8 at T_bar
  log_k_T   = 0.30,                        # strong rise in k with T
  m_beta0   = 1.5,
  m_beta1   = -0.15,
  T_bar     = 34
)
```



```

design <- tidyr::expand_grid(
  T    = temps,
  t    = durations,
  rep = seq_len(n_rep)
) |>
dplyr::mutate(
  log10_t = log10(t),
  T_c     = T - true_params_ext$T_bar,
  n       = n_per_rep
)

sim_bb_ext <- design |>
dplyr::mutate(
  ell    = 0.49 * plogis(true_params_ext$ell_raw),
  up     = 0.51 + 0.49 * plogis(true_params_ext$up_raw_0 +
                                true_params_ext$up_raw_T * T_c),
  k      = exp(true_params_ext$log_k_0 + true_params_ext$log_k_T * T_c),
  mid    = true_params_ext$m_beta0 + true_params_ext$m_beta1 * T_c,
  p_true = ell + (up - ell) / (1 + exp(k * (log10_t - mid))),
  p_draw = rbeta(dplyr::n(), p_true * phi, (1 - p_true) * phi),
  y      = rbinom(dplyr::n(), size = n, prob = p_draw)
)

```

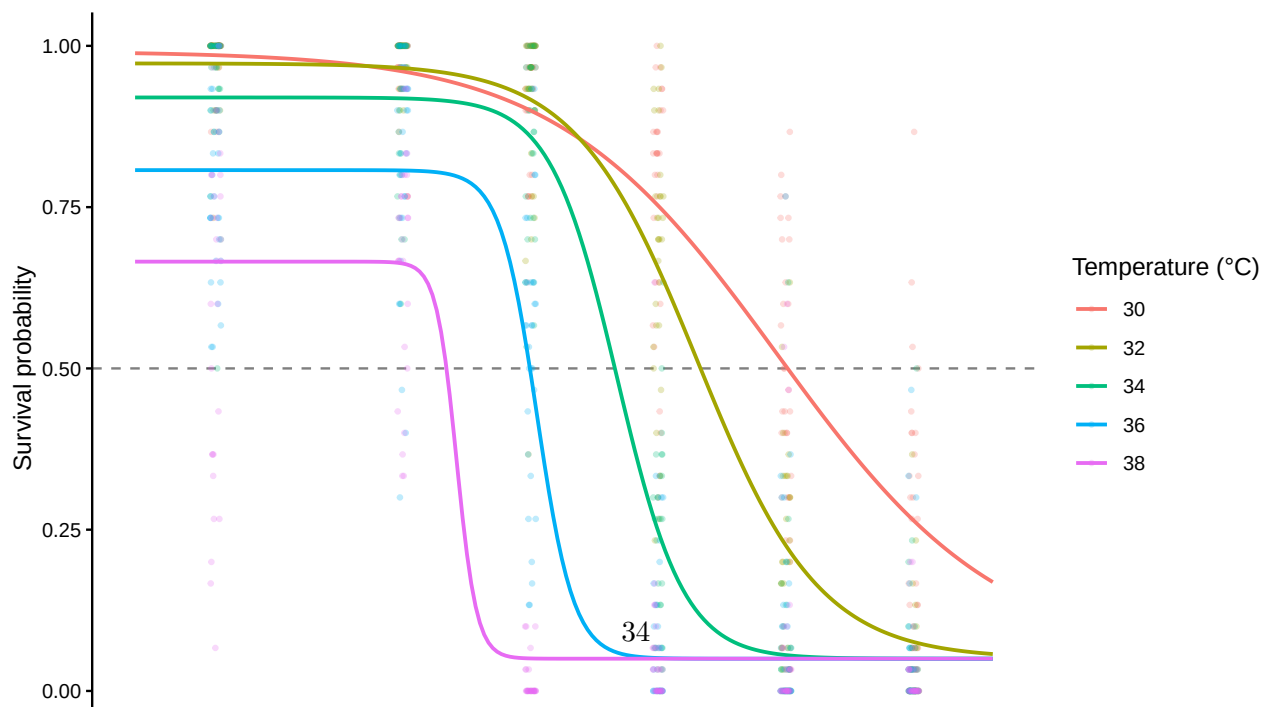
Here the lower asymptote remains at $\ell = 0.05$, while up falls from 0.99 at $T = 30$ to 0.67 at $T = 38$, and k rises from 2.4 to 26.6. The extreme-temperature curves therefore differ in shape, not merely position (Figure S5).

```

# Plot the true surface for the extended simulation example.
truth_grid_ext <- tidyr::expand_grid(
  T = temps,
  t = 10 ^ seq(log10(0.5), log10(800), length.out = 200)
) |>
dplyr::mutate(
  log10_t = log10(t),
  T_c     = T - true_params_ext$T_bar,
  ell     = 0.49 * plogis(true_params_ext$ell_raw),
  up      = 0.51 + 0.49 * plogis(true_params_ext$up_raw_0 +
                                true_params_ext$up_raw_T * T_c),
  k       = exp(true_params_ext$log_k_0 + true_params_ext$log_k_T * T_c),
  mid     = true_params_ext$m_beta0 + true_params_ext$m_beta1 * T_c,
  p_true  = ell + (up - ell) / (1 + exp(k * (log10_t - mid)))
)

ggplot2::ggplot() +
  ggplot2::geom_hline(yintercept = 0.5, linetype = "dashed",
                      colour = "grey50") +
  ggplot2::geom_jitter(
    data = sim_bb_ext,
    ggplot2::aes(x = log10_t, y = y / n, colour = factor(T)),
    width = 0.02, height = 0, alpha = 0.25, size = 0.7
  ) +
  ggplot2::geom_line(
    data = truth_grid_ext,
    ggplot2::aes(x = log10_t, y = p_true, colour = factor(T)),
    linewidth = 0.8
  ) +
  ggplot2::labs(x = expression(log[10]~exposure~time~(min)),
                y = "Survival probability",
                colour = "Temperature (°C)") +
  ggplot2::theme_classic()

```



Before fitting, we compute the analytical targets: $\log_{10} \text{LT50}(T)$, local $z(T)$ (negative reciprocal of the local slope of $\log_{10} \text{LT50}$ on T), and $CT_{max_{1hr}}$ (the temperature at which $\log_{10} \text{LT50}$ crosses $\log_{10} 60$).

```
# Derive the true TLS quantities for the extended simulation example.
# True log10(LT50) at any T from the T-varying 4PL: solve p(log10_t) = 0.5
# for log10_t. With the 4PL,
#   log10_t = mid + (1/k) * log((up - 0.5) / (0.5 - ell))
true_log10_LT50 <- function(T_value) {
  T_c <- T_value - true_params_ext$T_bar
  ell <- 0.49 * plogis(true_params_ext$ell_raw)
  up <- 0.51 + 0.49 * plogis(true_params_ext$up_raw_0 +
                             true_params_ext$up_raw_T * T_c)
  k <- exp(true_params_ext$log_k_0 + true_params_ext$log_k_T * T_c)
  mid <- true_params_ext$m_beta0 + true_params_ext$m_beta1 * T_c
  mid + (1 / k) * log((up - 0.5) / (0.5 - ell))
}

# Local z(T) via central finite difference of true log10(LT50)(T).
true_local_z <- function(T_at, half_step = 0.5) {
  -1 / ((true_log10_LT50(T_at + half_step) -
        true_log10_LT50(T_at - half_step)) / (2 * half_step))
}

# True CTmax_1hr: search the fine grid for the T at which log10(LT50)
# crosses log10(60).
T_grid_truth <- seq(20, 45, by = 0.001)
ll_grid_truth <- vapply(T_grid_truth, true_log10_LT50, numeric(1))
true_CTmax_1hr <- T_grid_truth[which.min(abs(ll_grid_truth - log10(60)))]

truth_ext <- tibble::tibble(
  T = temps,
  log10_LT50_true = vapply(temps, true_log10_LT50, numeric(1)),
  LT50_min_true = 10 ^ log10_LT50_true,
  local_z_true = vapply(temps, true_local_z, numeric(1))
)

true_mean_local_z <- mean(truth_ext$local_z_true)
```

Across the assay range the true $\log_{10} \text{LT50}(T)$ curve is bent: the implied local z varies from 6.05 °C at $T = 30$ to 6.53 °C at $T = 38$ (truth mean $z = 6.28$ °C). The truth implies $CT_{max_{1hr}} = 32.21$ °C. With straight-line $\log_{10} \text{LT50}(T)$ — the constant-shape case — local z would not vary across temperature at all.

We now fit with exactly the same `fit_4pl()` call as before. The default formula already puts a `temp_c` slope on each of the four 4PL sub-parameters, so the model has the capacity to recover the truth's temperature signal on `up` and `k` without any modification.

```

# Fit the default package 4PL to the extended simulation data.
std_ext <- standardize_data(sim_bb_ext,
                           temp = "T", duration = "t",
                           n_total = "n", n_surv = "y",
                           duration_unit = "minutes")

wf_ext <- fit_4pl(std_ext,
                 chains = 4, iter = 2000, cores = 4, seed = 123,
                 file     = file.path(models_dir,
                                     "sim_4pl_ext_betabin_DEFAULT_strong_v1"),
                 file_refit = "never") # see fit-4pl-tutorial note above

```

Table S10 reports recovery of the four 4PL sub-parameters on their natural scale at each assay temperature. The posterior keeps ℓ near 0.05, while recovering the decline in up and increase in k across the assay range.

```

# Format parameter recovery for the extended simulation example.
ext_post <- posterior::as_draws_df(get_brmsfit(wf_ext)) |> as.data.frame()
ext_meta <- wf_ext$meta
ext_bnds <- ext_meta$bounds # compute_4pl_bounds(0, 1)
ext_temp_mean <- ext_meta$temp_mean # what the fit centred on

ext_param_long <- purrr::map_dfr(temps, function(T_value) {
  T_c_fit <- T_value - ext_temp_mean
  T_c_true <- T_value - true_params_ext$T_bar

  draws_at_T <- tibble::tibble(
    ell = ext_bnds$low_min +
      stats::plogis(ext_post$b_lowraw_Intercept +
                    ext_post$b_lowraw_temp_c * T_c_fit) * ext_bnds$low_w,
    up = ext_bnds$up_min +
      stats::plogis(ext_post$b_upraw_Intercept +
                    ext_post$b_upraw_temp_c * T_c_fit) * ext_bnds$up_w,
    k = exp(ext_post$b_logk_Intercept + ext_post$b_logk_temp_c * T_c_fit),
    mid = ext_post$b_mid_Intercept + ext_post$b_mid_temp_c * T_c_fit
  )

  truth_at_T <- tibble::tibble(
    ell = 0.49 * plogis(true_params_ext$ell_raw),
    up = 0.51 + 0.49 * plogis(true_params_ext$up_raw_0 +
                              true_params_ext$up_raw_T * T_c_true),
    k = exp(true_params_ext$log_k_0 +
            true_params_ext$log_k_T * T_c_true),
    mid = true_params_ext$m_beta0 + true_params_ext$m_beta1 * T_c_true
  )

  purrr::map_dfr(c("ell", "up", "k", "mid"), function(p) {

```

```

q <- stats::quantile(draws_at_T[[p]], c(0.025, 0.5, 0.975),
                      na.rm = TRUE, names = FALSE)
tibble::tibble(
  Parameter      = p,
  `T (°C)`       = T_value,
  Truth          = truth_at_T[[p]],
  Median         = q[2],
  Lower          = q[1],
  Upper          = q[3]
)
})
})

ext_param_long |>
  tinytable::tt(digits = 3, escape = TRUE)

```

Now we apply the same `extract_tdt()` call as in Section 3.0.4 — there is no separate “T-varying” workflow because `extract_tdt()` already inverts the fitted surface numerically per draw:

```

# Extract TDT quantities from the extended simulation fit. We use the ABSOLUTE
# threshold here: with temperature effects on up/k the log10(LT50) curve is bent,
# so its local z varies with T and the asymmetry correction must be evaluated per
# draw. (The relative threshold reads mid(T), which is linear and would give a
# constant z that cannot match the bent truth.) z_local = TRUE additionally
# returns the per-temperature local z(T) used below.
et_ext <- extract_tdt(wf_ext, target_surv = "absolute", t_ref = 60, time_multiplier = 1,
                      ndraws = 1000, z_local = TRUE)

```

Table S11 compares the three classical TDT quantities from `extract_tdt()` to the analytical truth derived above.

```

# Format extracted TDT quantities for the extended simulation fit (z and CTmax;
# T_crit is omitted here -- it is the rate-multiplier prediction demonstrated in
# the tutorial, not the focus of this bent-curve recovery check).
tibble::tibble(
  Quantity      = c("z (°C)", "CTmax at 1 hr (°C)"),
  Truth         = c(round(true_mean_local_z, 3), round(true_CTmax_1hr, 3)),
  Median        = c(et_ext$z$summary$z_median, et_ext$CTmax$summary$temp_median),
  Lower         = c(et_ext$z$summary$z_lower, et_ext$CTmax$summary$temp_lower),
  Upper         = c(et_ext$z$summary$z_upper, et_ext$CTmax$summary$temp_upper)
) |>
  tinytable::tt(digits = 3, escape = TRUE)

```

Both posterior intervals cover their analytical truths. Under the absolute threshold the single-summary z is the per-draw local $z(T)$ averaged over the assay temperatures; it recovers the truth (mean local $z = 6.28$ °C; posterior median 6.19 °C, 95% CrI [5.78, 6.65] °C). Because the curve is

Table S10: Posterior recovery of the four 4PL sub-parameters on the natural scale at each assay temperature. The fit uses the disjoint-bounds reparameterisation (ℓ , up via `inv_logit`, k via `exp`); raw posterior columns are transformed back to the biological scale before summarising. Posterior medians and 95% credible intervals are reported alongside the analytical truth.

Parameter	T (°C)	Truth	Median	Lower	Upper
ell	30	0.05	0.0625	0.0293	0.1084
up	30	0.991	0.9935	0.989	0.9962
k	30	2.41	2.4043	2.1161	2.7386
mid	30	2.1	2.1065	2.0379	2.1701
ell	32	0.05	0.0536	0.0311	0.0797
up	32	0.973	0.9791	0.9697	0.9863
k	32	4.39	4.5723	4.1102	5.1149
mid	32	1.8	1.8061	1.7625	1.847
ell	34	0.05	0.0457	0.0328	0.0596
up	34	0.92	0.932	0.9166	0.9463
k	34	8	8.7122	7.1951	10.6985
mid	34	1.5	1.5056	1.4818	1.5292
ell	36	0.05	0.0391	0.0317	0.0475
up	36	0.807	0.8168	0.7885	0.842
k	36	14.577	16.5544	12.261	22.8179
mid	36	1.2	1.2049	1.1814	1.2328
ell	38	0.05	0.0335	0.0248	0.0444
up	38	0.666	0.6592	0.6201	0.6995
k	38	26.561	31.5023	20.6197	48.8663
mid	38	0.9	0.9041	0.8624	0.9533

Table S11: Classical TDT quantities from `extract_tdt(target_surv = "absolute")` on the T-varying fit, compared against analytical truth. Under the absolute threshold the single-summary z is the per-draw **local** $z(T)$ of the bent $\log_{10}$ LT50(T) curve, pooled (averaged) over the assay temperatures; the truth row reports the mean of the analytical local $z(T)$ at those temperatures. $CT_{max_{1hr}}$ comes from direct inversion of the fitted 4PL at 60 min per draw, including the per-draw asymmetry correction.

Quantity	Truth	Median	Lower	Upper
z (°C)	6.28	6.19	5.78	6.65
CTmax at 1 hr (°C)	32.21	32.27	32.04	32.47

bent, the *full* per-temperature local $z(T)$ profile varies across the assay range (examined below, and returned directly by `extract_tdt(target_surv = "absolute", z_local = TRUE)`) — unlike the relative threshold, which reads the linear $\text{mid}(T)$ and would give a single constant z . Direct 4PL inversion at the reference duration also recovers $CT_{\max_{1hr}}$ (truth 32.21 °C; posterior median 32.27 °C, 95% CrI [32.04, 32.47] °C), with the asymmetry correction evaluated per draw rather than assumed constant in T .

The bent $\log_{10} \text{LT50}(T)$ curve itself is the intermediate object both quantities are built from; it lives in `et_ext$lt50_curve` (the output of `derive_tdt_curve()`) and can be plotted directly (Figure S6).

```
# Plot the extended example LT50 curve.
truth_overlay <- {
  Ts <- seq(min(temps) - 2, max(temps) + 2, by = 0.05)
  tibble::tibble(temp = Ts,
                 duration_median = 10 ^ vapply(Ts, true_log10_LT50,
                                                numeric(1)))
}

plot_tdt_curve(et_ext$lt50_curve) +
  ggplot2::geom_line(data = truth_overlay,
                    ggplot2::aes(x = temp, y = duration_median),
                    linetype = "dashed", colour = "black",
                    inherit.aes = FALSE)
```

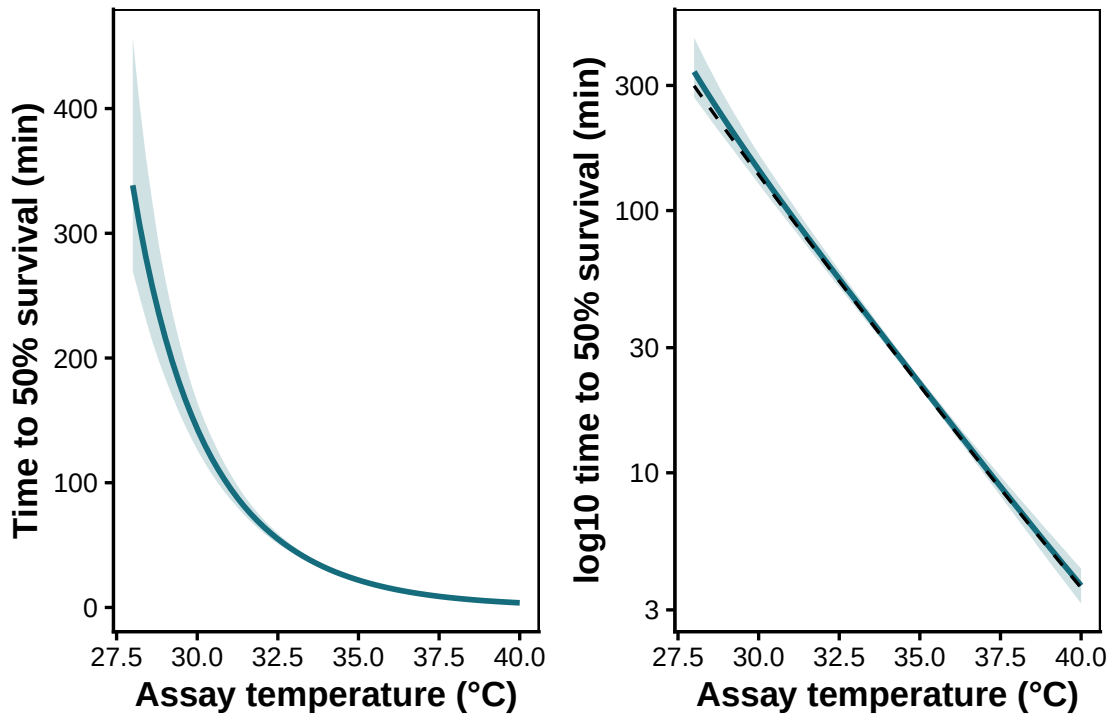


Figure S6: Posterior $\text{LT50}(T)$ curve (median + 95% CrI, blue) recovered by `derive_tdt_curve()` from the T -varying fit, with the analytical truth overlaid (dashed black). The curve is visibly bent rather than straight in T , and the posterior tracks the truth.

T (°C)	z local truth (°C)	Median	Lower	Upper
30	6.05	5.73	5.02	6.39
32	6.22	6.13	5.7	6.61
34	6.27	6.27	5.86	6.75
36	6.34	6.32	5.9	6.86
38	6.53	6.49	6.04	7.09

We do not need to hand-roll this: `extract_tdt(..., z_local = TRUE)` returns the per-draw **local** $z(T)$ at each assay temperature directly, in `et_ext$z$local` (a `$draws` slot plus a per-temperature `$summary`). We use that summary below (Figure [S7](#)).

```
# Per-temperature local z(T) straight from extract_tdt(z_local = TRUE).
# et_ext$z$local$summary has columns temp / z_median / z_lower / z_upper; we
# rename to the names the figure below expects and join the analytical truth.
local_z_summary <- et_ext$z$local$summary |>
  dplyr::transmute(T_at      = temp,
                   z_local_median = z_median,
                   z_local_lower  = z_lower,
                   z_local_upper  = z_upper)

local_z_summary |>
  dplyr::left_join(
    tibble::tibble(T_at = temps, z_local_truth = truth_ext$local_z_true),
    by = "T_at"
  ) |>
  dplyr::select(T_at, z_local_truth, z_local_median,
                z_local_lower, z_local_upper) |>
  dplyr::rename(
    `T (°C)`      = T_at,
    `z local truth (°C)` = z_local_truth,
    Median        = z_local_median,
    Lower         = z_local_lower,
    Upper         = z_local_upper
  ) |>
  tinytable::tt(digits = 3, escape = TRUE)
```



```
# Plot how local z changes across temperature in the extended example.
truth_curve_z <- {
  Ts <- seq(min(temps), max(temps), by = 0.1)
  tibble::tibble(T_at = Ts,
                 z_local_truth = vapply(Ts, true_local_z, numeric(1)))
}

ggplot2::ggplot(local_z_summary,
               ggplot2::aes(x = T_at, y = z_local_median)) +
  ggplot2::geom_ribbon(ggplot2::aes(ymin = z_local_lower,
                                   ymax = z_local_upper),
                    fill = "#146C7C", alpha = 0.2) +
  ggplot2::geom_line(colour = "#146C7C", linewidth = 1) +
  ggplot2::geom_line(data = truth_curve_z,
                    ggplot2::aes(x = T_at, y = z_local_truth),
                    linetype = "dashed", colour = "black",
                    inherit.aes = FALSE) +
  ggplot2::labs(x = "Assay temperature (°C)",
               y = expression(local~italic(z)~"("°C per 10-fold change in time)")) +
  ggplot2::theme_classic()
```

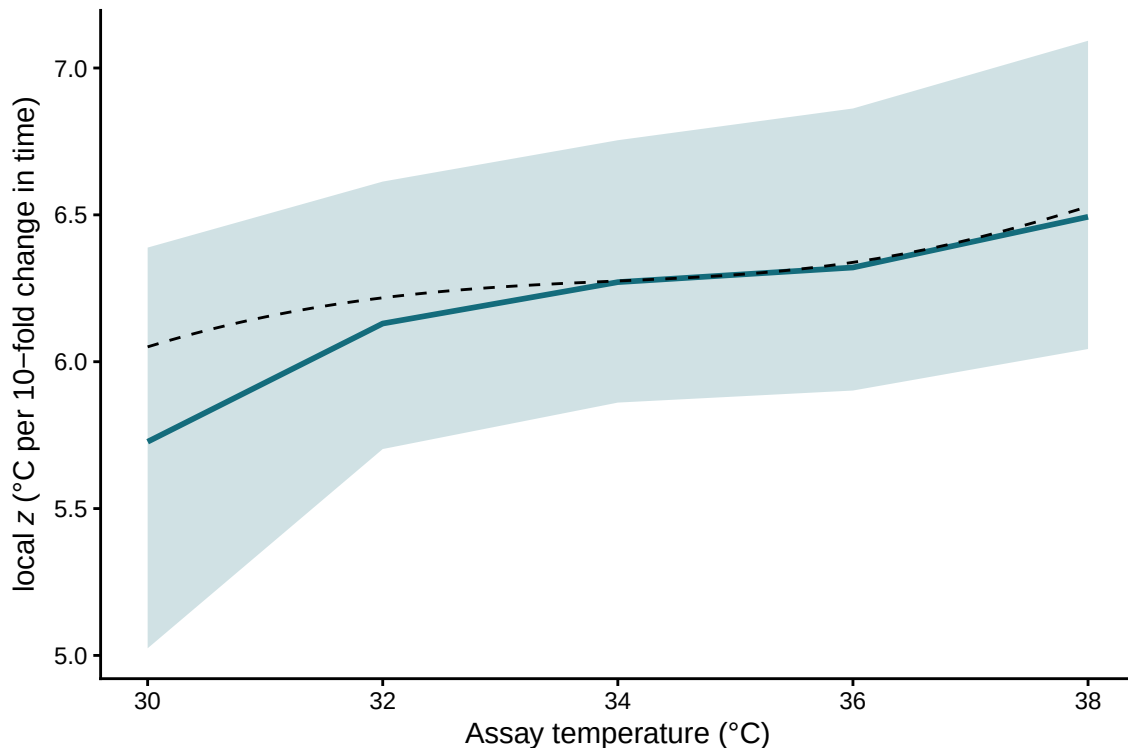


Figure S7: Local $z(T)$ recovered from the per-draw $\log_{10}$ LT50(T) curve (blue median + 95% CrI), with the analytical truth overlaid (dashed black). Where the constant-shape simple-case fit would give the same z at every assay temperature, here local z varies with T in lock-step with the bent $\log_{10}$ LT50(T) curve.

The same `fit_4pl()` and `extract_tdt()` calls recover the bent $\log_{10} LT_{50}(T)$ curve, average z across the assay grid, and per-draw $CT_{max_{1hr}}$. Under the constant-shape truth in Section 3.0.4, near-zero slopes on up , ℓ , and k reduce this calculation to the closed-form case.

3.0.7 Heat injury accumulation and survival under a fluctuating trace

TLS models can translate dynamic field temperatures into predicted heat-injury (HI) accumulation toward a threshold such as LT_{50} (Arnold *et al.* 2025; Jørgensen *et al.* 2021; Noble *et al.* 2026; Ørsted *et al.* 2022; Rezende *et al.* 2020). The joint 4PL posterior propagates uncertainty in z and $CT_{max_{1hr}}$ through the HI integral and uses the fitted surface to predict survival fraction $S_{pred}(t)$ along the same trajectory (Equation 9 of the manuscript). `predict_heat_injury()` returns both trajectories in one call.

3.0.7.1 A simulated temperature profile across 5 days

To mimic conditions for a moderately heat-tolerant ectotherm, we generate 5 days of hourly temperatures with a sinusoidal diurnal cycle, ranging from 12 °C overnight to a midday peak of 24 °C. We set the net-accumulation threshold T_c to 20 °C, below which HI does not increase (Jørgensen *et al.* 2021; Ørsted *et al.* 2022). The simulated organism has $CT_{max_{1hr}} \approx 32.1$ °C and $z \approx 6.7$ °C, so the daily peak remains below the 1-hour critical limit and generates gradual sublethal accumulation.

```
# Create example temperature traces for heat-injury prediction.
hi_trace <- tibble::tibble(
  time = 0:(24 * 5 - 1),
  temp  = 18 + 6 * sin(2 * pi * (0:(24 * 5 - 1) - 6) / 24) # range 12-24 °C
)
hi_T_c <- 20 # damage threshold (°C)
```

```
# Plot the example temperature traces used for heat-injury prediction.
ggplot2::ggplot(hi_trace, ggplot2::aes(time, temp)) +
  ggplot2::geom_line(colour = "darkred", linewidth = 0.6) +
  ggplot2::geom_hline(yintercept = hi_T_c, linetype = "dashed", colour = "grey40") +
  ggplot2::geom_hline(yintercept = true_CTmax, linetype = "dotted", colour = "grey40") +
  ggplot2::annotate("text", x = max(hi_trace$time), y = hi_T_c, label = "T_c",
    hjust = 1, vjust = -0.4, size = 3, colour = "grey40") +
  ggplot2::annotate("text", x = max(hi_trace$time), y = true_CTmax,
    label = "CTmax_1hr",
    hjust = 1, vjust = -0.4, size = 3, colour = "grey40") +
  ggplot2::ylim(10, 45) +
  ggplot2::labs(x = "Hour", y = "Temperature (°C)") +
  ggplot2::theme_classic()
```

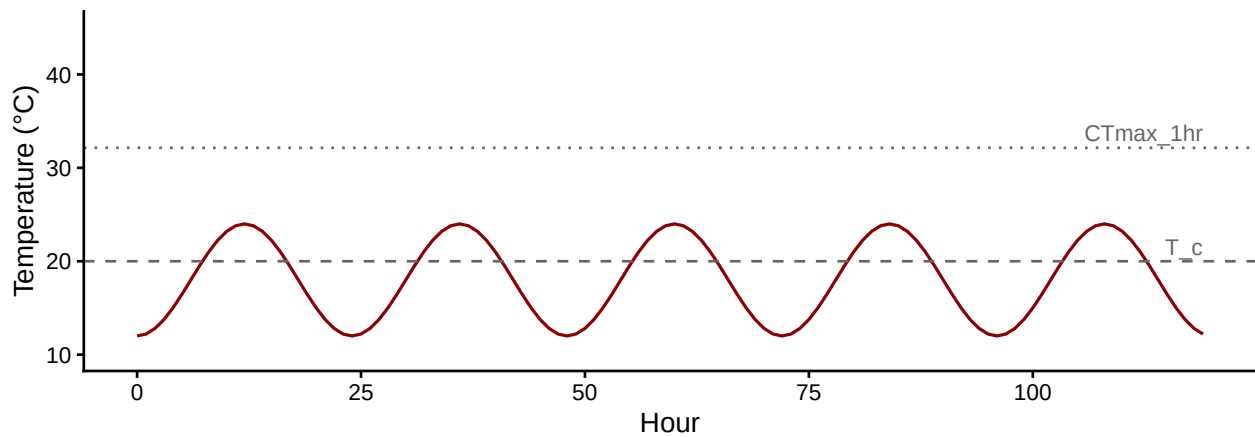


Figure S8: 5-day diurnal temperature trace. The dashed line is the damage-accumulation threshold $T_c = 20^\circ\text{C}$; the dotted line is the simulated organism's $CT_{max_{1hr}} \approx 32.1^\circ\text{C}$ — well above the daily peaks, so accumulation is sub-lethal and gradual.

3.0.7.2 Posterior HI and survival via `predict_heat_injury()`

`predict_heat_injury()` takes a temperature trace plus a fitted workflow, propagates the full posterior of $(\ell, \text{up}, k, \beta_0, \beta_1)$ through the HI integral, and returns the median and 95% credible band of cumulative HI and predicted survival at every time point. We supply `T_c` to enforce zero damage accumulation below the threshold (matching Equation 7 of the manuscript). The model time units here are minutes (because `standardize_data()` was called with `duration_unit = "minutes"`), so the trace's `time` is in hours and damage rates are computed accordingly internally.

```
# Predict heat injury and survival for the example temperature traces.
# The trace's `time` is in hours; `predict_heat_injury()` reconciles it against
# the model's own time unit (minutes here) internally, so we pass `trace_unit`
# rather than rescaling the trace by hand.
hi <- predict_heat_injury(
  trace      = hi_trace,
  workflow   = wf_bb,
  target_surv = 0.5,
```

```

T_c      = hi_T_c,
trace_unit = "hours",
ndraws   = 500
)

```

```

# Plot heat-injury and survival trajectories for the example traces.
# `plot_heat_injury()` returns the package's canonical two-panel figure:
# cumulative HI on top (with the 100% = one-LT50 reference line) over the
# predicted survival fraction, both with 95% credible bands.
plot_heat_injury(hi)

```

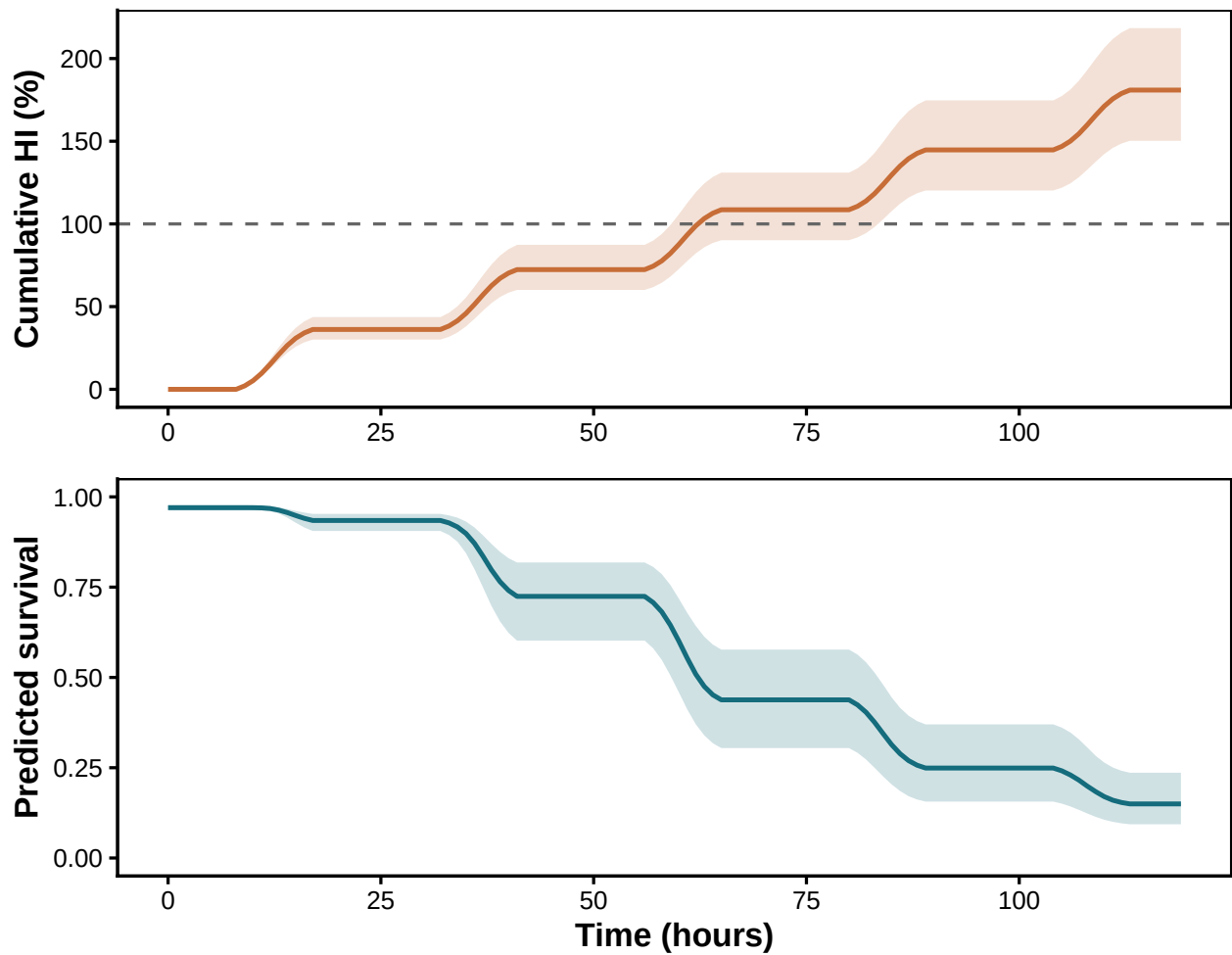


Figure S9: Posterior heat-injury accumulation (top) and predicted survival fraction (bottom) under the 5-day field trace, drawn by `plot_heat_injury()`. Solid lines are posterior medians across 500 draws from `wf_bb`; shaded bands are 95% credible intervals. The dashed line marks the LT50 threshold (HI = 100%). The two trajectories tell the same story: $S_{\text{pred}}(t)$ falls as $HI(t)$ crosses 100%.

By the end of day 5 the posterior median HI is 181% (95% CI: 150% to 218%), and median predicted survival is 0.15 (95% CI: 0.09 to 0.24).

3.0.8 Validation: recovering known doses

Before trusting `predict_heat_injury()` on real field data, we validate it against analytical truth on three reference traces with known dosing structure. `make_temperature_scenarios()` builds the traces; `planted_dose_from_trace()` implements the classical HI integral (Equation 7) directly given known `z` and `CTmax_1hr`. Posterior medians from `predict_heat_injury()` should sit near the analytical truth, and the 95% credible intervals should cover it.

```
# Validate heat-injury predictions against an analytical planted-dose calculation.
# Build validation traces with known planted doses.
val_scens <- make_temperature_scenarios(
  baseline      = 20,
  spike_temp    = 28,
  n_hours       = 96,
  spike_times_single = 24,
  spike_times_multi  = c(24, 48, 72)
)
val_T_c <- 24

# Compute analytical truth from the generating parameters.
planted <- purrr::map(val_scens, planted_dose_from_trace,
  z = true_z, CTmax_1hr = true_CTmax, T_c = val_T_c)

# Predict the same doses from the fitted beta-binomial workflow.
predicted <- purrr::map(val_scens, function(sc) {
  predict_heat_injury(
    trace      = sc,
    workflow   = wf_bb,
    target_surv = 0.5,
    T_c        = val_T_c,
    trace_unit  = "hours",
    ndraws     = 300
  )
})

# Compare planted and posterior-predicted final heat injury.
val_table <- tibble::tibble(
  Scenario      = c("Flat", "Single spike", "Multi spike"),
  `Planted HI (%)` = c(tail(planted$flat$hi_cumulative, 1),
    tail(planted$single_spike$hi_cumulative, 1),
    tail(planted$multi_spike$hi_cumulative, 1)),
  `Posterior median (%)` = c(tail(predicted$flat$summary$hi_median, 1),
    tail(predicted$single_spike$summary$hi_median, 1),
    tail(predicted$multi_spike$summary$hi_median, 1)),
  `Lower (%)` = c(tail(predicted$flat$summary$hi_lower, 1),
    tail(predicted$single_spike$summary$hi_lower, 1),
    tail(predicted$multi_spike$summary$hi_lower, 1)),
  `Upper (%)` = c(tail(predicted$flat$summary$hi_upper, 1),
```

Scenario	Planted HI (%)	Posterior median (%)	Lower (%)	Upper (%)	Truth in CrI?
Flat	0	0	0	0	TRUE
Single spike	24	23	21	26	TRUE
Multi spike	72	70	62	78	TRUE

```

        tail(predicted$single_spike$summary$hi_upper, 1),
        tail(predicted$multi_spike$summary$hi_upper, 1))
)
val_table$`Truth in CrI?` <- with(val_table,
                                `Lower (%)` <= `Planted HI (%)` &
                                `Planted HI (%)` <= `Upper (%)`)
tinytable::tt(val_table, digits = 2, escape = TRUE)

```

The flat trace stays below T_c for its entire duration and so accrues zero HI, matching the analytical truth exactly. The single- and multi-spike traces deliver known doses analytically; `predict_heat_injury()` recovers each with the truth lying inside its credible interval.

4 Case Study 1: Zebrafish heat tolerance across an oxygen gradient

Saruhashi *et al.* (2026) tested the oxygen- and capacity-limited thermal tolerance (OCLTT) hypothesis — that oxygen availability sets the upper thermal limit — by scoring survival of zebrafish (*Danio rerio*) larvae across assay temperatures (38–40 °C) and exposure durations (7.5–240 min) under different oxygen treatments. We take the diploid larvae and compare **normoxia** with **hyperoxia**, letting both the critical temperature (CT_{max}) and the thermal sensitivity (z) depend on oxygen inside one joint 4PL. The dual purpose here is methodological *and* a validation: if the joint model is doing its job it should recover the OCLTT prediction the authors report — that added oxygen raises heat tolerance.

4.1 Data and standardisation

```

# Bundled Saruhashi et al. (2026) survival data; diploid larvae, normoxia vs
# hyperoxia (we drop hypoxia, whose z is weakly identified by these durations).
data(zebrafish_o2)
zf_raw <- zebrafish_o2 |>
  dplyr::filter(ploidy == "diploid", oxygen %in% c("normoxia", "hyperoxia")) |>
  dplyr::mutate(oxygen = droplevels(oxygen))

# Counts -> the bayesTLS modelling schema. `oxygen` is carried through as the moderator.
zf_std <- standardize_data(zf_raw, temp = "temp", duration = "duration_min",
                          n_total = "n_total", n_surv = "n_surv",
                          duration_unit = "minutes")
str(zf_std)

```

```
tibble [380 x 15] (S3: tbl_df/tbl/data.frame)
 $ cohort      : chr [1:380] "ZB100" "ZB100" "ZB100" "ZB100" ...
 $ ploidy      : Factor w/ 2 levels "diploid","triploid": 1 1 1 1 1 1 1 1 1 1 ...
 $ oxygen      : Factor w/ 2 levels "normoxia","hyperoxia": 1 1 1 1 1 1 1 1 1 1 ...
 $ o2_nominal  : int [1:380] 100 100 100 100 100 100 100 100 100 100 ...
 $ o2_measured : num [1:380] 96 105.4 96 100 97.8 ...
 $ temp       : num [1:380] 26 38 26 38 26 38 26 38 26 38 ...
 $ temp_measured: num [1:380] 26.3 37.7 26.3 38.2 26.4 38.3 26.4 37.8 26.4 37.8 ...
 $ duration_min : num [1:380] 50 50 50 50 50 50 89 89 89 89 ...
 $ n_total     : int [1:380] 5 4 4 4 4 4 5 5 6 6 ...
 $ n_surv      : num [1:380] 5 4 4 4 4 4 5 5 6 6 ...
 $ duration    : num [1:380] 50 50 50 50 50 50 89 89 89 89 ...
 $ logd       : num [1:380] 1.7 1.7 1.7 1.7 1.7 ...
 $ n_dead      : num [1:380] 0 0 0 0 0 0 0 0 0 0 ...
 $ survival    : num [1:380] 1 1 1 1 1 1 1 1 1 1 ...
 $ temp_c      : num [1:380] -8.58 3.42 -8.58 3.42 -8.58 ...
- attr(*, "tdt_meta")=List of 5
 ..$ temp_mean      : num 34.6
 ..$ duration_unit  : chr "minutes"
 ..$ random_effects : NULL
 ..$ response_type  : chr "count"
 ..$ response_var   : NULL
```

4.2 Joint 4PL with oxygen on CT_{max} and z

We fit CT_{max} and z directly as functions of oxygen using cell-means coding ($\sim 0 + \text{oxygen}$), so each treatment gets its own coefficient and the asymptotes and steepness are shared. The CT_{max}/z priors are coding-aware, so this returns the same per-group estimates as treatment coding ($\sim \text{oxygen}$) — cell-means simply reads most directly.

```
wf_zf <- fit_4pl(zf_std, ctmx = ~ 0 + oxygen, z = ~ 0 + oxygen, t_ref = 60,
               control = list(adapt_delta = 0.95),
               file_refit = "on_change",
               file       = file.path(models_dir, "zf_oxygen_normhyper"))
summary(wf_zf)
```

```
Family: beta_binomial
Links: mu = identity
Formula: n_surv | trials(n_total) ~ low + (up - low)/(1 + exp(exp(logk) * (logd - mid)))
        low ~ (0.001 + inv_logit(lowraw) * 0.498)
        up  ~ (0.501 + inv_logit(upraw) * 0.498)
        mid ~ (1.77815 - (temp_c - CTmaxdev)/exp(logz))
        lowraw ~ 0 + oxygen + temp_c:oxygen
        upraw  ~ 0 + oxygen + temp_c:oxygen
        logk   ~ 0 + oxygen + temp_c:oxygen
        CTmaxdev ~ 0 + oxygen
```

```

logz ~ 0 + oxygen
Data: data (Number of observations: 380)
Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
total post-warmup draws = 8000

```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat
lowraw_oxygennormoxia	-2.09	0.91	-3.87	-0.33	1.00
lowraw_oxygenhyperoxia	-3.30	0.99	-5.21	-1.36	1.00
lowraw_oxygennormoxia:temp_c	0.85	0.35	0.17	1.53	1.00
lowraw_oxygenhyperoxia:temp_c	-0.13	0.40	-1.01	0.54	1.00
upraw_oxygennormoxia	4.42	0.71	3.14	5.89	1.00
upraw_oxygenhyperoxia	3.40	0.98	1.50	5.31	1.00
upraw_oxygennormoxia:temp_c	-0.26	0.19	-0.65	0.06	1.00
upraw_oxygenhyperoxia:temp_c	0.17	0.42	-0.60	1.05	1.00
logk_oxygennormoxia	2.95	0.51	2.01	4.05	1.00
logk_oxygenhyperoxia	1.18	0.72	-0.26	2.54	1.00
logk_oxygennormoxia:temp_c	0.08	0.10	-0.15	0.25	1.00
logk_oxygenhyperoxia:temp_c	0.27	0.17	-0.05	0.60	1.00
CTmaxdev_oxygennormoxia	4.18	0.22	3.68	4.55	1.00
CTmaxdev_oxygenhyperoxia	4.80	0.09	4.61	4.97	1.00
logz_oxygennormoxia	1.76	0.26	1.17	2.18	1.00
logz_oxygenhyperoxia	0.93	0.10	0.73	1.12	1.00
	Bulk_ESS	Tail_ESS			
lowraw_oxygennormoxia	4982	4305			
lowraw_oxygenhyperoxia	6536	5892			
lowraw_oxygennormoxia:temp_c	3188	2123			
lowraw_oxygenhyperoxia:temp_c	5804	5628			
upraw_oxygennormoxia	6670	4990			
upraw_oxygenhyperoxia	7178	4738			
upraw_oxygennormoxia:temp_c	5035	3476			
upraw_oxygenhyperoxia:temp_c	6144	5541			
logk_oxygennormoxia	3103	2595			
logk_oxygenhyperoxia	4109	3857			
logk_oxygennormoxia:temp_c	2790	2385			
logk_oxygenhyperoxia:temp_c	3953	4300			
CTmaxdev_oxygennormoxia	2649	2058			
CTmaxdev_oxygenhyperoxia	5172	4379			
logz_oxygennormoxia	3495	1916			
logz_oxygenhyperoxia	5772	4504			

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
phi	0.57	0.10	0.40	0.79	1.00	8915	5802

Draws were sampled using `sample(hmc)`. For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

4.3 Per-treatment thermal limits and OCLTT check

`tls(by = "oxygen")` derives z , $CT_{max_{1hr}}$ and T_{crit} per treatment from the shared posterior, so each quantity carries full uncertainty and T_{crit} pairs CT_{max} and z per draw.

```
zf_tls <- tls(wf_zf, by = "oxygen", lethal = TRUE)
zf_tls$summary
```

```
# A tibble: 6 x 5
  oxygen    quantity median lower upper
  <fct>    <chr>      <dbl> <dbl> <dbl>
1 normoxia z          5.94  3.23  8.87
2 normoxia CTmax      38.8  38.3  39.1
3 normoxia Tcrit      24.1  16.0  30.8
4 hyperoxia z          2.55  2.07  3.07
5 hyperoxia CTmax      39.4  39.2  39.6
6 hyperoxia Tcrit      33.0  31.1  34.6
```

The joint fit recovers the OCLTT direction Saruhashi *et al.* (2026) report: hyperoxia raises $CT_{max_{1hr}}$ to 39.4 °C, above the normoxic 38.8 °C, and lifts T_{crit} markedly (33 °C vs 24.1 °C) — added oxygen buys heat tolerance. Saruhashi *et al.* report this ordering qualitatively rather than as tabulated z/CT_{max} , so the validation here is the recovered *direction*, not a number-for-number match. (Normoxia’s z is wide — survival saturates over the tested durations — while hyperoxia’s is tight; the CT_{max}/T_{crit} ordering is the well-identified signal.)

```
pzc_zf <- predict_survival_curves(wf_zf)
plot_survival_curves(pzc_zf, observed = zf_std, log_time = TRUE)
```

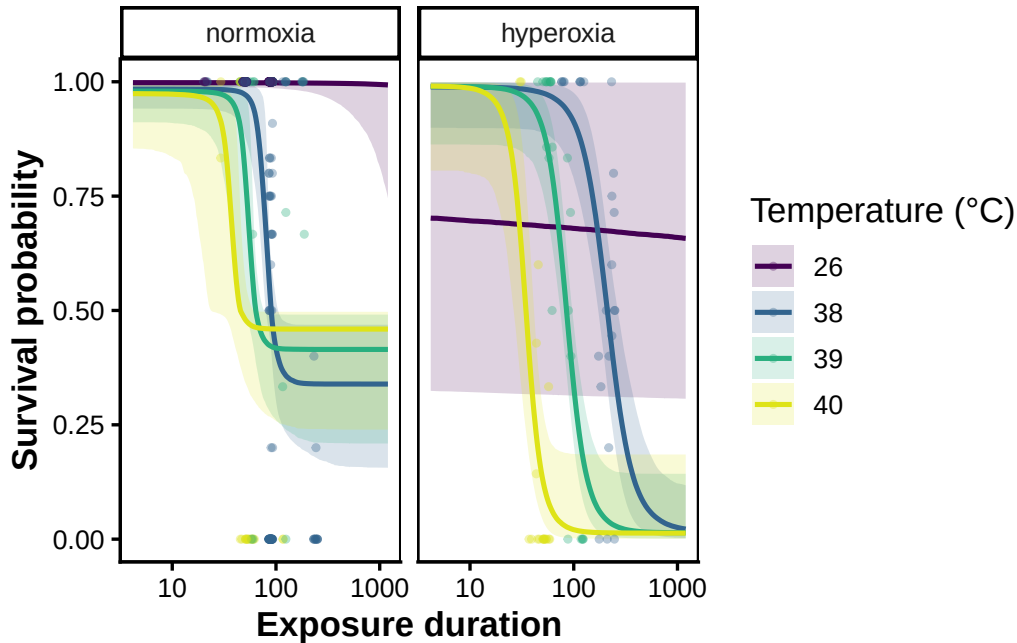


Figure S10: Zebrafish survival surface. Fitted joint-4PL survival probability against exposure duration for normoxia and hyperoxia (posterior median and 95% credible band), with the observed cell proportions overlaid.

5 Case Study 2: Cereal aphids — comparing three species

Li *et al.* (2023) measured survival of three cereal-aphid species — *Metopolophium dirhodum*, *Sitobion avenae* and *Rhopalosiphum padi* — across temperatures and exposure durations, and linked their thermal tolerance landscapes to field abundance. We compare the three species on the heat branch at a single age (6 days), modelling CT_{max} and z as functions of species in one joint 4PL so the species contrasts fall straight out of the posterior. This case study does double duty: it shows a **three-group** comparison, validates the fit against Li et al.’s published estimates, and (below) demonstrates `predict_heat_injury()` across multiple groups under a real field temperature trace.

5.1 Data and standardisation

```
# Bundled Li et al. (2023) survival data; heat branch, 6-day-old aphids.
data(aphid_tdt)
aphid_raw <- aphid_tdt |>
  dplyr::filter(branch == "heat", age == "6") |>
  dplyr::mutate(species = droplevels(species))

aphid_std <- standardize_data(aphid_raw, temp = "temp", duration = "duration_min",
                             n_total = "n_total", n_surv = "n_surv",
```

```

duration_unit = "minutes") # keeps `species`
str(aphid_std)

```

```

tibble [436 x 12] (S3: tbl_df/tbl/data.frame)
 $ species      : Factor w/ 3 levels "M_dirhodum","S_avenae",...: 1 1 1 1 1 1 1 1 1 1 ...
 $ age          : Factor w/ 3 levels "2","6","12": 2 2 2 2 2 2 2 2 2 2 ...
 $ branch       : Factor w/ 2 levels "heat","cold": 1 1 1 1 1 1 1 1 1 1 ...
 $ temp         : num [1:436] 40 40 40 40 40 40 40 40 40 40 ...
 $ duration_min : num [1:436] 2 2 4 4 4 5 5 5 6 6 ...
 $ n_total      : int [1:436] 4 10 9 10 10 10 10 10 10 9 ...
 $ n_surv       : num [1:436] 4 9 9 10 10 8 6 5 2 6 ...
 $ duration     : num [1:436] 2 2 4 4 4 5 5 5 6 6 ...
 $ logd        : num [1:436] 0.301 0.301 0.602 0.602 0.602 ...
 $ n_dead       : num [1:436] 0 1 0 0 0 2 4 5 8 3 ...
 $ survival     : num [1:436] 1 0.9 1 1 1 ...
 $ temp_c       : num [1:436] 3.13 3.13 3.13 3.13 3.13 ...
 - attr(*, "tdt_meta")=List of 5
 ..$ temp_mean      : num 36.9
 ..$ duration_unit  : chr "minutes"
 ..$ random_effects : NULL
 ..$ response_type  : chr "count"
 ..$ response_var   : NULL

```

5.2 Joint 4PL with species on CT_{max} and z

```

wf_aphid <- fit_4pl(aphid_std, ctmax = ~ 0 + species, z = ~ 0 + species, t_ref = 60,
                    file_refit = "on_change",
                    file        = file.path(models_dir, "aphid_species"))
summary(wf_aphid)

```

```

Family: beta_binomial
Links: mu = identity
Formula: n_surv | trials(n_total) ~ low + (up - low)/(1 + exp(exp(logk) * (logd - mid)))
        low ~ (0.001 + inv_logit(lowraw) * 0.498)
        up  ~ (0.501 + inv_logit(upraw) * 0.498)
        mid ~ (1.77815 - (temp_c - CTmaxdev)/exp(logz))
        lowraw ~ 0 + species + temp_c:species
        upraw  ~ 0 + species + temp_c:species
        logk   ~ 0 + species + temp_c:species
        CTmaxdev ~ 0 + species
        logz   ~ 0 + species
Data: data (Number of observations: 436)
Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 8000

```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat
lowraw_speciesM_dirhodum	-4.26	0.73	-5.79	-2.93	1.00
lowraw_speciesS_avenae	-3.33	0.67	-4.75	-2.11	1.00
lowraw_speciesR_padi	-3.46	0.63	-4.74	-2.29	1.00
lowraw_speciesM_dirhodum:temp_c	-0.05	0.37	-0.78	0.67	1.00
lowraw_speciesS_avenae:temp_c	-0.45	0.73	-1.33	1.07	1.02
lowraw_speciesR_padi:temp_c	1.25	0.45	-0.40	1.86	1.01
upraw_speciesM_dirhodum	2.00	0.74	0.86	3.73	1.00
upraw_speciesS_avenae	1.07	0.66	0.16	2.74	1.01
upraw_speciesR_padi	3.18	0.88	1.64	5.01	1.00
upraw_speciesM_dirhodum:temp_c	0.04	0.24	-0.46	0.48	1.00
upraw_speciesS_avenae:temp_c	0.28	0.36	-0.50	0.90	1.02
upraw_speciesR_padi:temp_c	-0.21	0.33	-0.82	0.46	1.00
logk_speciesM_dirhodum	2.48	0.13	2.25	2.74	1.00
logk_speciesS_avenae	2.98	0.31	2.40	3.56	1.02
logk_speciesR_padi	2.39	0.15	2.11	2.70	1.00
logk_speciesM_dirhodum:temp_c	0.06	0.05	-0.04	0.16	1.00
logk_speciesS_avenae:temp_c	-0.12	0.27	-0.49	0.39	1.02
logk_speciesR_padi:temp_c	0.26	0.09	0.10	0.44	1.00
CTmaxdev_speciesM_dirhodum	-1.58	0.12	-1.83	-1.35	1.00
CTmaxdev_speciesS_avenae	-0.32	0.05	-0.44	-0.24	1.01
CTmaxdev_speciesR_padi	0.11	0.05	0.02	0.22	1.00
logz_speciesM_dirhodum	1.53	0.03	1.46	1.60	1.00
logz_speciesS_avenae	1.25	0.03	1.20	1.32	1.01
logz_speciesR_padi	1.26	0.08	1.17	1.50	1.00

	Bulk_ESS	Tail_ESS
lowraw_speciesM_dirhodum	7947	6145
lowraw_speciesS_avenae	1560	3082
lowraw_speciesR_padi	5982	5620
lowraw_speciesM_dirhodum:temp_c	7140	5495
lowraw_speciesS_avenae:temp_c	236	996
lowraw_speciesR_padi:temp_c	845	253
upraw_speciesM_dirhodum	3264	4251
upraw_speciesS_avenae	372	1050
upraw_speciesR_padi	4173	5027
upraw_speciesM_dirhodum:temp_c	3961	4340
upraw_speciesS_avenae:temp_c	260	865
upraw_speciesR_padi:temp_c	4115	5909
logk_speciesM_dirhodum	3330	5321
logk_speciesS_avenae	286	1288
logk_speciesR_padi	3006	3881
logk_speciesM_dirhodum:temp_c	4401	5520
logk_speciesS_avenae:temp_c	235	885
logk_speciesR_padi:temp_c	3936	4597
CTmaxdev_speciesM_dirhodum	2598	4585
CTmaxdev_speciesS_avenae	604	1161

CTmaxdev_speciesR_padi	4930	2859
logz_speciesM_dirhodum	3000	4089
logz_speciesS_avenae	346	1284
logz_speciesR_padi	813	294

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
phi	9.70	1.65	7.01	13.44	1.00	6448	5861

Draws were sampled using `sample(hmc)`. For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

5.3 Per-species thermal limits and validation against Li et al.

```
aphid_tls <- tls(wf_aphid, by = "species", lethal = TRUE)
aphid_tls$summary
```

```
# A tibble: 9 x 5
  species    quantity median lower upper
  <fct>      <chr>      <dbl> <dbl> <dbl>
1 M_dirhodum z          4.62  4.32  4.94
2 M_dirhodum CTmax      35.3  35.0  35.5
3 M_dirhodum Tcrit      23.7  21.1  26.2
4 R_padi     z          3.47  3.21  4.49
5 R_padi     CTmax      37.0  36.9  37.1
6 R_padi     Tcrit      28.2  25.3  30.1
7 S_avenae   z          3.48  3.31  3.75
8 S_avenae   CTmax      36.6  36.4  36.6
9 S_avenae   Tcrit      27.8  25.9  29.6
```

Our per-species estimates fall inside the ranges Li et al. report (their Table 1: z 3.2–4.7 °C, $CT_{max_{1hr}}$ 35.2–37.3 °C): here z spans 3.5–4.6 °C and $CT_{max_{1hr}}$ spans 35.3–37 °C. The joint fit also reproduces their species *ordering* — *M. dirhodum* is the least heat-sensitive (highest z , 4.6 °C) and *R. padi* the most heat-tolerant (highest $CT_{max_{1hr}}$, 37 °C) — so the model recovers both the magnitudes and the ranking from a single posterior, without per-temperature regressions.

```
psc_aphid <- predict_survival_curves(wf_aphid)
plot_survival_curves(psc_aphid, observed = aphid_std, log_time = TRUE)
```

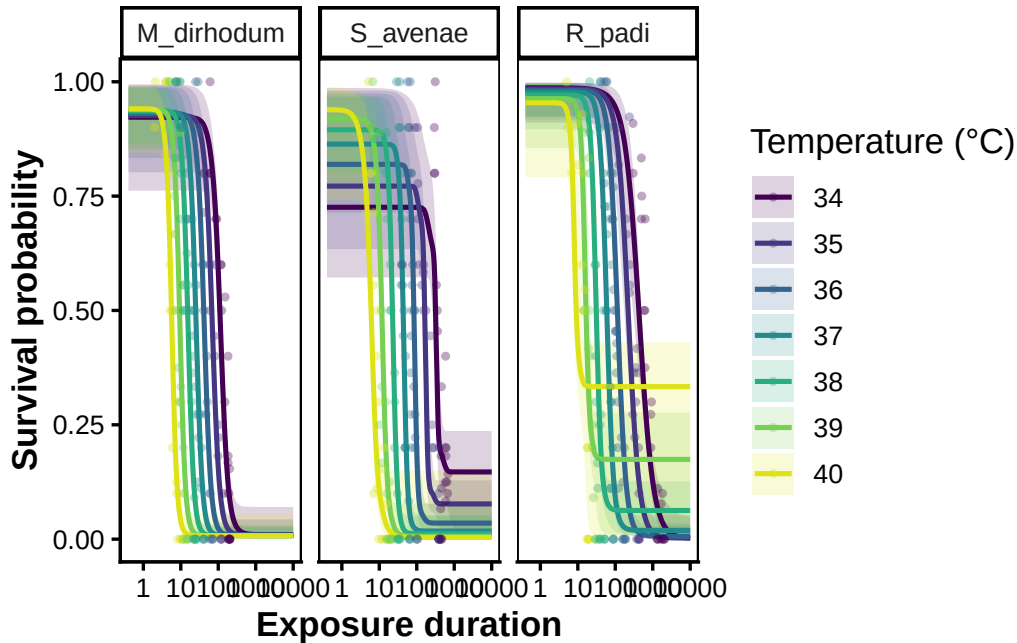


Figure S11: Aphid survival surfaces. Fitted joint-4PL survival probability against exposure duration for the three species (posterior median and 95% credible band), with the observed cell proportions overlaid.

5.4 Heat injury across species under a field temperature trace

With the per-species TDT surface in hand, `predict_heat_injury()` accumulates damage along an hourly field temperature trace and converts it to predicted survival. We use hourly 2 m air temperature for Wuhan (the hottest of Li et al.'s three Chinese wheat sites), May–June 2016, re-sourced from the Open-Meteo ERA5 archive and shipped with the package (built by `data-raw/fetch_aphid_temp_trace.R`). We restrict the window to May–June so the injury scale stays readable — over the full May–August record the most heat-vulnerable species (*M. dirhodum*, the lowest T_{crit}) accrues injury into the tens of thousands of percent of an LT_{50} dose. The trace is used **unmodified** (no warming offset), and by `= "species"` projects all three species at once, demonstrating multi-group heat injury. Li et al. drew their trace from a different weather product and used their own TDT parameters, so this reproduces the early-season *shape* of their Fig. 2 within our posterior framework rather than its exact values.

```
# Hourly (time, temp) trace per city ships in inst/extdata; take Wuhan, trimmed
# to May-June (over the full May-Aug record the most vulnerable species runs to
# ~tens of thousands of % of an LT50 dose, swamping the axis -- see prose).
aphid_trace <- readr::read_csv(
  here::here("inst", "extdata", "data_temp_trace_aphid_summer2016.csv"),
  show_col_types = FALSE) |>
  dplyr::filter(city == "Wuhan",
    as.integer(format(datetime, "%m")) <= 6) |> # May-June only
  dplyr::transmute(time = time_h, temp = temp_c) # hours, degrees C
```

```
aphid_hi <- predict_heat_injury(aphid_trace, wf_aphid, target_surv = "relative",
                               trace_unit = "hours", by = "species")
plot_heat_injury(aphid_hi)
```

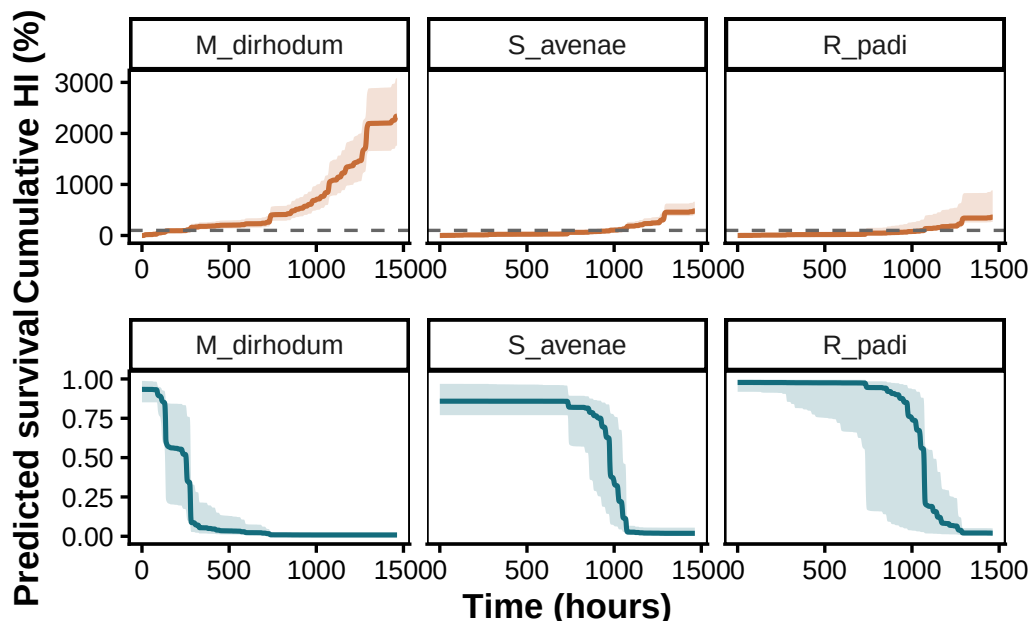


Figure S12: Multi-group heat-injury projection. Cumulative heat injury (100% = one LT50 dose) and predicted survival for the three aphid species under the unmodified Wuhan May–June 2016 hourly temperature trace, from a single `predict_heat_injury(by = "species")` call.

The accumulated injury and end-of-window survival differ across species in line with their thermal limits, and the same single call would map the south-to-north climate gradient (Wuhan hottest, Beijing coolest) onto a corresponding gradient in projected survival — the link Li et al. draw between thermal tolerance and field abundance.

6 Case Study 3: Snow gum leaf PSII — light vs dark recovery

Arnold *et al.* (2026) developed a thermal-load-sensitivity (TLS) protocol for photosystem II (PSII) and asked, among other things, whether the light environment a leaf experiences *after* heat exposure changes its apparent heat tolerance. Here we take the snow gum (*Eucalyptus pauciflora*) arm of their Experiment 1: excised leaf sections from six mature trees were heated across a grid of assay temperatures (30–56 °C) and exposure durations (5–120 min), then held either in **darkness** or under **90 min of moderate light** before F_v/F_m was re-measured 16–24 h later. The response — retained PSII function, post-exposure F_v/F_m divided by its pre-exposure value — is a **continuous proportion in [0, 1] with no count denominator**, so the 4PL uses a **Beta** likelihood rather than the beta-binomial of the count case studies. Loss of PSII function is **sublethal**, so we report z and CT_{max} but not T_{crit} .

We let both CT_{max} and z depend on the recovery condition inside one joint 4PL — a two-group comparison that is also a validation. Arnold *et al.* (2026) report that post-heat light *lowers* apparent heat tolerance while leaving thermal sensitivity broadly unchanged, so the joint model should place the Light CT_{max} below the Dark one with similar z .

6.1 Data and standardisation

```
# Bundled Arnold et al. (2026) snow gum PSII data (Experiment 1, light vs dark
# recovery). Two raw rows with post/pre Fv/Fm marginally above 1 (measurement
# noise) are already excluded upstream, leaving 394 rows.
```

```
data(snowgum_psii)
str(snowgum_psii)
```

```
'data.frame':  394 obs. of  8 variables:
 $ Temp      : num  30 30 30 30 30 30 30 30 30 30 30 ...
 $ Time      : num  15 15 15 15 15 15 15 15 15 15 15 ...
 $ recovery   : Factor w/ 2 levels "Dark","Light": 1 1 1 1 1 1 2 2 2 2 ...
 $ plant     : Factor w/ 6 levels "1","2","3","4",...: 1 2 3 4 5 6 1 2 3 4 ...
 $ meas_day  : Factor w/ 2 levels "1","2": 2 2 2 2 2 2 2 2 2 2 ...
 $ initial_fvfm: num  0.862 0.837 0.851 0.831 0.835 0.842 0.859 0.851 0.856 0.85 ...
 $ final_fvfm : num  0.794 0.79 0.807 0.781 0.805 0.809 0.79 0.767 0.798 0.775 ...
 $ fvfm_prop  : num  0.921 0.944 0.948 0.94 0.964 ...
```

```
# `proportion =` (not n_total/n_surv) flags a continuous response: standardize_data()
# records response_type = "proportion" and clamps to (0.001, 0.999) for the Beta
# density. `recovery` (Dark/Light) and `plant` (six trees) pass through as the
# moderator and the random-effect grouping.
```

```
leaf_std <- standardize_data(snowgum_psii, temp = "Temp", duration = "Time",
                             proportion = "fvfm_prop", duration_unit = "minutes")
str(leaf_std)
```

```
tibble [394 x 13] (S3: tbl_df/tbl/data.frame)
```

```
 $ Temp      : num [1:394] 30 30 30 30 30 30 30 30 30 30 30 ...
 $ Time      : num [1:394] 15 15 15 15 15 15 15 15 15 15 15 ...
 $ recovery   : Factor w/ 2 levels "Dark","Light": 1 1 1 1 1 1 2 2 2 2 ...
 $ plant     : Factor w/ 6 levels "1","2","3","4",...: 1 2 3 4 5 6 1 2 3 4 ...
 $ meas_day  : Factor w/ 2 levels "1","2": 2 2 2 2 2 2 2 2 2 2 ...
 $ initial_fvfm: num [1:394] 0.862 0.837 0.851 0.831 0.835 0.842 0.859 0.851 0.856 0.85 ...
 $ final_fvfm : num [1:394] 0.794 0.79 0.807 0.781 0.805 0.809 0.79 0.767 0.798 0.775 ...
 $ fvfm_prop  : num [1:394] 0.921 0.944 0.948 0.94 0.964 ...
 $ temp      : num [1:394] 30 30 30 30 30 30 30 30 30 30 30 ...
 $ duration   : num [1:394] 15 15 15 15 15 15 15 15 15 15 15 ...
 $ logd      : num [1:394] 1.18 1.18 1.18 1.18 1.18 1.18 ...
 $ survival   : num [1:394] 0.921 0.944 0.948 0.94 0.964 ...
 $ temp_c     : num [1:394] -12.6 -12.6 -12.6 -12.6 -12.6 ...
```


Table S12: Summary of the snow gum leaf PSII dataset after standardisation. Temperature centre is the grand mean used to mean-centre the temperature covariate.

n observations	n trees (plant)	Recovery conditions	Temperature range (°C)	Duration range (min)	Temp
394	6	Dark, Light	30–56	5–120	43

```
- attr(*, "tdt_meta")=List of 5
..$ temp_mean      : num 42.6
..$ duration_unit  : chr "minutes"
..$ random_effects : NULL
..$ response_type  : chr "proportion"
..$ response_var   : chr "survival"
```

Response model and zeros. Unlike the count-based case studies, the leaf response is a continuous proportion with no denominator, so we use a Beta likelihood. The Beta density is undefined at exactly 0 or 1, but 89 of 394 observations (23%) are zeros — complete loss of measurable PSII function at the hottest, longest exposures. `standardize_data()` clamps these to 0.001 so a single Beta model covers the whole surface. A zero-inflated Beta that separately modelled complete shutdown would represent these structural zeros more faithfully, at the cost of an extra dose-dependent component.

```
leaf_summary <- tibble::tibble(
  `n observations`      = nrow(leaf_std),
  `n trees (plant)`    = nlevels(leaf_std$plant),
  `Recovery conditions` = paste(levels(leaf_std$recovery), collapse = ", "),
  `Temperature range (°C)` = paste(range(leaf_std$temp), collapse = "-"),
  `Duration range (min)` = paste(round(range(leaf_std$duration), 0), collapse = "-"),
  `Temperature centre (°C)` = round(attr(leaf_std, "tdt_meta")$temp_mean, 1)
)
tinytable::tt(leaf_summary, digits = 2, escape = TRUE)
```

6.2 Joint 4PL with recovery on CT_{max} and z

We fit CT_{max} and z directly as functions of **recovery** using cell-means coding (`~ 0 + recovery`), so each condition gets its own coefficient. A random intercept on **plant** absorbs among-tree variation in the threshold; **meas_day** has only two levels, too few to estimate a variance, so it is omitted. With the shape parameters left to inherit the same per-condition structure, each recovery condition has its own full dose-response surface, coupled through the shared tree random intercept. The CT_{max}/z priors are coding-aware, so cell-means returns the same per-group estimates as treatment coding while reading most directly.

```
# Beta family is auto-picked from response_type; named here for clarity. The 89
# clamped structural zeros sit on the Beta boundary and induce a small number of
# divergent transitions (~1%); adapt_delta is raised to keep them low. The
# posterior is otherwise well-mixed (Rhat <= 1.003, bulk ESS > 2000).
```

```

wf_leaf <- fit_4pl(
  leaf_std,
  ctmax = ~ 0 + recovery + (1 | plant),
  z      = ~ 0 + recovery,
  family = brms::Beta(link = "identity"),
  t_ref  = 60,
  iter = 6000, warmup = 3000, seed = 123,
  control = list(adapt_delta = 0.999, max_treedepth = 15),
  file_refit = "on_change",
  file = file.path(models_dir, "fit_leaf_recovery_4pl")
)
summary(wf_leaf)

```

```

Family: beta
Links: mu = identity
Formula: survival ~ low + (up - low)/(1 + exp(exp(logk) * (logd - mid)))
        low ~ (0.001 + inv_logit(lowraw) * 0.498)
        up  ~ (0.501 + inv_logit(upraw) * 0.498)
        mid ~ (1.77815 - (temp_c - CTmaxdev)/exp(logz))
        lowraw ~ 0 + recovery + temp_c:recovery
        upraw  ~ 0 + recovery + temp_c:recovery
        logk   ~ 0 + recovery + temp_c:recovery
        CTmaxdev ~ 0 + recovery + (1 | plant)
        logz   ~ 0 + recovery
Data: data (Number of observations: 394)
Draws: 4 chains, each with iter = 6000; warmup = 3000; thin = 1;
       total post-warmup draws = 12000

```

Multilevel Hyperparameters:
~plant (Number of levels: 6)

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS
sd(CTmaxdev_Intercept)	0.55	0.25	0.19	1.16	1.00	2174
						Tail_ESS
sd(CTmaxdev_Intercept)						5581

Regression Coefficients:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS
lowraw_recoveryDark	-3.48	0.98	-5.36	-1.50	1.00	10866
lowraw_recoveryLight	-1.93	0.52	-2.96	-0.97	1.00	2359
lowraw_recoveryDark:temp_c	-0.38	0.33	-1.13	0.13	1.00	5600
lowraw_recoveryLight:temp_c	-0.11	0.07	-0.24	0.01	1.00	2364
upraw_recoveryDark	1.26	0.16	0.96	1.58	1.00	7225
upraw_recoveryLight	1.17	0.17	0.85	1.53	1.00	3834
upraw_recoveryDark:temp_c	-0.02	0.02	-0.06	0.03	1.00	7890
upraw_recoveryLight:temp_c	-0.01	0.02	-0.06	0.03	1.00	4944
logk_recoveryDark	1.92	0.21	1.51	2.32	1.00	5385
logk_recoveryLight	3.06	0.73	1.63	4.36	1.00	2110

Table S13: Sampling diagnostics for the leaf 4PL fit. Healthy targets: Rhat < 1.01, high ESS. A small residual divergence count comes from the clamped structural zeros at the Beta boundary.

rhat_max	ess_bulk_min	ess_tail_min	divergences	treedepth_max	treedepth_hits	bfmi_min	rhat
1	2110	2293	135	15	0	0.835	TRU
logk_recoveryDark:temp_c		-0.10	0.03	-0.14	-0.04	1.00	4509
logk_recoveryLight:temp_c		-0.11	0.11	-0.27	0.14	1.00	2234
CTmaxdev_recoveryDark		3.09	0.30	2.46	3.68	1.00	4000
CTmaxdev_recoveryLight		1.40	0.33	0.82	2.17	1.00	2274
logz_recoveryDark		1.38	0.07	1.24	1.51	1.00	9124
logz_recoveryLight		1.36	0.07	1.18	1.48	1.00	4036
		Tail_ESS					
lowraw_recoveryDark		9318					
lowraw_recoveryLight		6012					
lowraw_recoveryDark:temp_c		5160					
lowraw_recoveryLight:temp_c		6629					
upraw_recoveryDark		7599					
upraw_recoveryLight		4402					
upraw_recoveryDark:temp_c		7559					
upraw_recoveryLight:temp_c		4540					
logk_recoveryDark		4972					
logk_recoveryLight		2850					
logk_recoveryDark:temp_c		3275					
logk_recoveryLight:temp_c		2774					
CTmaxdev_recoveryDark		5572					
CTmaxdev_recoveryLight		2293					
logz_recoveryDark		7273					
logz_recoveryLight		3519					

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
phi	9.43	0.76	8.00	11.02	1.00	9195	8583

Draws were sampled using `sample(hmc)`. For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
tinytable::tt(diagnose_tdt_fit(wf_leaf), digits = 3, escape = TRUE)
```

6.3 Per-condition thermal limits

`tls(by = "recovery")` derives z and $CT_{max_{1hr}}$ for each condition from the shared posterior, so each quantity carries full uncertainty. Loss of PSII function is sublethal, so `lethal = FALSE` and

Table S14: Per-condition TDT summaries for snow gum leaf PSII (relative midpoint threshold), posterior medians with 95% credible intervals. z is the temperature slope on the midpoint ($z = -1/\beta_1$); $CT_{max_{1hr}}$ is the temperature at which the retained-function threshold is reached after 1 h.

Recovery	Quantity	Median	Lower	Upper
Dark	z (°C)	3.98	3.46	4.54
Dark	CTmax_1hr (°C)	45.74	45.10	46.32
Light	z (°C)	3.94	3.26	4.39
Light	CTmax_1hr (°C)	44.00	43.46	44.81

no T_{crit} is returned.

```
leaf_tls <- tpls(wf_leaf, by = "recovery", lethal = FALSE)
leaf_tls$summary
```

```
# A tibble: 4 x 5
  recovery quantity median lower upper
  <fct>    <chr>    <dbl> <dbl> <dbl>
1 Dark      z         3.98  3.46  4.54
2 Dark    CTmax    45.7  45.1  46.3
3 Light     z         3.94  3.26  4.39
4 Light    CTmax    44.0  43.5  44.8
```

```
leaf_tls_tab <- leaf_tls$summary |>
  dplyr::mutate(quantity = dplyr::recode(quantity,
                                         z = "z (°C)", CTmax = "CTmax_1hr (°C)")) |>
  dplyr::transmute(`Recovery` = recovery, Quantity = quantity,
                   Median = round(median, 2),
                   Lower = round(lower, 2), Upper = round(upper, 2))
tinytable::tt(leaf_tls_tab, escape = TRUE)
```

The joint fit reproduces the direction Arnold *et al.* (2026) report: 90 min of post-heat light lowers $CT_{max_{1hr}}$ to 44 °C, below the dark-recovery 45.7 °C, while thermal sensitivity is essentially unchanged (z 3.9 °C under light vs 4 °C in the dark). Post-heat light therefore depresses the apparent tolerance limit without altering how steeply tolerance time falls with temperature — matching their summary that light following heat alters tolerance but not sensitivity. Both quantities fall out of a single posterior, with no per-temperature regressions.

6.4 Retained PSII function curves with observed overlay

`predict_survival_curves()` evaluates the fitted 4PL on a grid of assay temperatures and durations, here returning posterior retained-function proportions for each recovery condition; overlaying the observed proportions checks how closely the fitted surfaces track the data.

```

psc_leaf <- predict_survival_curves(wf_leaf)
plot_survival_curves(psc_leaf, observed = leaf_std, log_time = TRUE) +
  ggplot2::labs(x = "Exposure duration (minutes)",
                y = "Retained PSII function")

```

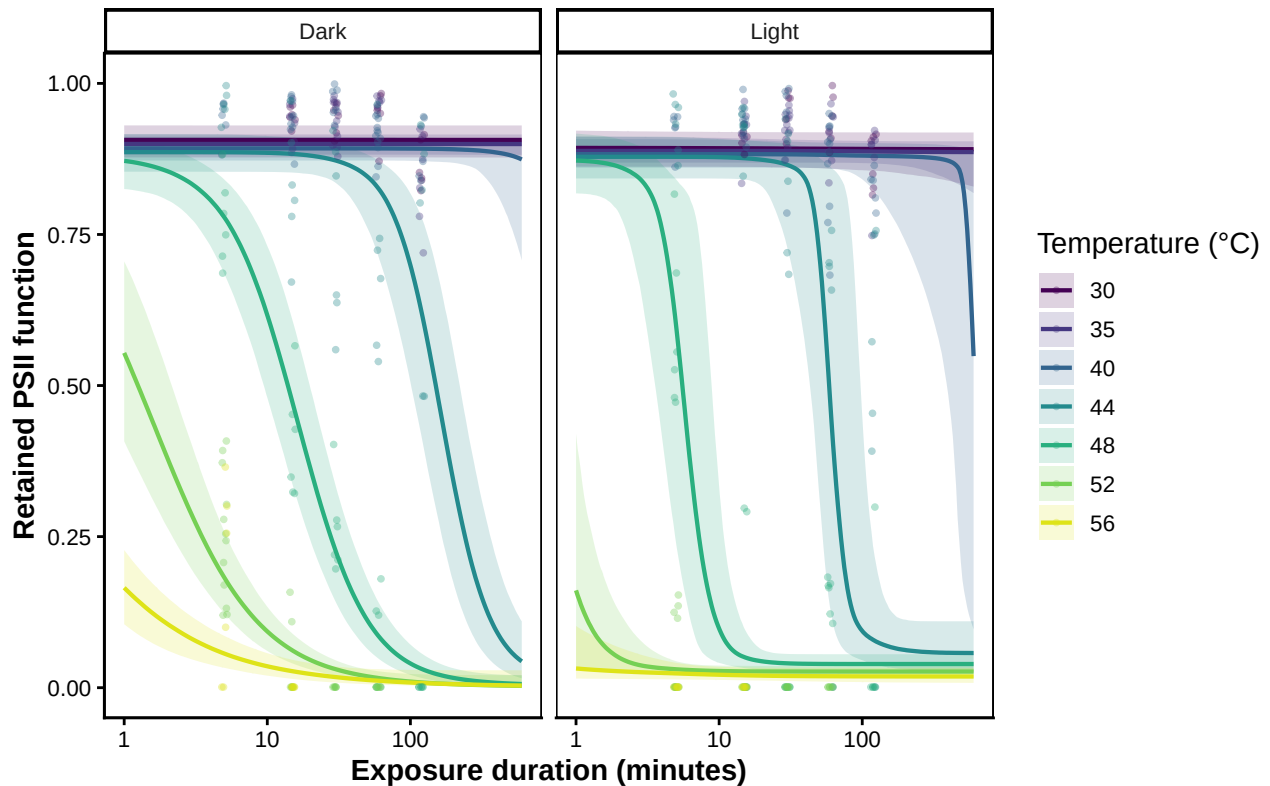


Figure S13: Fitted retained-PSII-function surfaces for snow gum under dark and light post-heat recovery (posterior median and 95% credible band), with observed per-replicate proportions overlaid. The light-recovery curves cross the midpoint at shorter durations, the visual signature of the lower CT_{max} .

6.5 Model fit: Bayesian R^2

Loss of PSII function is described by a single joint 4PL (Beta likelihood), so there is one model to report (Table S15). `bayes_R2_tls()` wraps `brms::bayes_R2()` for a fitted workflow and returns a tidy posterior median with its 95% credible interval.

```

bayes_R2_tls(wf_leaf) |>
  dplyr::transmute(
    Model      = "Leaf PSII (4PL, Beta)",
    `Bayesian R²` = format_interval(estimate, lower, upper, digits = 3)
  ) |>
  tinytable::tt(escape = TRUE)

```

Table S15: Bayesian R^2 for the snow gum leaf PSII 4PL fit, from `bayes_R2_t1s()` (a tidy wrapper around `brms::bayes_R2()`). Posterior median with 95% credible interval.

Model	Bayesian R^2
Leaf PSII (4PL, Beta)	0.935 [0.93, 0.94]

7 Case Study 4: Vinegar fly (*Drosophila suzukii*) across sexes

We'll now turn our attention to a paper by Ørsted *et al.* (2024), who measured TDTs for productivity (offspring number; a measure of fertility), time to coma (sublethal) and mortality using *Drosophila suzukii*.

We'll first load the data and start with the mortality data:

7.0.0.1 Load the data

```
# Load the Drosophila data and summarise mortality counts by sex.
data(dsuzukii)

# Count deaths in each temperature-by-time-by-sex group.
mort_dros <- dsuzukii |>
  dplyr::group_by(temp, time, sex) |>
  dplyr::summarise(n_total = dplyr::n(), n_dead = sum(dead), .groups = "drop") |>
  arrange(temp, time, sex)

# Keep sex for the separate and joint models below.
```

7.0.0.2 Fitting the 4PL model

Now that we have the data set up we'll fit the 4PL model:

```
# Fit separate mortality 4PL models for female and male Drosophila.
# Split mortality counts by sex.
mort_dros_F <- dplyr::filter(mort_dros, sex == "F")
mort_dros_M <- dplyr::filter(mort_dros, sex == "M")

# Standardise each sex's counts for the modelling schema.
std_mort_F <- standardize_data(mort_dros_F, temp = "temp", duration = "time",
                              n_total = "n_total", n_dead = "n_dead",
                              duration_unit = "minutes")
std_mort_M <- standardize_data(mort_dros_M, temp = "temp", duration = "time",
                              n_total = "n_total", n_dead = "n_dead",
                              duration_unit = "minutes")

# Fit one 4PL per sex.
wf_dros_mort_F <- fit_4pl(
```

```

std_mort_F,
chains = 4, iter = 8000, warmup = 4000, cores = 4, seed = 123,
control = list(adapt_delta = 0.99, max_treedepth = 12),
file = file.path(models_dir, "fit_dsuzukii_mortality_4pl_F")
)

wf_dros_mort_M <- fit_4pl(
  std_mort_M,
  chains = 4, iter = 8000, warmup = 4000, cores = 4, seed = 123,
  control = list(adapt_delta = 0.99, max_treedepth = 12),
  file = file.path(models_dir, "fit_dsuzukii_mortality_4pl_M")
)

# Show brms summaries and diagnostics for each fit.
summary(wf_dros_mort_F)

```

```

Family: beta_binomial
Links: mu = identity
Formula: n_surv | trials(n_total) ~ (0.001 + inv_logit(lowraw) * 0.498) + ((0.501 + inv_logit(
  lowraw ~ temp_c
  upraw ~ temp_c
  logk ~ temp_c
  mid ~ temp_c
Data: data (Number of observations: 45)
Draws: 4 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 16000

```

Regression Coefficients:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
lowraw_Intercept	-4.08	0.78	-5.71	-2.66	1.00	12930	11013
lowraw_temp_c	0.08	0.40	-0.70	0.86	1.00	14664	11307
upraw_Intercept	0.86	0.32	0.26	1.52	1.00	10127	9695
upraw_temp_c	-0.13	0.20	-0.54	0.26	1.00	11649	11209
logk_Intercept	2.88	0.21	2.47	3.32	1.00	10048	11037
logk_temp_c	0.05	0.13	-0.21	0.29	1.00	11240	10922
mid_Intercept	2.10	0.02	2.06	2.13	1.00	9791	9418
mid_temp_c	-0.33	0.01	-0.35	-0.31	1.00	10708	10260

Further Distributional Parameters:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
phi	21.20	9.95	8.32	46.61	1.00	13520	11797

Draws were sampled using `sample(hmc)`. For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
summary(wf_dros_mort_M)
```

```
Family: beta_binomial
Links: mu = identity
Formula: n_surv | trials(n_total) ~ (0.001 + inv_logit(lowraw) * 0.498) + ((0.501 + inv_logit(
  lowraw ~ temp_c
  upraw ~ temp_c
  logk ~ temp_c
  mid ~ temp_c
Data: data (Number of observations: 49)
Draws: 4 chains, each with iter = 8000; warmup = 4000; thin = 1;
       total post-warmup draws = 16000
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
lowraw_Intercept	-3.59	0.85	-5.36	-2.03	1.00	13433	10665
lowraw_temp_c	0.17	0.53	-0.85	1.18	1.00	12294	11927
upraw_Intercept	0.94	0.26	0.44	1.47	1.00	12223	11112
upraw_temp_c	-0.25	0.16	-0.58	0.05	1.00	13551	10456
logk_Intercept	2.82	0.22	2.40	3.26	1.00	11760	10646
logk_temp_c	0.01	0.13	-0.23	0.27	1.00	12607	10667
mid_Intercept	2.08	0.02	2.04	2.11	1.00	11907	10407
mid_temp_c	-0.31	0.01	-0.34	-0.29	1.00	11758	10919

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
phi	17.09	7.88	7.28	37.10	1.00	13420	10731

Draws were sampled using `sample(hmc)`. For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

The per-sex Bayesian R^2 for the two mortality fits:

```
# Format Bayesian R2 for the Drosophila mortality fits.
bayes_r2_table(list(
  Female = wf_dros_mort_F,
  Male   = wf_dros_mort_M
)) |> tibble:::tt(escape = TRUE)
```

Now that we have the models we can extract the TDT for each sex. Note that the models have converged well with base don the Rhat, but we can also look at the MCMC chains if we need do as per below:

Table S16: Bayesian R^2 for the per-sex *D. suzukii* mortality 4PL fits, via `bayes_R2_t1s()` (a tidy wrapper around `brms::bayes_R2()`). Posterior median with 95% credible interval.

Model	Bayesian R^2
Female	0.896 [0.86, 0.915]
Male	0.785 [0.729, 0.82]

7.0.0.3 Extracting z and $CT_{max_{1hr}}$ for males and females

We can extract the TDT parameters now as we have done in other case studies:

```
# Extract z, CTmax and Tcrit from the separate Drosophila mortality fits.
# Use the absolute LT50 threshold and the 4-hour reference exposure.
tdt_dros_m <- extract_tdt(wf_dros_mort_M, target_surv = "absolute", t_ref = 60*4, lethal = T
tdt_dros_f <- extract_tdt(wf_dros_mort_F, target_surv = "absolute", t_ref = 60*4, lethal = T

# Summarise z.
z_m <- get_z_summary(tdt_dros_m)
z_f <- get_z_summary(tdt_dros_f)

# Summarise CTmax at 4 hours.
ctmax4_m <- get_ctmax_summary(tdt_dros_m)
ctmax4_f <- get_ctmax_summary(tdt_dros_f)

# Summarise Tcrit.
tcrit_m <- get_tcrit_summary(tdt_dros_m)
tcrit_f <- get_tcrit_summary(tdt_dros_f)

# Format the Drosophila sex TDT table.
dros_sex_tdt <- data.frame(
  Sex = c("Female", "Male"),
  `z (°C)` = c(
    format_interval(z_f$z_median, z_f$z_lower, z_f$z_upper),
    format_interval(z_m$z_median, z_m$z_lower, z_m$z_upper)
  ),
  `CTmax (4 h, °C)` = c(
    format_interval(ctmax4_f$temp_median, ctmax4_f$temp_lower, ctmax4_f$temp_upper),
    format_interval(ctmax4_m$temp_median, ctmax4_m$temp_lower, ctmax4_m$temp_upper)
  ),
  `T_crit (°C)` = c(
    format_interval(tcrit_f$temp_median, tcrit_f$temp_lower, tcrit_f$temp_upper),
    format_interval(tcrit_m$temp_median, tcrit_m$temp_lower, tcrit_m$temp_upper)
  ),
  check.names = FALSE
)

tinytable::tt(dros_sex_tdt, escape = TRUE)
```

Table S17: Thermal-death-time parameters for *D. suzukii* mortality, estimated from a separate joint Bayesian 4PL fit per sex (absolute LT_{50} threshold, reference exposure $t_{ref} = 4$ h). Cells are the posterior median with the 95% credible interval in brackets. T_{crit} uses the rate-multiplier definition and is available because the endpoint is lethal.

Sex	z (°C)	CTmax (4 h, °C)	T_crit (°C)
Female	3.04 [2.84, 3.26]	35.13 [34.99, 35.25]	29.35 [27.73, 30.87]
Male	3.16 [2.93, 3.43]	35.16 [35.02, 35.26]	29.12 [27.4, 30.75]

The per-sex TDT parameters are summarised in Table S17. We see that our z values (their Table 1: 3.28 for males, 3.03 for females) as well as $CT_{max_{4hr}}$ (35.2 °C for both sexes) pretty much match Ørsted *et al.* (2024) quite well for mortality.

We could ask whether males and females are significantly different or not from each other by simply comparing the posterior distributions. Since we have the full posetrior draws, this is easy to do:

```
# Compare female and male posterior z distributions directly.
# Draw z posteriors for each sex.
z_draws_m <- get_z_draws(tdt_dros_m)
z_draws_f <- get_z_draws(tdt_dros_f)

# Form the female-minus-male contrast.
diff_sex_z <- z_draws_f$z - z_draws_m$z # difference the z columns, not the tibbles

# pMCMC is twice the smaller posterior tail probability.
pmcmc_sex_z <- 2 * min(mean(diff_sex_z > 0), mean(diff_sex_z < 0))
```

The median difference in z between males and females is -0.12 °C (95% CI: -0.45 to 0.2 °C; pMCMC = 0.45). The interval spans zero, so the sexes' thermal sensitivities are not clearly distinguishable.

7.0.0.4 Fitting a model to compare sexes directly

We can also fit the sex data in a single model, which is a more powerful approach. To do that, we need to add **sex** as a categorical predictor.

```
# Fit a single Drosophila mortality model with sex on the midpoint.
# Prepare one dataset with both sexes and default priors.
std_sex <- standardize_data(mort_dros, temp = "temp", duration = "time",
                           n_total = "n_total", n_dead = "n_dead",
                           duration_unit = "minutes")
priors_sex <- make_4pl_priors(std_sex)

# Put sex on `mid` so z and CTmax can differ by sex.
joint_sex_fit <- fit_4pl(std_sex, mid = ~ sex * temp_c, prior = priors_sex,
                        t_ref = 60 * 4,
                        chains = 4, iter = 3000, warmup = 1000, cores = 4, seed = 123,
```

```

control = list(adapt_delta = 0.95),
file = file.path(models_dir, "fit_dsuzukii_mortality_sex_joint"),
file_refit = "on_change")
summary(joint_sex_fit)

```

```

Family: beta_binomial
Links: mu = identity
Formula: n_surv | trials(n_total) ~ (0.001 + inv_logit(lowraw) * 0.498) + ((0.501 + inv_logit(
  lowraw ~ temp_c
  upraw ~ temp_c
  logk ~ temp_c
  mid ~ sex * temp_c
Data: std_sex (Number of observations: 94)
Draws: 4 chains, each with iter = 3000; warmup = 1000; thin = 1;
       total post-warmup draws = 8000

```

Regression Coefficients:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
lowraw_Intercept	-4.11	0.74	-5.69	-2.76	1.00	8455	5001
lowraw_temp_c	0.16	0.42	-0.67	0.98	1.00	9237	5819
upraw_Intercept	0.80	0.19	0.43	1.18	1.00	8749	5724
upraw_temp_c	-0.20	0.12	-0.45	0.05	1.00	10011	6401
logk_Intercept	2.91	0.15	2.63	3.21	1.00	7502	6173
logk_temp_c	0.04	0.09	-0.14	0.23	1.00	8361	5396
mid_Intercept	2.08	0.01	2.05	2.11	1.00	6820	5212
mid_sexM	0.02	0.02	-0.01	0.06	1.00	7899	5721
mid_temp_c	-0.33	0.01	-0.35	-0.31	1.00	7308	5569
mid_sexM:temp_c	0.01	0.01	-0.01	0.04	1.00	7953	5072

Further Distributional Parameters:

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
phi	18.70	6.77	9.67	35.63	1.00	7984	5226

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

We can see here that the interaction mid_sexM:temp_c spans zero, supporting little evidence for sex differences in z .

We can now do some manual manipulation to show how to recover the same z that `tls()` and the individual per-sex models return. We do this in two steps: **Step 1** computes the *relative* z (the slope of the midpoint line, $-1/\text{slope}$), and **Step 2** computes the *absolute* z (adding the asymmetry correction so the line follows the full-4PL LT_{50}). Each printout below is the posterior median bracketed by its 2.5% and 97.5% quantiles — i.e. 2.88 3.06 3.26 means a median of 3.06 with a 95% CI of [2.88, 3.26].

```
# Derive relative and absolute z from the joint Drosophila sex model.
# Extract posterior coefficients from the underlying brms fit.
post <- as_draws_df(get_brmsfit(joint_sex_fit)) # workflow -> underlying brmsfit

# Females are the reference; males add the sex-by-temperature interaction.
slope_F <- post$b_mid_temp_c
slope_M <- post$b_mid_temp_c + post[["b_mid_sexM:temp_c"]]

# Relative z.
# Relative z is the inverse midpoint-temperature slope.
z_F <- -1 / slope_F
z_M <- -1 / slope_M

# Print posterior 2.5%, 50% and 97.5% quantiles.
round(quantile(z_F, c(0.025, 0.5, 0.975)), 2)
```

```
2.5%    50%  97.5%
2.88   3.06  3.26
```

```
round(quantile(z_M, c(0.025, 0.5, 0.975)), 2)
```

```
2.5%    50%  97.5%
3.01   3.21  3.42
```

```
# Absolute z.
# Absolute z follows the corrected full-4PL LT50 curve.
lt50 <- function(tc, slope, intercept) {
  low <- 0.001 + plogis(post$b_lowraw_Intercept + post$b_lowraw_temp_c * tc) * 0.498
  up   <- 0.501 + plogis(post$b_upraw_Intercept + post$b_upraw_temp_c * tc) * 0.498
  k    <- exp(post$b_logk_Intercept + post$b_logk_temp_c * tc)
  (intercept + slope * tc) + log((up - 0.5) / (0.5 - low)) / k # natural log
}

# Estimate the local absolute slope over a 1 C step at the mean temperature.
int_F <- post$b_mid_Intercept
int_M <- post$b_mid_Intercept + post$b_mid_sexM
z_F_abs <- -1 / (lt50(0.5, slope_F, int_F) - lt50(-0.5, slope_F, int_F))
z_M_abs <- -1 / (lt50(0.5, slope_M, int_M) - lt50(-0.5, slope_M, int_M))

round(quantile(z_F_abs, c(0.025, 0.5, 0.975)), 2)
```

```
2.5%    50%  97.5%
2.86   3.04  3.24
```

```
round(quantile(z_M_abs, c(0.025, 0.5, 0.975)), 2)
```

```
2.5%    50%  97.5%
2.98    3.18  3.40
```

We can also calculate the $CT_{max_{4hr}}$ in similar light:

```
# Derive relative and absolute CTmax from the joint Drosophila sex model.
# Use the 4-hour reference exposure.
t_ref <- 60 * 4
T_bar <- mean(std_sex$temp - std_sex$temp_c) # centring temperature

# Relative CTmax solves the midpoint line for t_ref.
ctmax_F <- T_bar + (log10(t_ref) - int_F) / slope_F
ctmax_M <- T_bar + (log10(t_ref) - int_M) / slope_M
round(quantile(ctmax_F, c(0.025, 0.5, 0.975)), 2)
```

```
2.5%    50%  97.5%
35.07   35.19 35.29
```

```
round(quantile(ctmax_M, c(0.025, 0.5, 0.975)), 2)
```

```
2.5%    50%  97.5%
35.11   35.22 35.32
```

```
# Absolute CTmax solves the corrected LT50 curve.
slope_abs <- function(slope, intercept) lt50(0.5, slope, intercept) - lt50(-0.5, slope, intercept)
ctmax_F_abs <- T_bar + (log10(t_ref) - lt50(0, slope_F, int_F)) / slope_abs(slope_F, int_F)
ctmax_M_abs <- T_bar + (log10(t_ref) - lt50(0, slope_M, int_M)) / slope_abs(slope_M, int_M)
round(quantile(ctmax_F_abs, c(0.025, 0.5, 0.975)), 2)
```

```
2.5%    50%  97.5%
35.02   35.14 35.24
```

```
round(quantile(ctmax_M_abs, c(0.025, 0.5, 0.975)), 2)
```

```
2.5%    50%  97.5%
35.06   35.17 35.26
```

Everything above - z , $CT_{max_{4hr}}$ and T_{crit} , at either the relative or the absolute threshold — can be calculated using a single `tls()` call on any custom `brms` 4PL: it evaluates each sub-parameter at the requested moderator grid with `posterior_linpred()` (just like you would predict with any `brms` object). Moderators on *any* 4PL parameter (and random effects) are handled automatically. We show both thresholds below.

Table S18: Thermal-death-time parameters for *D. suzukii* mortality from the single joint 4PL model with sex on the midpoint (`joint_sex_fit`), bundled by one `tls()` call at the **relative** midpoint threshold (`target_surv = "relative"`, reference exposure $t_{ref} = 4$ h), matching the Step-1 by-hand relative calculation. Cells are the posterior median with the 95% credible interval in brackets. Compare with the separate-fit estimates in the table above.

Sex	z (°C)	CTmax (4 h, °C)	T_crit (°C)
Female	3.06 [2.88, 3.26]	35.19 [35.07, 35.29]	29.36 [27.81, 30.88]
Male	3.21 [3.01, 3.42]	35.22 [35.11, 35.32]	29.14 [27.46, 30.73]

```
# Extract relative TLS summaries by sex from the joint Drosophila model.
set.seed(123)
# One call bundles z, CTmax and T_crit for every level of `by` into a tidy,
# long summary (one row per sex x quantity). target_surv = "relative" uses the
# midpoint (low+up)/2 threshold, matching the Step-1 by-hand calculation.
tls_sex_summary <- tls(joint_sex_fit, by = "sex", target_surv = "relative",
                      lethal = TRUE, t_ref = 60 * 4)$summary

# Format the Drosophila sex joint table.
tls_sex_table <- tls_sex_summary |>
  dplyr::transmute(
    Sex      = dplyr::recode(as.character(sex), F = "Female", M = "Male"),
    quantity = factor(quantity,
                      levels = c("z", "CTmax", "Tcrit"),
                      labels = c("z (°C)", "CTmax (4 h, °C)", "T_crit (°C)")),
    cell     = format_interval(median, lower, upper)
  ) |>
  dplyr::arrange(Sex) |> # Female before Male, matching the table above
  tidyr::pivot_wider(names_from = quantity, values_from = cell)

tinytable::tt(tls_sex_table, escape = TRUE)
```

Switching to the **absolute** LT_{50} threshold — matching the Step-2 by-hand calculation — needs only `target_surv = "absolute"`:

```
# Extract absolute TLS summaries by sex from the joint Drosophila model.
set.seed(123)
# target_surv = "absolute" reads the curve at the 0.5 absolute-survival crossing
# (adding the asymmetry correction), matching the Step-2 by-hand calculation.
tls_sex_summary_abs <- tls(joint_sex_fit, by = "sex", target_surv = "absolute",
                          lethal = TRUE, t_ref = 60 * 4)$summary

# Format the Drosophila sex joint absolute table.
tls_sex_table_abs <- tls_sex_summary_abs |>
```

Table S19: As Table S18 but at the **absolute** LT_{50} threshold (`target_surv = "absolute"`), matching the Step-2 by-hand absolute calculation. Cells are the posterior median with the 95% credible interval in brackets.

Sex	z (°C)	CTmax (4 h, °C)	T_{crit} (°C)
Female	3.04 [2.86, 3.24]	35.14 [35.02, 35.24]	29.35 [27.8, 30.86]
Male	3.18 [2.98, 3.41]	35.17 [35.06, 35.26]	29.12 [27.45, 30.7]

```
dplyr::transmute(
  Sex      = dplyr::recode(as.character(sex), F = "Female", M = "Male"),
  quantity = factor(quantity,
                     levels = c("z", "CTmax", "Tcrit"),
                     labels = c("z (°C)", "CTmax (4 h, °C)", "T_crit (°C)")),
  cell     = format_interval(median, lower, upper)
) |>
dplyr::arrange(Sex) |>
tidyr::pivot_wider(names_from = quantity, values_from = cell)

tinytable::tt(tls_sex_table_abs, escape = TRUE)
```

The two `tls()` calls reproduce the by-hand calculations: the relative-threshold summary (Table S18) matches the Step-1 relative values (z 3.06 / 3.21, $CT_{max_{4hr}}$ 35.2 / 35.2 °C for female / male), and the absolute-threshold summary (Table S19) matches the Step-2 absolute values (z 3.04 / 3.18, $CT_{max_{4hr}}$ 35.1 / 35.2 °C); T_{crit} is ~29 °C under both.

7.0.0.5 Analysing sublethal measures

The *Drosophila* dataset also includes two sublethal endpoints: heat-coma time and reproductive output.

Heat-coma time can be analysed in two equivalent ways. The classical route fits a Bayesian hierarchical regression of $\log_{10}(\text{time to coma})$ on temperature and derives $z = -1/\beta_1$ and $CT_{max_{1hr}} = \bar{T} + (\log_{10} 60 - \beta_0)/\beta_1$ from the fitted coefficients. The joint route reformulates the same observations as proportion-awake counts and fits the 4PL, recovering the same quantities from the posterior. Below, we fit both models with `sex` as a moderator so one model estimates female- and male-specific parameters.

7.0.0.5.1 Knockdown: time to coma

Each fly was exposed for a fixed duration, scored as awake or in heat coma, and assigned a coma time if it lost the righting response during the exposure. Flies still awake at the end of the exposure had not yet reached the endpoint, so their coma times are **right-censored** at the exposure duration. Overall, 41% of flies are censored.

Censoring matters because dropping censored flies would select for individuals that entered coma early and potentially bias the z estimate. We retain all flies in two complementary analyses. The

linear model fits a right-censored log-time regression, treating the exposure duration of censored flies as a lower bound on their unobserved coma time. The **4PL** model instead analyses the binary awake/coma outcome as proportion-awake counts across exposure durations, matching the `awake ~ time` dose-response used by Ørsted *et al.* (2024).

```
# Prepare right-censored Drosophila heat-coma data for modelling.
# Use exposure duration as the right-censoring bound for awake flies.
coma_tbar <- mean(dsuzukii$temp)
coma_dat <- dsuzukii |>
  dplyr::mutate(
    in_coma      = !is.na(t_coma),
    coma_min     = ifelse(in_coma, t_coma, time), # observed coma time, else exposure time
    log10_coma   = log10(coma_min),
    cens         = ifelse(in_coma, "none", "right"),
    temp_c       = temp - coma_tbar,
    # the assay block shared by flies measured together (one temperature x
    # exposure-level x sex group); used as a random intercept below
    block        = interaction(temp, lvl, sex, drop = TRUE)
  )
```

The classical sublethal TDT model is a hierarchical log-time regression, with a censoring term and a `sex` interaction so females and males have separate slopes (z) and intercepts (CT_{max}). The `(1 | block)` random intercept absorbs variation shared by the flies assayed together in one temperature $\times$ exposure-level $\times$ sex block, to control for non-independence within a block:

```
# Fit the censored log-time heat-coma model by sex.
priors_coma_lin <- c(
  brms::prior(normal(2, 1.5), class = "Intercept"),
  brms::prior(normal(-1, 1), class = "b"),
  brms::prior(exponential(2), class = "sd"),
  brms::prior(exponential(2), class = "sigma")
)

fit_coma_lin <- brms::brm(
  brms::bf(log10_coma | cens(cens) ~ temp_c * sex + (1 | block)),
  data      = coma_dat,
  prior     = priors_coma_lin,
  chains    = 4, iter = 6000, warmup = 3000, cores = 4, seed = 123,
  control   = list(adapt_delta = 0.95),
  backend   = "cmdstanr", silent = 2, refresh = 0,
  file      = file.path(models_dir, "fit_dsuzukii_coma_linear"),
  file_refit = "on_change"
)
```

```
# Derive z and CTmax from the censored heat-coma model.
# Females are the reference; males add the sex interaction.
post_coma_lin <- posterior::as_draws_df(fit_coma_lin)
```



```

slope_F_c <- post_coma_lin$b_temp_c
slope_M_c <- post_coma_lin$b_temp_c + post_coma_lin[["b_temp_c:sexM"]]
int_F_c   <- post_coma_lin$b_Intercept
int_M_c   <- post_coma_lin$b_Intercept + post_coma_lin$b_sexM

z_coma_lin_F <- -1 / slope_F_c
z_coma_lin_M <- -1 / slope_M_c
# The fitted model uses temp_c = temp - coma_tbar. The CTmax equation solves
# for temp_c, so add coma_tbar to express the threshold in degrees C.
ct_coma_lin_F <- coma_tbar + (log10(60) - int_F_c) / slope_F_c
ct_coma_lin_M <- coma_tbar + (log10(60) - int_M_c) / slope_M_c

```

The 4PL route reformulates the same data as proportion-awake counts for each temperature-by-exposure-by-sex block. We fit one joint model with `sex` on the midpoint — the same structure as the mortality `joint_sex_fit` above, but here we write the `brms` model out **by hand** (rather than via `fit_4pl(mid = ...)`) to show that `tls()` derives the per-sex quantities from *any* `brms` 4PL, not only from `fit_4pl()` workflows:

```

# Convert heat-coma observations into proportion-awake counts for the 4PL.
coma_panel <- dsuzukii |>
  dplyr::group_by(temp, lvl, sex) |>
  dplyr::summarise(n_total = dplyr::n(),
                  n_awake = sum(is.na(t_coma)), # awake = not (yet) in coma
                  duration = dplyr::first(time),
                  .groups = "drop") |>
  dplyr::filter(duration > 0)

std_coma <- standardize_data(coma_panel, temp = "temp", duration = "duration",
                             n_total = "n_total", n_surv = "n_awake",
                             duration_unit = "minutes")

priors_coma_4pl <- make_4pl_priors(std_coma)

# Build the same sex-on-midpoint 4PL BY HAND with brms::bf() (rather than
# fit_4pl(mid = ...)) to show tls()/extract_tdt() work on ANY brms 4PL.
coma_4pl_bf <- brms::bf(
  n_surv | trials(n_total) ~
    (0.001 + inv_logit(lowraw) * 0.498) +
    ((0.501 + inv_logit(upraw) * 0.498) - (0.001 + inv_logit(lowraw) * 0.498)) /
    (1 + exp(exp(logk) * (logd - mid))),
  lowraw ~ temp_c, upraw ~ temp_c, logk ~ temp_c,
  mid ~ sex * temp_c, # sex on midpoint => sex-specific z & CTmax
  nl = TRUE, family = brms::beta_binomial(link = "identity")
)

fit_coma_4pl <- brms::brm(
  coma_4pl_bf, data = std_coma, prior = priors_coma_4pl,

```

Table S20: Thermal sensitivity z and the 1-hour critical temperature $CT_{max_{1hr}}$ for *D. suzukii* heat coma, by sex, from the censored linear log-time regression and the joint 4PL on proportion-awake counts, compared with the two-stage drc estimates of Ørsted *et al.* (2024) (their static $CT_{max,1h}$). Cells are the posterior median with the 95% credible interval in brackets; the Ørsted values are point estimates.

Method	z (F)	z (M)	CTmax 1h (F)	CTmax 1h (M)
Linear (censored log-time)	2.34 [2.22, 2.48]	2.34 [2.23, 2.47]	36.47 [36.39, 36.55]	36.27 [36.19, 36.35]
Joint 4PL (proportion awake)	2.41 [2.25, 2.58]	2.4 [2.26, 2.56]	36.48 [36.36, 36.58]	36.27 [36.16, 36.37]
Ørsted et al. (2024), drc	2.23	2.33	36.36	36.31

```

chains = 4, iter = 4000, warmup = 2000, cores = 4, seed = 123,
control = list(adapt_delta = 0.95),
backend = "cmdstanr", silent = 2, refresh = 0,
file = file.path(models_dir, "fit_dsuzukii_coma_4pl"),
file_refit = "on_change"
)

# Extract sex-specific heat-coma TLS summaries from the 4PL.
set.seed(123)
# Report z and 1-hour CTmax by sex. fit_coma_4pl is a raw brmsfit (no bayesTLS
# metadata), so pass temp_mean explicitly -- tls() needs the centring temperature.
coma_4pl_tls <- tls(fit_coma_4pl, by = "sex", target_surv = "relative",
  t_ref = 60, time_multiplier = 1,
  temp_mean = mean(std_coma$temp - std_coma$temp_c))$summary

# Format the Drosophila coma comparison table.
ctls <- function(q, sx) {
  r <- coma_4pl_tls[coma_4pl_tls$quantity == q & coma_4pl_tls$sex == sx, ]
  format_interval(r$median, r$lower, r$upper)
}
fmt <- function(x) format_interval(stats::median(x),
  stats::quantile(x, .025, names = FALSE),
  stats::quantile(x, .975, names = FALSE))

coma_compare <- tibble::tribble(
  ~Method, ~`z (F)`, ~`z (M)`, ~`CTmax 1h (F)`,
  "Linear (censored log-time)", fmt(z_coma_lin_F), fmt(z_coma_lin_M), fmt(ct_coma_lin_F),
  "Joint 4PL (proportion awake)", ctls("z","F"), ctls("z","M"), ctls("CTmax","F"),
  "Ørsted et al. (2024), drc", "2.23", "2.33", "36.36",
)

tinytable::tt(coma_compare, escape = TRUE)

```

Both Bayesian routes agree and recover similar estimates to Ørsted *et al.* (2024) within their

credible intervals (Table S20). Heat-coma sensitivity is $z \approx 2.3\text{-}2.4$ °C and $CT_{max_{1hr}} \approx 36.3\text{-}36.5$ °C for both sexes, with little evidence for sex differences.

7.0.0.5.2 Productivity: reproductive output

For reproductive output, the response has two parts: whether a fly reproduced at all, and how many offspring it produced if it did reproduce. We therefore model productivity with a **hurdle** model rather than a 4PL. The hurdle component estimates the probability of zero offspring, while the positive component models offspring number among reproducing flies. This separates the two biological responses and allows them to vary differently with thermal stress.

As with the 4PL endpoints, productivity changes with exposure duration and temperature. Instead of fitting separate dose-response curves at each temperature and then regressing thresholds, we derive the thermal-death-time quantities directly from the single joint model using the duration and temperature coefficients for each hurdle component.

We compare a Gamma and a log-normal positive part by leave-one-out predictive fit and keep the better one:

```
# Prepare Drosophila productivity data for the hurdle models.
prod_dat <- dsuzukii |>
  dplyr::mutate(logd = log10(time), temp_c = temp - mean(temp), sex = factor(sex))
prod_tbar <- mean(dsuzukii$temp)

# Fit and compare hurdle-Gamma and hurdle-lognormal productivity models.
# Allow both hurdle components to vary with exposure, temperature and sex.
prod_form <- brms::bf(prod ~ logd * temp_c * sex,
  hu ~ logd * temp_c * sex)
prod_args <- list(formula = prod_form, data = prod_dat,
  chains = 4, iter = 3000, warmup = 1000, cores = 4, seed = 123,
  control = list(adapt_delta = 0.95), backend = "cmdstanr",
  silent = 2, refresh = 0, file_refit = "on_change")

fit_prod_ga <- do.call(brms::brm, c(prod_args,
  list(family = brms::hurdle_gamma(),
    file = file.path(models_dir, "fit_dsuzukii_prod_hurdle_gamma"))))
fit_prod_ln <- do.call(brms::brm, c(prod_args,
  list(family = brms::hurdle_lognormal(),
    file = file.path(models_dir, "fit_dsuzukii_prod_hurdle_lognormal"))))

fit_prod_ga <- brms::add_criterion(fit_prod_ga, "loo")
fit_prod_ln <- brms::add_criterion(fit_prod_ln, "loo")
prod_loo <- brms::loo_compare(fit_prod_ga, fit_prod_ln)

# Compare productivity models by leave-one-out predictive fit.
prod_loo      # Gamma positive part is preferred (elpd difference / SE below)
```

	model	elpd_diff	se_diff	p_worse	diag_diff	diag_elpd
fit_prod_ga		0.0	0.0	NA		
fit_prod_ln		-48.0	6.8	1.00		

```
# Row 2 is the worse model because loo_compare sorts best first.
prod_loo_dz <- abs(prod_loo[2, "elpd_diff"]) / prod_loo[2, "se_diff"]
```

The Gamma positive component predicts the data better than the log-normal by 48 expected log predictive-density units (7 SE), so we use `fit_prod_ga` below.

We derive the reproductive summaries directly from the joint posterior. For reproduction incidence (did you reproduce or not?), this gives threshold durations and their temperature slopes; for conditional clutch size, it gives the duration-sensitivity of positive offspring production. Both are calculated analytically from the fitted coefficients for each posterior draw.

```
# Define helpers for reading productivity TDT summaries from posterior coefficients.
prod_post <- posterior::as_draws_df(fit_prod_ga)
prod_tbar <- mean(dsuzukii$temp)
prod_temps <- sort(unique(dsuzukii$temp))
prod_tc <- prod_temps - prod_tbar

# Extract effective coefficients for each sex and hurdle component.
prod_eff <- function(pre, male) {
  pick <- function(s) { nm <- paste0(pre, s)
    if (nm %in% names(prod_post)) prod_post[[nm]] else 0 }
  list(I = pick("Intercept") + (if (male) pick("sexM") else 0),
        T = pick("temp_c") + (if (male) pick("temp_c:sexM") else 0),
        D = pick("logd") + (if (male) pick("logd:sexM") else 0),
        DT = pick("logd:temp_c") + (if (male) pick("logd:temp_c:sexM") else 0))
}
```

```
# Derive productivity incidence and clutch-size thermal summaries.
# Incidence has a natural 50% threshold, so z and CTmax come from the hurdle part.
inc_tdt <- function(male) {
  cc <- prod_eff("b_hu_", male)
  logd50 <- function(t) -(cc$I + cc$T * t) / (cc$D + cc$DT * t)
  eps <- 1e-3
  zT <- vapply(prod_tc, function(t)
    -1 / ((logd50(t + eps) - logd50(t - eps)) / (2 * eps)),
    numeric(nrow(prod_post)))
  tcg <- seq(min(prod_tc) - 2, max(prod_tc) + 2, length.out = 400)
  ldm <- vapply(tcg, logd50, numeric(nrow(prod_post)))
  ctm <- apply(ldm, 1, function(r) {
    k <- which(diff(sign(r - log10(60))) != 0)[1]
    if (is.na(k)) NA_real_
    # Add the centring temperature to report CTmax on the observed scale.
    else prod_tbar + stats::approx(r[k:(k + 1)], tcg[k:(k + 1)], log10(60))$y
  })
}
```

Table S21: Reproductive **incidence** thermal-death-time parameters for *D. suzukii* (whether a fly reproduces at all; the hurdle component of the hurdle-Gamma model), read analytically from the joint posterior, by sex, with the two-stage drc fecundity estimates of Ørsted *et al.* (2024) (static $CT_{max,1h}$) for comparison. z is the mean local z over the assay range. Cells are the posterior median with the 95% credible interval; Ørsted values are point estimates.

Source	z (F)	z (M)	CTmax 1h (F)	CTmax 1h (M)
Incidence (this study)	3.18 [2.9, 3.55]	2.88 [2.65, 3.17]	36.1 [35.9, 36.28]	36.48 [36.3, 36.66]
Ørsted et al. (2024), drc	3.27	3.46	36.27	36.66

```

})
list(z = rowMeans(zT), ct = ctm)      # z = mean local z over the assay range
}
prod_inc_F <- inc_tdt(FALSE)
prod_inc_M <- inc_tdt(TRUE)

# Magnitude has no natural LT50, so report duration-sensitivity b(T).
mag_b <- function(male) {
  cc <- prod_eff("b_", male)
  vapply(prod_tc, function(t) (cc$D + cc$DT * t) / log(10), numeric(nrow(prod_post)))
}
mag_b_F <- mag_b(FALSE)
mag_b_M <- mag_b(TRUE)

# Format the Drosophila productivity incidence table.
fmt_q <- function(x) format_interval(stats::median(x, na.rm = TRUE),
                                     stats::quantile(x, .025, names = FALSE, na.rm = TRUE),
                                     stats::quantile(x, .975, names = FALSE, na.rm = TRUE))

prod_inc_tbl <- tibble::tribble(
  ~Source,                ~`z (F)`,                ~`z (M)`,                ~`CTmax 1h (F)`,
  "Incidence (this study)",  fmt_q(prod_inc_F$z),  fmt_q(prod_inc_M$z),  fmt_q(prod_inc_F$ct),
  "Ørsted et al. (2024), drc", "3.27",                "3.46",                "36.27",
)
tinytable::tt(prod_inc_tbl, escape = TRUE)

```

The incidence component gives a conventional reproduction TDT. The estimated z is approximately 3.2 °C for females and 2.9 °C for males, with $CT_{max,1hr}$ from 36.1 to 36.5 °C, close to the fecundity estimates reported by Ørsted *et al.* (2024) (Table S21). This works because reproduction incidence is bounded and has a natural 50% threshold: whether half of flies reproduce. The duration-by-temperature interaction makes the TDT line slightly curved, so we report the mean local z across the assay range.

The magnitude component answers a different biological question. Conditional clutch size is modelled as an unbounded positive response, with no upper asymptote and no natural 50% threshold.

Table S22: Duration-sensitivity of the conditional clutch (offspring among flies that reproduced), $b(T) = d \log_{10}(\text{clutch}) / d \log_{10}(\text{exposure})$, read from the hurdle-Gamma μ posterior at each assay temperature, by sex (posterior median). $P(\text{decline})$ is the posterior probability that $b(T) < 0$. The clutch is insensitive to exposure at 34 °C ($b \approx 0$) and increasingly reduced by it toward 38 °C.

Temp (°C)	b (F)	P(decline) (F)	b (M)	P(decline) (M)
34	0.02	0.45	-0.04	0.67
35	-0.14	0.90	-0.13	0.98
36	-0.30	1.00	-0.23	1.00
37	-0.46	1.00	-0.32	1.00
38	-0.62	1.00	-0.42	1.00

Unlike survival or reproduction incidence, there is therefore no single threshold from which to define a conventional z . What the joint model does identify is the duration-sensitivity of clutch size, $b(T) = d \log_{10}(\text{clutch}) / d \log_{10}(\text{exposure})$, which changes strongly with temperature (Table S22):

```
# Format the Drosophila productivity magnitude table.
prod_mag_tbl <- tibble::tibble(
  `Temp (°C)`      = prod_temps,
  `b (F)`          = round(apply(mag_b_F, 2, stats::median), 2),
  `P(decline) (F)` = round(apply(mag_b_F, 2, function(b) mean(b < 0)), 2),
  `b (M)`          = round(apply(mag_b_M, 2, stats::median), 2),
  `P(decline) (M)` = round(apply(mag_b_M, 2, function(b) mean(b < 0)), 2)
)
tinytable::tt(prod_mag_tbl, escape = TRUE)
```

Among flies that reproduce, clutch size is essentially unaffected by exposure at 34 °C ($b \approx 0$; $P(\text{decline}) = 0.45$ for females), but declines increasingly steeply at higher temperatures, reaching $b \approx -0.62$ at 38 °C.

The hurdle model reveals two effects that a single fecundity curve combines. Heat reduces the probability of reproducing at every tested temperature, giving a clear incidence-based $z \approx 3$ °C. Among flies that still reproduce, however, heat reduces clutch size mainly at higher temperatures (≥ 36 °C). Reporting the two components separately makes this biological split visible and provides a more nuanced understanding of temperature/time effects on reproduction.

8 Extended Simulation Results

This section provides more detail on the simulation results from the various scenarios we explored to understand bias and coverage of z and CT_{max} . The main manuscript figures summarize the two shape-perturbing scenarios (drifting upper asymptote **up**, varying slope **k**) and the bias results from the two partial-mortality sweeps.

Here we document the remaining scenarios we explored: the strict-equivalence baseline, likelihood misspecification alone, symmetric asymptote compression, and the design-impact — plus the coverage panels for the two sweeps reported as bias-only in the main text.

For each scenario we also tabulate the root-mean-square error (RMSE, in °C) of z and $CT_{max,1h}$ — the typical magnitude of the estimation error, combining bias and variance in a single number on the parameter’s own scale — for the four distinct point estimators. Because RMSE depends only on the point estimates, the Normal- and t -quantile two-stage CI variants share a value and are reported once each.

8.1 *Strict equivalence baseline*

This scenario is the strict-equivalence baseline. The data-generating curve spans the full survival range (up = 0.999, low = 0.001), has a shared slope ($k = 8$), and has a midpoint that declines linearly with temperature (slope -0.15) under a binomial likelihood with no overdispersion. We crossed this scenario with two replicate designs, $n_{\text{reps}} \in \{3, 5\}$, with $N_{\text{sim}} = 1000$ datasets per cell.

We included this scenario to check whether the estimators agree when the classical two-stage assumptions are satisfied. As expected, all methods recovered z and $CT_{max,1h}$ with little bias. The main difference was interval calibration: two-stage intervals based on Normal quantiles did not have nominal coverage, whereas the t -quantile delta-method intervals were close to the nominal 95% coverage. Again, it is important to recognise that we have ensured error from the two-stage is correctly propagated for comparison but often this is ignored.

Of the 1,000 datasets per cell, models in the two-stage analytical approach failed to fit or resulted in extreme bias — here and throughout, $|z|$ bias above 20 °C or $|CT_{max,1h}|$ bias above 10 °C, far outside any plausible value — for 0 binomial and 29–30 (2.9–3.0%) beta-binomial (glmmTMB) Stage-1 fits, versus 0 for the joint 4PL, across the simulations. The overdispersion-corrected two-stage thus fails to converge on a sensible estimate even in this most benign scenario, whereas the joint 4PL does not.

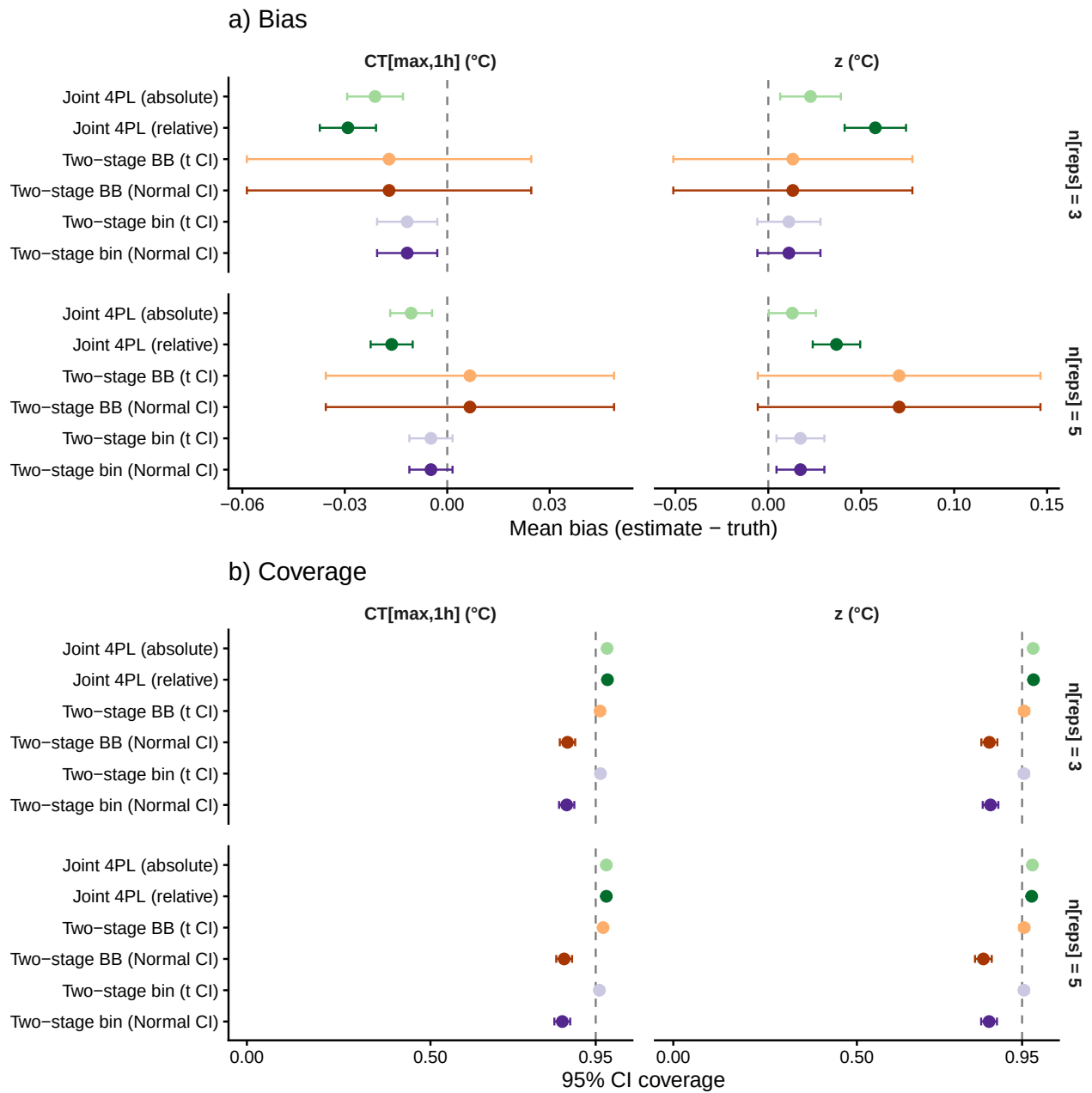


Figure S14: Bias and coverage at the strict-equivalence baseline (binomial DGP, asymptotes near 0/1, single slope k). Rows: $n_{reps} = 3$ (top) and $n_{reps} = 5$ (bottom). Columns: z (left) and $CT_{max,1h}$ (right). Dashed reference lines at zero bias and nominal 0.95 coverage.

```
# Format the RMSE table for simulation scenario 1.
.load_screened("scen1") |>
  dplyr::mutate(Cell = factor(sub(".*_n([0-9]+)$", "\\1", scenario),
                                levels = c("3", "5"))) |>
  .rmse_table(cell_name = "n_reps")
```


Table S23: RMSE ($^{\circ}\text{C}$) of z and $CT_{max,1h}$ at the strict-equivalence baseline, by point estimator and replication budget ($N_{sim} = 1000$ per cell). Lower is better; RMSE combines bias and precision on the parameter’s own scale.

n_reps	Method	z ($^{\circ}\text{C}$)	CTmax ($^{\circ}\text{C}$)
3	Two-stage (binomial)	0.27	0.14
3	Two-stage (beta-binomial)	1.02	0.66
3	Joint 4PL (relative)	0.27	0.14
3	Joint 4PL (absolute)	0.26	0.13
5	Two-stage (binomial)	0.21	0.10
5	Two-stage (beta-binomial)	1.21	0.67
5	Joint 4PL (relative)	0.21	0.10
5	Joint 4PL (absolute)	0.21	0.10

8.2 Likelihood misspecification alone

This scenario keeps the same mean dose-response shape as the strict-equivalence baseline, but generates counts from a beta-binomial likelihood with overdispersion $\phi = 5$. We again used two replicate budgets, $n_{\text{reps}} \in \{3, 5\}$, with $N_{\text{sim}} = 1000$ datasets per cell.

We included this scenario to isolate the cost of likelihood misspecification: the mean structure is correct for all estimators, but the binomial two-stage variant ignores overdispersion at Stage 1. Point estimates remained largely unbiased, showing that overdispersion alone did not shift the target estimands. The effect was on uncertainty rather than location, and it was modest here: overdispersion alone barely separated the binomial and beta-binomial two-stage fits, in either point estimates or calibration. Both produced Normal-quantile intervals that under-covered and were brought close to nominal by the t -quantile correction, while the joint 4PL was well calibrated throughout.

Of the 1,000 datasets per cell, models in the two-stage analytical approach failed to fit or resulted in extreme bias for 0 binomial and 2–7 (0.2–0.7%) beta-binomial (glmmTMB) Stage-1 fits, versus 0 for the joint 4PL, across the simulations.

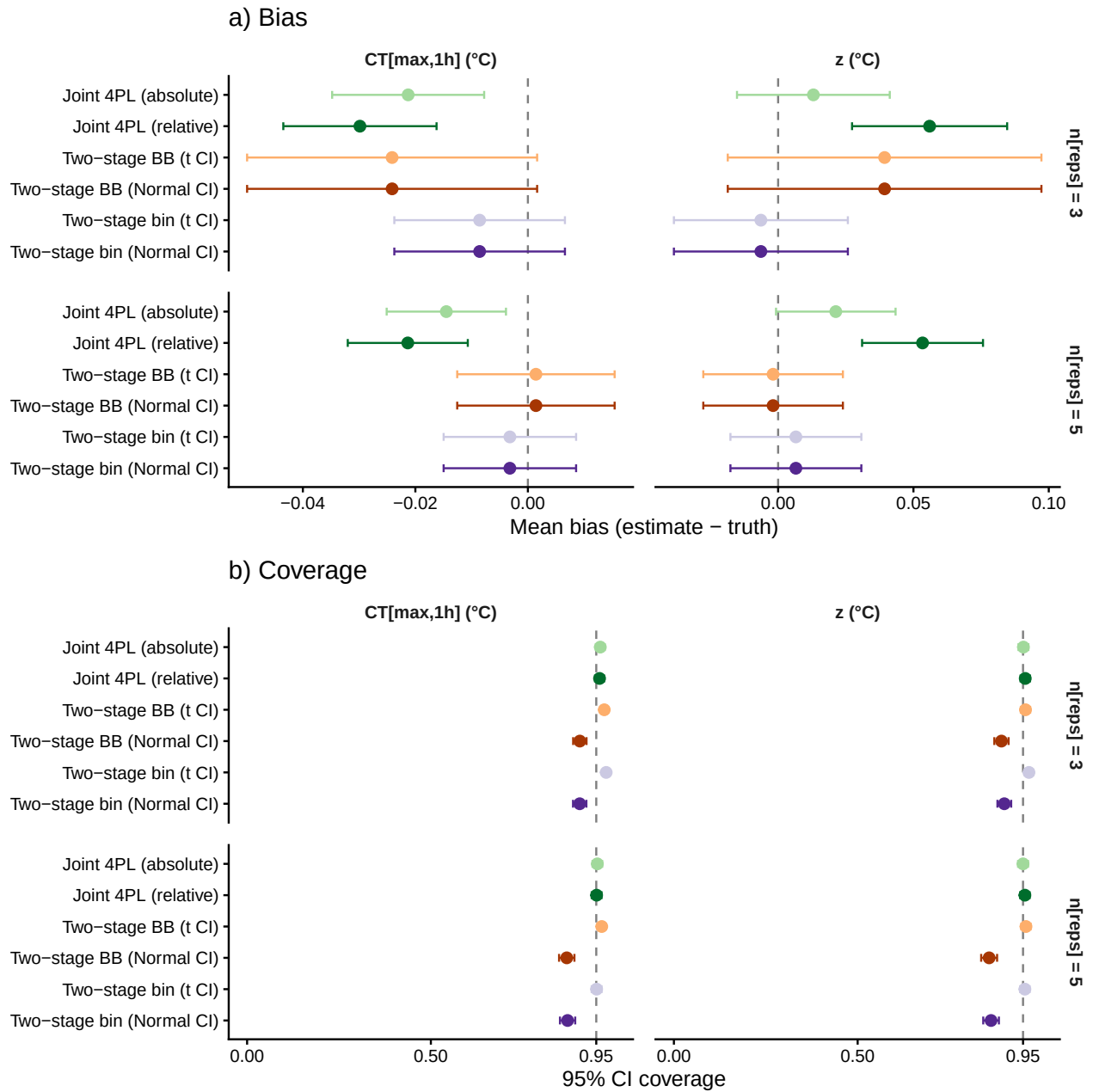


Figure S15: Bias and coverage with likelihood misspecification only (beta-binomial DGP with $\phi = 5$; curve shape as in the strict-equivalence baseline). Rows: $n_{reps} = 3$ (top) and $n_{reps} = 5$ (bottom).

```
# Format the RMSE table for simulation scenario 2.
.load_screened("scen2") |>
  dplyr::mutate(Cell = factor(sub(".*_n([0-9]+)$", "\\1", scenario),
                              levels = c("3", "5"))) |>
  .rmse_table(cell_name = "n_reps")
```

Table S24: RMSE ($^{\circ}\text{C}$) of z and $CT_{max,1h}$ under likelihood misspecification only (beta-binomial DGP, $\phi = 5$; curve shape as in the baseline), by point estimator and replication budget.

n_reps	Method	z ($^{\circ}\text{C}$)	CTmax ($^{\circ}\text{C}$)
3	Two-stage (binomial)	0.52	0.24
3	Two-stage (beta-binomial)	0.93	0.41
3	Joint 4PL (relative)	0.47	0.22
3	Joint 4PL (absolute)	0.46	0.22
5	Two-stage (binomial)	0.39	0.19
5	Two-stage (beta-binomial)	0.42	0.23
5	Joint 4PL (relative)	0.36	0.17
5	Joint 4PL (absolute)	0.36	0.17

8.3 Asymptotes compress symmetrically

In this scenario, both asymptotes move toward each other as temperature increases: **up** declines by 0.01 per $^{\circ}\text{C}$ from 0.92, while low increases by 0.01 per $^{\circ}\text{C}$ from 0.05. This preserves the approximate logit-difference but compresses the response range. Although this symmetric compression is not biologically realistic, it is useful as a diagnostic test: it changes the dose-response range while largely preserving the midpoint-temperature relationship. We used two replicate budgets, $n_{\text{reps}} \in \{3, 5\}$, with $N_{\text{sim}} = 1000$ datasets per cell.

The two-stage fits showed clear bias here — negative in $CT_{max,1h}$ and positive in z — together with under-coverage, because the classical model cannot represent changing asymptotes directly. The joint 4PL remained close to unbiased with near-nominal coverage.

Of the 1,000 datasets per cell, models in the two-stage analytical approach failed to fit or resulted in extreme bias for 0 binomial and 0 beta-binomial (glmmTMB) Stage-1 fits, versus 0 for the joint 4PL, across the simulations.

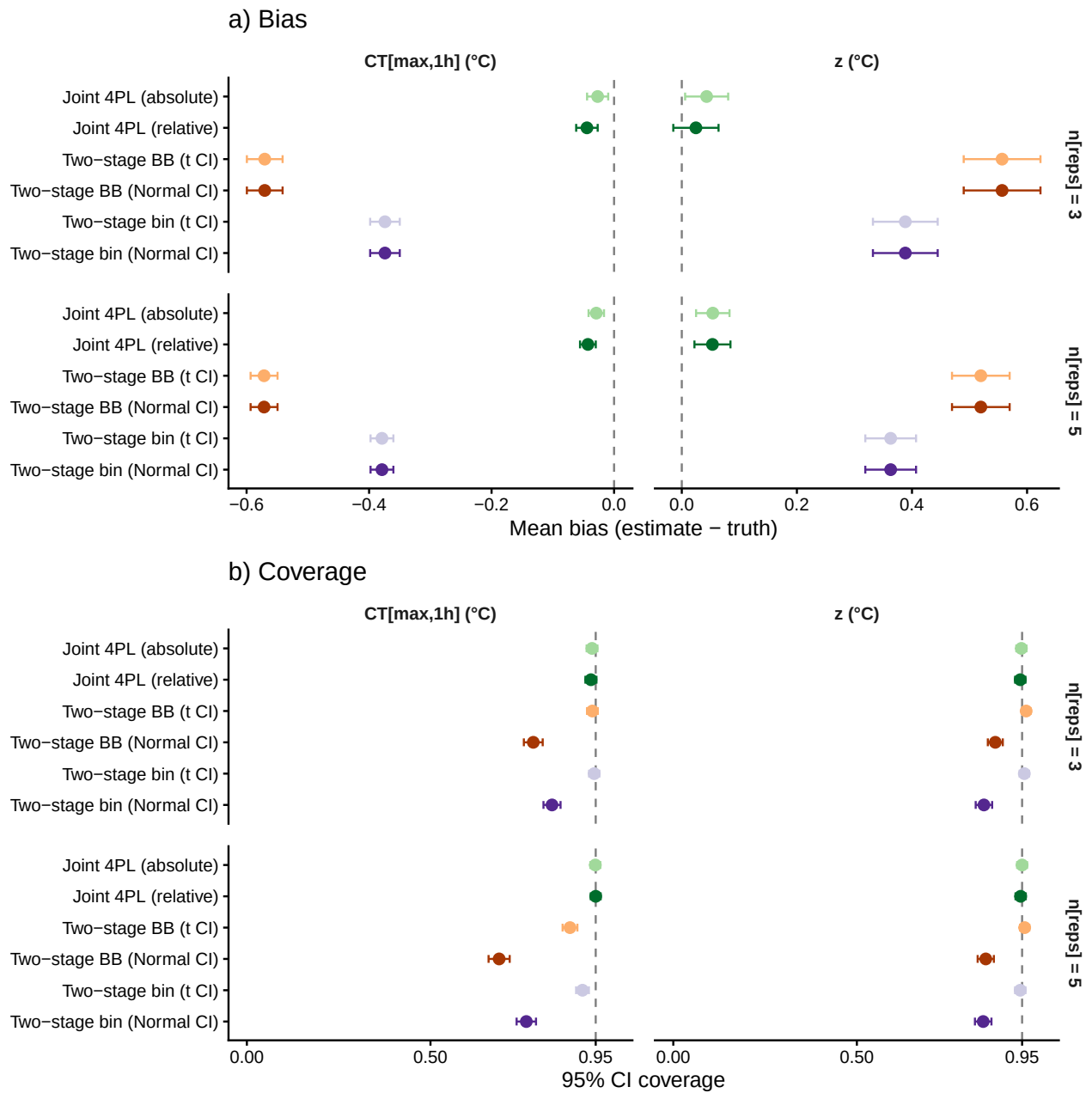


Figure S16: Bias and coverage when both asymptotes compress symmetrically with temperature.
Rows: $n_{reps} = 3$ (top) and $n_{reps} = 5$ (bottom).

```
# Format the RMSE table for simulation scenario 4.
.load_screened("scen4") |>
  dplyr::mutate(Cell = factor(sub(".*_n([0-9]+)$", "\\1", scenario),
                                levels = c("3", "5"))) |>
  .rmse_table(cell_name = "n_reps")
```

Table S25: RMSE ($^{\circ}\text{C}$) of z and $CT_{max,1h}$ when both asymptotes compress symmetrically with temperature, by point estimator and replication budget.

n_reps	Method	z ($^{\circ}\text{C}$)	CTmax ($^{\circ}\text{C}$)
3	Two-stage (binomial)	0.99	0.54
3	Two-stage (beta-binomial)	1.21	0.74
3	Joint 4PL (relative)	0.63	0.29
3	Joint 4PL (absolute)	0.60	0.28
5	Two-stage (binomial)	0.80	0.48
5	Two-stage (beta-binomial)	0.96	0.67
5	Joint 4PL (relative)	0.51	0.21
5	Joint 4PL (absolute)	0.47	0.21

8.4 Coverage for drifting up across temperatures (β_{up})

This complements the main-text β_{up} bias sweep with the matching coverage panels.

Of the 1,000 datasets per cell, models in the two-stage analytical approach failed to fit or resulted in extreme bias for 0 binomial and 0–1 (0.0–0.1%) beta-binomial (glmmTMB) Stage-1 fits, versus 0 for the joint 4PL, across the simulations.

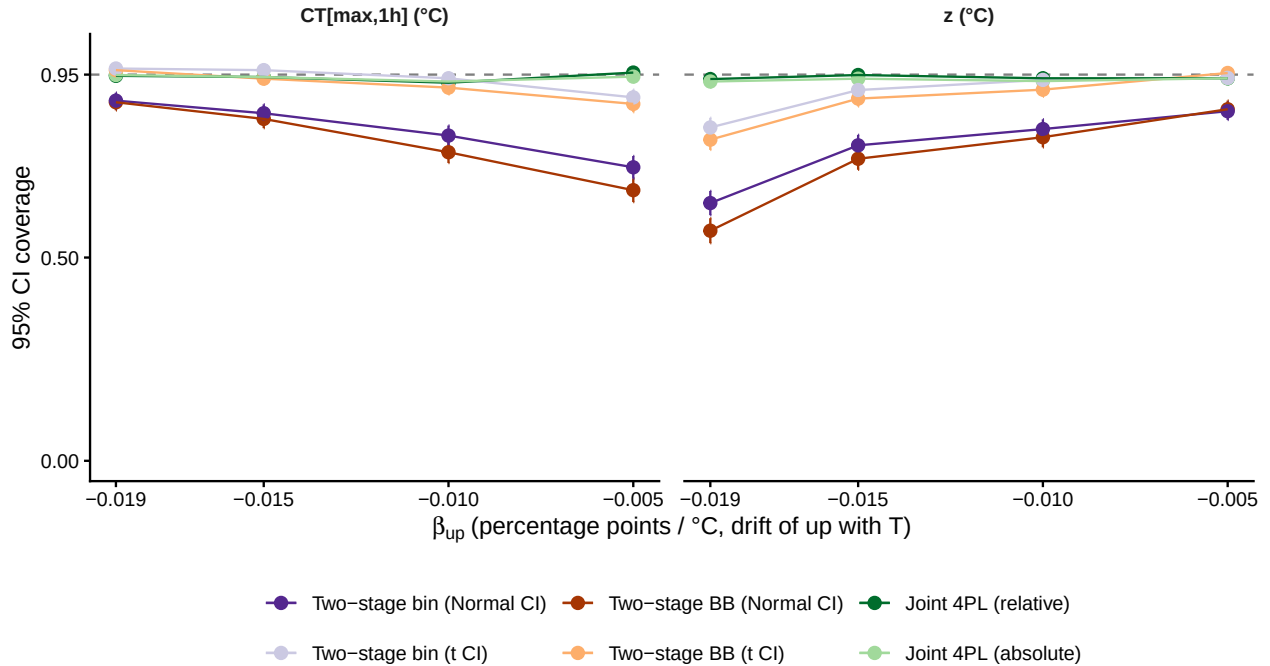


Figure S17: 95% CI coverage across the β_{up} sweep (Scen 6). Left: z . Right: $CT_{max,1h}$. Dashed line at nominal 0.95. Same DGP and method set as the main-text β_{up} bias sweep, with coverage shown for all six methods. Normal- and t-quantile two-stage CIs differ in coverage even though their point estimates are identical.

Table S26: RMSE ($^{\circ}\text{C}$) of z and $CT_{max,1h}$ across the β_{up} drift sweep (Scen 6), by point estimator. β_{up} is the per- $^{\circ}\text{C}$ drift of the upper asymptote (percentage points per $^{\circ}\text{C}$).

beta_up (pp/ $^{\circ}\text{C}$)	Method	z ($^{\circ}\text{C}$)	CTmax ($^{\circ}\text{C}$)
-0.019	Two-stage (binomial)	0.69	0.25
-0.019	Two-stage (beta-binomial)	0.78	0.28
-0.019	Joint 4PL (relative)	0.52	0.22
-0.019	Joint 4PL (absolute)	0.47	0.20
-0.015	Two-stage (binomial)	0.60	0.30
-0.015	Two-stage (beta-binomial)	0.66	0.35
-0.015	Joint 4PL (relative)	0.49	0.22
-0.015	Joint 4PL (absolute)	0.45	0.21
-0.010	Two-stage (binomial)	0.59	0.37
-0.010	Two-stage (beta-binomial)	0.63	0.45
-0.010	Joint 4PL (relative)	0.50	0.23
-0.010	Joint 4PL (absolute)	0.46	0.22
-0.005	Two-stage (binomial)	0.65	0.49
-0.005	Two-stage (beta-binomial)	0.71	0.62
-0.005	Joint 4PL (relative)	0.52	0.23
-0.005	Joint 4PL (absolute)	0.47	0.23

```
# Format the RMSE table for simulation scenario 6.
.load_screened("scen6") |>
  dplyr::mutate(bu = -as.numeric(sub("scen6_ub_m", "", scenario)) / 1000,
    Cell = factor(sprintf("%.3f", bu),
      levels = sprintf("%.3f", sort(unique(bu))))) |>
  .rmse_table(cell_name = "beta_up (pp/ $^{\circ}\text{C}$ )")
```

8.5 Coverage for constant-but-reduced up (up_0)

This complements the main-text up_0 bias sweep. With a low up_0 , the two-stage Stage-1 GLMs miscalibrate per temperature, the Stage-2 OLS inherits that bias, and CI coverage on $CT_{max,1h}$ collapses well below nominal even after the t-quantile small-sample correction.

Of the 1,000 datasets per cell, models in the two-stage analytical approach failed to fit or resulted in extreme bias for 0–89 (0.0–8.9%) binomial and 0–101 (0.0–10.1%) beta-binomial (glmmTMB) Stage-1 fits, versus 0 for the joint 4PL, across the simulations.

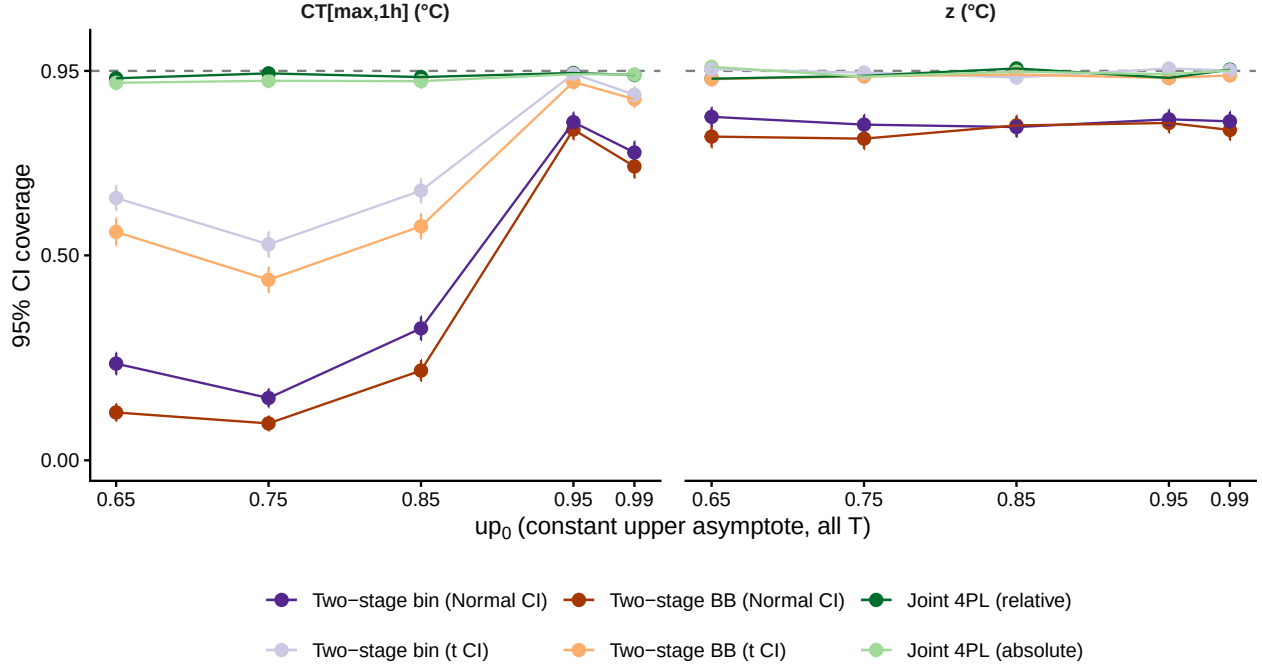


Figure S18: 95% CI coverage across the up_0 sweep (Scen 7). Same DGP and method set as the main-text up_0 bias sweep. Two-stage CTmax coverage collapses as up_0 falls; the t-correction widens CIs without recovering nominal calibration. Joint 4PL coverage holds at 92–96% throughout.

```
# Format the RMSE table for simulation scenario 7.
.load_screened("scen7") |>
  dplyr::mutate(u0 = as.numeric(sub("scen7_u0_", "", scenario)) / 100,
               Cell = factor(sprintf("%.2f", u0),
                             levels = sprintf("%.2f", sort(unique(u0))))) |>
  .rmse_table(cell_name = "up_0")
```

8.6 Experimental Design Impacts

8.6.0.1 Different temperature time duration levels

This scenario varies experimental design while holding the data-generating process fixed. We crossed two temperature-duration grids (full: 5 temperatures $\times$ 6 durations; sparse: 3 temperatures $\times$ 4 durations) with three levels of replication, $n_{\text{reps}} \in \{1, 3, 5\}$. Here, n_{reps} is the number of replicate cups per temperature-by-duration cell; each cup contained $N \sim \text{Uniform}\{10, \dots, 20\}$ organisms. The DGP used the baseline beta-binomial shape ($up = 0.92$, $low = 0.05$, $k = 8$, midpoint linear in T , $\phi = 5$); the sparsest design had only 12 ‘cups’ total.

We included this scenario to test how much information each estimator needs before the two-stage reduction becomes unstable. Performance improved with denser designs and more replication, but the sparse one-replicate design was especially problematic for the two-stage pipeline. The joint 4PL was more stable across the design gradient because it shares information across temperatures and durations rather than estimating each temperature-specific curve separately.

Table S27: RMSE ($^{\circ}\text{C}$) of z and $CT_{max,1h}$ across the up_0 sweep (Scen 7), by point estimator. up_0 is the constant upper asymptote held across all temperatures.

up_0	Method	z ($^{\circ}\text{C}$)	CT_{max} ($^{\circ}\text{C}$)
0.65	Two-stage (binomial)	7.90	9.28
0.65	Two-stage (beta-binomial)	3.15	5.94
0.65	Joint 4PL (relative)	1.03	0.60
0.65	Joint 4PL (absolute)	1.39	0.73
0.75	Two-stage (binomial)	1.86	2.94
0.75	Two-stage (beta-binomial)	2.15	3.54
0.75	Joint 4PL (relative)	0.78	0.37
0.75	Joint 4PL (absolute)	0.86	0.48
0.85	Two-stage (binomial)	1.17	1.38
0.85	Two-stage (beta-binomial)	1.40	1.78
0.85	Joint 4PL (relative)	0.58	0.30
0.85	Joint 4PL (absolute)	0.57	0.31
0.95	Two-stage (binomial)	0.73	0.35
0.95	Two-stage (beta-binomial)	0.87	0.43
0.95	Joint 4PL (relative)	0.47	0.23
0.95	Joint 4PL (absolute)	0.45	0.22
0.99	Two-stage (binomial)	0.61	0.29
0.99	Two-stage (beta-binomial)	0.76	0.34
0.99	Joint 4PL (relative)	0.43	0.21
0.99	Joint 4PL (absolute)	0.41	0.20

Of the 1,000 datasets per cell, models in the two-stage analytical approach failed to fit or resulted in extreme bias for 0–37 (0.0–3.7%) binomial and 0–63 (0.0–6.3%) beta-binomial (glmmTMB) Stage-1 fits, versus 0 for the joint 4PL, across the simulations.

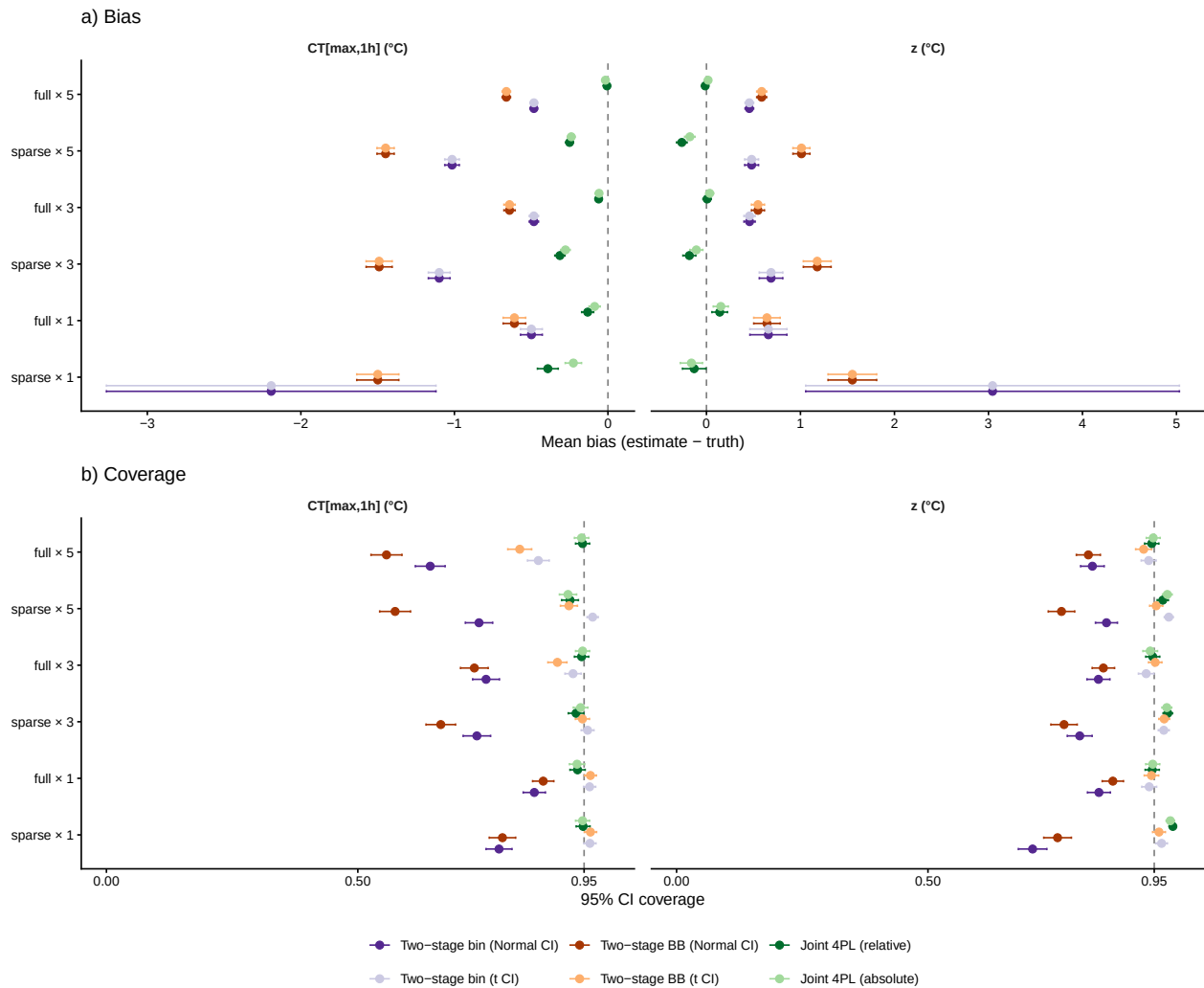


Figure S19: Bias and coverage across six design × replication cells (Scen 8). Two-stage at sparse × 1 produces large, unstable estimates with implausibly wide t-corrected CIs; joint 4PL returns coherent intervals of appropriate width across all six cells.

```
# Format the RMSE table for simulation scenario 8.
.load_screened("scen8") |>
  dplyr::mutate(design = ifelse(grepl("full", scenario), "full", "sparse"),
    n_reps = as.integer(sub(".*_n([0-9]+)$", "\\1", scenario)),
    Cell = factor(sprintf("%s x %d", design, n_reps),
      levels = c("sparse x 1", "full x 1",
        "sparse x 3", "full x 3",
        "sparse x 5", "full x 5"))) |>
  .rmse_table(cell_name = "Design x reps")
```

Table S28: RMSE ($^{\circ}\text{C}$) of z and $CT_{max,1h}$ across the six design $\times$ replication cells (Scen 8), by point estimator. The sparse, single-replicate design is the most demanding for the two-stage reduction.

Design x reps	Method	z ($^{\circ}\text{C}$)	CTmax ($^{\circ}\text{C}$)
sparse x 1	Two-stage (binomial)	32.04	17.36
sparse x 1	Two-stage (beta-binomial)	4.32	2.61
sparse x 1	Joint 4PL (relative)	2.00	1.16
sparse x 1	Joint 4PL (absolute)	1.91	0.88
full x 1	Two-stage (binomial)	3.24	1.25
full x 1	Two-stage (beta-binomial)	2.34	1.32
full x 1	Joint 4PL (relative)	1.34	0.64
full x 1	Joint 4PL (absolute)	1.30	0.59
sparse x 3	Two-stage (binomial)	2.13	1.58
sparse x 3	Two-stage (beta-binomial)	2.65	2.02
sparse x 3	Joint 4PL (relative)	1.12	0.62
sparse x 3	Joint 4PL (absolute)	1.11	0.56
full x 3	Two-stage (binomial)	1.07	0.68
full x 3	Two-stage (beta-binomial)	1.24	0.87
full x 3	Joint 4PL (relative)	0.70	0.34
full x 3	Joint 4PL (absolute)	0.67	0.33
sparse x 5	Two-stage (binomial)	1.28	1.27
sparse x 5	Two-stage (beta-binomial)	1.77	1.71
sparse x 5	Joint 4PL (relative)	0.94	0.48
sparse x 5	Joint 4PL (absolute)	0.90	0.46
full x 5	Two-stage (binomial)	0.86	0.61
full x 5	Two-stage (beta-binomial)	1.03	0.79
full x 5	Joint 4PL (relative)	0.50	0.24
full x 5	Joint 4PL (absolute)	0.48	0.23

8.6.0.2 Performance when estimating under short timeframes

Thermal-tolerance assays are often limited by how long they can run. Long exposures at cooler assay temperatures are costly, so experiments may stop before those treatments reach their LT50. This scenario isolates that cost. We held the data-generating process fixed at the baseline beta-binomial shape (up = 0.92, low = 0.05, $k = 8$, midpoint linear in T , $\phi = 5$), used the full five-temperature design (30–38 °C) with $n_{\text{reps}} = 5$, and varied only the maximum exposure duration. Each design used six log-spaced durations capped at 60, 120, 240, or 405 min; the 405-min cap reproduces the full grid. Because the truth is identical across caps, changes in bias, coverage, and RMSE reflect only how well each design recovers the same z and $CT_{\text{max}_{1\text{hr}}}$.

Shorter windows remove the long-duration observations that anchor the cooler-temperature LT50s. At the shortest caps, the coolest temperatures may never reach 50% mortality, so the LT50 must be extrapolated rather than observed. This scenario therefore tests whether each estimator degrades gracefully as duration information is withdrawn, and whether the joint 4PL retains calibration better than the two-stage reduction by borrowing strength across temperatures.

Across the 1,000 datasets per cap, the two-stage Stage-1 fits failed or produced extreme bias for 0–281 (0.0–28.1%) binomial and 0–287 (0.0–28.7%) beta-binomial (glmmTMB) fits, compared with 0 for the joint 4PL.

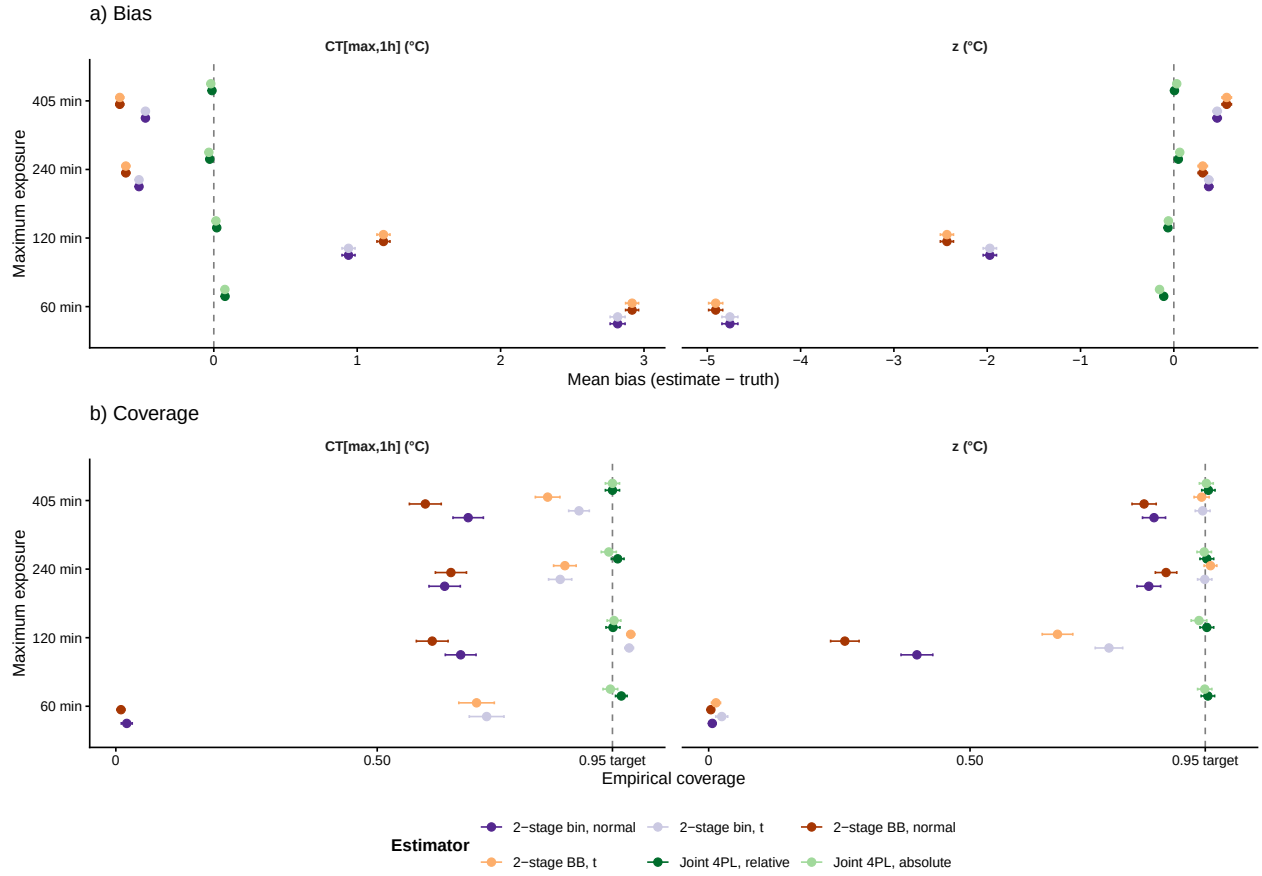


Figure S20: Effect of maximum exposure duration on estimator performance (Scen 9). Rows show mean bias (a; estimate minus truth) and 95% interval coverage (b) for z and $CT_{max_{1hr}}$. The 405-min cap is the full design; shorter caps progressively remove the long-duration observations that anchor LT_{50} at cooler assay temperatures.

```
# Format the RMSE table for simulation scenario 9.
.load_screened("scen9") |>
  dplyr::mutate(tmax = as.integer(sub(".*_tmax_([0-9]+)$", "\\1", scenario)),
    Cell = factor(sprintf("%d min", tmax),
      levels = c("60 min", "120 min",
        "240 min", "405 min"))) |>
  .rmse_table(cell_name = "Max exposure")
```

9 Manuscript Figure 5: cross-case-study summary

This figure compiles the joint Bayesian 4PL posterior distributions of the TDT quantities (z and CT_{max}) from the four case studies into a single multi-panel display. Every panel is a group comparison drawn from a single joint fit: three cereal-aphid species (panel a), snow gum under dark vs light post-heat recovery (panel b), zebrafish across two oxygen treatments (panel c), and the vinegar fly by sex (panel d). The CT_{max} reference exposure follows each case study's own analysis

Table S29: RMSE ($^{\circ}\text{C}$) of z and $CT_{max,1h}$ by maximum exposure duration (Scen 9). Lower RMSE indicates more accurate point estimates. The 60-min cap is the most demanding because the coolest assay temperatures may not reach 50% mortality within the observation window.

Max exposure	Method	z ($^{\circ}\text{C}$)	CTmax ($^{\circ}\text{C}$)
60 min	Two-stage (binomial)	4.90	2.90
60 min	Two-stage (beta-binomial)	5.02	2.98
60 min	Joint 4PL (relative)	0.57	0.28
60 min	Joint 4PL (absolute)	0.55	0.28
120 min	Two-stage (binomial)	2.28	1.19
120 min	Two-stage (beta-binomial)	2.68	1.39
120 min	Joint 4PL (relative)	0.49	0.22
120 min	Joint 4PL (absolute)	0.46	0.20
240 min	Two-stage (binomial)	0.79	0.64
240 min	Two-stage (beta-binomial)	0.86	0.76
240 min	Joint 4PL (relative)	0.51	0.22
240 min	Joint 4PL (absolute)	0.48	0.21
405 min	Two-stage (binomial)	0.85	0.60
405 min	Two-stage (beta-binomial)	0.99	0.79
405 min	Joint 4PL (relative)	0.51	0.24
405 min	Joint 4PL (absolute)	0.48	0.24

— 1 h for aphids, leaf and zebrafish, 4 h for the vinegar fly — so the right-column strips are labelled $CT_{max_{1hr}}$ and $CT_{max_{4hr}}$ respectively; the panels carry independent x-axes, so the columns are not on a shared scale.

```
# Assemble the manuscript cross-case posterior-density figure from the four joint
# 4PL case-study fits. Per-group z and CTmax draws come from the tidy-long tls()
# output (aphids, leaf, zebrafish) or the per-sex extract_tdt() objects (fly).
suppressPackageStartupMessages({
  library(ggplot2); library(dplyr); library(tibble); library(ggribes)
  library(patchwork); library(ggtext); library(cowplot); library(magick); library(here)
})

# A tls() result -> fig4 long draws + summary, relabelling CTmax to the study's
# reference (1 h here). `modcol` is the moderator column name in the tls output.
tls_to_fig4 <- function(tls_obj, modcol, case, ctmax_lab = "CTmax_1hr") {
  relab <- function(q) ifelse(q == "z", "z", ctmax_lab)
  dr <- tls_obj$draws |>
    dplyr::filter(quantity %in% c("z", "CTmax")) |>
    dplyr::transmute(case_study = case, group = as.character(.data[[modcol]]),
                      quantity = relab(quantity), value, .draw)
  sm <- tls_obj$summary |>
    dplyr::filter(quantity %in% c("z", "CTmax")) |>
    dplyr::transmute(case_study = case, group = as.character(.data[[modcol]]),
                      quantity = relab(quantity), median, lower, upper)
  list(draws = dr, summary = sm)
}

mk_row <- function(cs, grp, q, med, lo, hi) {
  tibble::tibble(case_study = cs, group = grp, quantity = q,
                  median = med, lower = lo, upper = hi)
}

aphid_f4 <- tls_to_fig4(aphid_tls, "species", "Cereal aphids")
leaf_f4 <- tls_to_fig4(leaf_tls, "recovery", "Snow gum leaf PSII")
zf_f4 <- tls_to_fig4(zf_tls, "oxygen", "Zebrafish (oxygen)")

# Vinegar fly mortality, by sex: separate per-sex fits (absolute 4 h threshold).
fly_draws <- dplyr::bind_rows(
  tibble::tibble(case_study = "Vinegar fly (mortality)", group = "Female",
                  quantity = "z", value = get_z_draws(tdt_dros_f)$z,
                  .draw = get_z_draws(tdt_dros_f)$draw),
  tibble::tibble(case_study = "Vinegar fly (mortality)", group = "Female",
                  quantity = "CTmax_4hr", value = get_ctmax_draws(tdt_dros_f)$CTmax,
                  .draw = get_ctmax_draws(tdt_dros_f)$draw),
  tibble::tibble(case_study = "Vinegar fly (mortality)", group = "Male",
                  quantity = "z", value = get_z_draws(tdt_dros_m)$z,
                  .draw = get_z_draws(tdt_dros_m)$draw),
  tibble::tibble(case_study = "Vinegar fly (mortality)", group = "Male",
                  quantity = "CTmax_4hr", value = get_ctmax_draws(tdt_dros_m)$CTmax,
```

```

        .draw = get_ctmax_draws(tdt_dros_m)$draw)
)
fly_summary <- dplyr::bind_rows(
  mk_row("Vinegar fly (mortality)", "Female", "z", z_f$z_median, z_f$z_lower, z_f$z_upper),
  mk_row("Vinegar fly (mortality)", "Female", "CTmax_4hr", ctmax4_f$temp_median, ctmax4_f$temp_upper),
  mk_row("Vinegar fly (mortality)", "Male", "z", z_m$z_median, z_m$z_lower, z_m$z_upper),
  mk_row("Vinegar fly (mortality)", "Male", "CTmax_4hr", ctmax4_m$temp_median, ctmax4_m$temp_upper)
)

fig4_draws <- dplyr::bind_rows(aphid_f4$draws, leaf_f4$draws, zf_f4$draws, fly_draws) |>
  dplyr::filter(is.finite(value))
# Panel a only: trim the aphid display tails to the central 98% per group so its
# axes stay tight and the contrast brackets sit cleanly (the 95% CIs are still
# drawn explicitly). Other case studies (b/c/d) are left untouched.
fig4_draws <- fig4_draws |>
  dplyr::group_by(case_study, group, quantity) |>
  dplyr::filter(case_study != "Cereal aphids" |
    (value <= stats::quantile(value, 0.99) &
     value >= stats::quantile(value, 0.01))) |>
  dplyr::ungroup()
fig4_summary <- dplyr::bind_rows(aphid_f4$summary, leaf_f4$summary, zf_f4$summary, fly_summary)

# Persist the per-group z/CTmax medians so the manuscript can report figures
# (e.g. the CTmax_1hr range across case studies) as inline code, not magic
# numbers. Read back in ms/ms.qmd.
dir.create(here::here("output", "data"), showWarnings = FALSE, recursive = TRUE)
saveRDS(fig4_summary, here::here("output", "data", "fig4_summary.rds"))

# Facet order: z (left), CTmax (right); fly uses the 4 h CTmax reference.
fig4_draws <- mutate(fig4_draws, quantity = factor(quantity, levels = c("z", "CTmax_1hr", "CTmax_4hr")))
fig4_summary <- mutate(fig4_summary, quantity = factor(quantity, levels = c("z", "CTmax_1hr", "CTmax_4hr")))

# Group display order. Colour by level; the CI bar + point use a darker tone of
# the fill. Aphid species reuse the original 3-level viridis palette; snow gum
# keeps its olive, the fly its green/orange, zebrafish a blue oxygen pair.
sp_lv <- c("M_dirhodum", "S_avenae", "R_padi")
rec_lv <- c("Dark", "Light")
oxy_lv <- c("normoxia", "hyperoxia")
vz <- viridisLite::viridis(3, begin = 0.15, end = 0.85)
fig4_pal <- c(setNames(vz, sp_lv),
  "Dark" = "#4F7942", "Light" = "#8FB07A",
  "normoxia" = "#2C7FB8", "hyperoxia" = "#7FCDBB",
  "Female" = "#009E73", "Male" = "#E69F00")
darken <- function(col, f = 0.34) { v <- grDevices::col2rgb(col) / 255 * (1 - f); grDevices::rgb(v[1], v[2], v[3]) }
fig4_pal_dark <- sapply(fig4_pal, darken)
qlab <- as_labeller(c(z = "italic(z)~(degree*C)", CTmax_1hr = "CT[max[1*hr]]~(degree*C)",
  CTmax_4hr = "CT[max[4*hr]]~(degree*C)", default = label_parsed)

```

```

P <- function(x) here("pics", x)

# Patrice's v2 illustrations are vector SVGs. Rasterise each to a high-resolution
# temporary PNG so the 600-dpi figure stays crisp and the existing
# draw_image()/<img> paths (which expect a raster file) work unchanged. Cached so
# each SVG is rasterised once per session.
svg_png <- local({
  cache <- new.env(parent = emptyenv())
  function(svg, width = 1600) {
    key <- paste0(svg, "@", width)
    if (is.null(cache[[key]])) {
      o <- tempfile(fileext = ".png")
      magick::image_write(magick::image_read_svg(P(svg), width = width), o)
      cache[[key]] <- o
    }
    cache[[key]]
  }
})

# Trim white margins so the aphid species illustrations are size-comparable
# (mirrors the original danio-stage y-label treatment for the 3-group panel).
aphid_src <- c(M_dirhodum = "aphid_Mdirhodum_V2.png", S_avenae = "aphid_Savenae_V2.png", R_pa
aphid_trim <- vapply(aphid_src, function(f) {
  o <- tempfile(fileext = ".png"); image_write(image_trim(image_read(P(f)), fuzz = 8), o); o
}, character(1))
imgH <- function(p, h) sprintf("<img src='%s' height='%d' />", p, h)
imgW <- function(p, w) sprintf("<img src='%s' width='%d' />", p, w)
sym <- function(ch, pt) sprintf("<span style='font-size:%dpt'>%s</span>", pt, ch)

# pMCMC = twice the smaller posterior tail probability of the paired difference.
pmcmc <- function(d) { n <- length(d); max(2 * min(sum(d > 0) / n, sum(d < 0) / n), 1 / n) }
fmtp <- function(p) ifelse(p < 0.001, "p<0.001", paste0("p=", sub("^0", "", sprintf("%.3f", p)
make_brackets <- function(case, glevels, base = 0.10, step = 0.20, cbase = base, cstep = step)
  ylev <- rev(glevels); ypos <- setNames(seq_along(ylev), ylev); prs <- combn(glevels, 2, simpl
  sp_all <- sapply(prs, function(pr) abs(ypos[[pr[1]]] - ypos[[pr[2]]])); maxsp <- max(sp_all)
  # x-offset keyed to the bracket's VERTICAL SPAN: adjacent-pair (inner) brackets
  # share one thin x-column and the widest (outer) bracket steps right by `step`.
  # INNER p-values print horizontally to the LEFT of their bar (tucked toward the
  # data, in the between-row gaps); the OUTER p-value prints to the RIGHT. `base`
  # offsets the inner column past the CI labels.
  dc <- filter(fig4_draws, case_study == case); rows <- list()
  for (q in as.character(unique(dc$quantity))) {
    b <- if (grepl("^CTmax", q)) cbase else base
    s <- if (grepl("^CTmax", q)) cstep else step
    dq <- filter(dc, quantity == q); rng <- range(dq$value); w <- diff(rng); tick <- 0.04 * w
    for (i in seq_along(prs)) {
      pr <- prs[[i]]; ya <- ypos[[pr[1]]]; yb <- ypos[[pr[2]]]; span <- abs(ya - yb)

```



```

xbar <- rng[2] + w * (b + s * (span - 1))
m <- inner_join(filter(dq, group == pr[1])[, c(".draw", "value")],
                filter(dq, group == pr[2])[, c(".draw", "value")],
                by = ".draw", suffix = c("_a", "_b"))
p <- pmcmc(m$value_a - m$value_b)
side <- if (span == maxsp) "R" else "L"
rows[[length(rows) + 1]] <- tibble(quantity = factor(q, levels = levels(fig4_draws$quantities)),
                                   x = xbar, xt = xbar - tick, side = side,
                                   xtext = if (side == "R") xbar + 0.04 * w else xbar - tick - 0.04 * w,
                                   y = min(ya, yb), yend = max(ya, yb), label = fntp(p))
}
}
bind_rows(rows)
}
add_brackets <- function(p, case, glevels, base = 0.10, step = 0.20, cbase = base, cstep = step) {
  br <- make_brackets(case, glevels, base, step, cbase, cstep)
  p + geom_segment(data = br, aes(x = x, xend = x, y = y, yend = yend), inherit.aes = FALSE, linetype = "solid",
                  geom_segment(data = br, aes(x = xt, xend = x, y = y, yend = y), inherit.aes = FALSE, linetype = "solid",
                  geom_segment(data = br, aes(x = xt, xend = x, y = yend, yend = yend), inherit.aes = FALSE, linetype = "solid",
                  geom_text(data = subset(br, side == "L"), aes(x = xtext, y = (y + yend) / 2, label = label), inherit.aes = FALSE,
                  geom_text(data = subset(br, side == "R"), aes(x = xtext, y = (y + yend) / 2, label = label), inherit.aes = FALSE)
}
fig4_panel <- function(cs, glevels, ylabels, title, sci, tag,
                      texp = 1.15, xexp_r = 0.05, xexp_l = 0.18, ridge_scale = 0.65,
                      ridge_scale_ctmax = ridge_scale) {
  d <- filter(fig4_draws, case_study == cs) |> mutate(group = factor(group, levels = rev(glevels)))
  s <- filter(fig4_summary, case_study == cs) |> mutate(group = factor(group, levels = rev(glevels)))
  ggplot(d, aes(value, group, fill = group)) +
    geom_density_ridges(
      data = dplyr::filter(d, !as.character(quantity) %in% c("CTmax_1hr", "CTmax_4hr")),
      colour = "white", linewidth = .2, alpha = .9, scale = ridge_scale,
      rel_min_height = .005
    ) +
    geom_density_ridges(
      data = dplyr::filter(d, as.character(quantity) %in% c("CTmax_1hr", "CTmax_4hr")),
      colour = "white", linewidth = .2, alpha = .9, scale = ridge_scale_ctmax,
      rel_min_height = .005
    ) +
    geom_linerange(data = s, aes(y = group, xmin = lower, xmax = upper, colour = ci_col), inherit.aes = FALSE,
    geom_point(data = s, aes(median, group, fill = group), inherit.aes = FALSE, shape = 21, colour = ci_col),
    geom_label(data = s, aes(median, group, label = sprintf("%.1f [%.1f, %.1f]", median, lower, upper), inherit.aes = FALSE,
    facet_wrap(~ quantity, scales = "free_x", nrow = 1, strip.position = "bottom", labeller = function(facets) {
    scale_fill_manual(values = fig4_pal, guide = "none") + scale_colour_identity() +
    scale_x_continuous(expand = expansion(mult = c(xexp_l, xexp_r))) +
    scale_y_discrete(labels = ylabels, expand = expansion(mult = c(.06, texp))) +
    labs(x = NULL, y = NULL, title = title, subtitle = sci, tag = tag) +
    theme_classic(base_size = 13) +

```

```

    theme(strip.background = element_blank(), strip.placement = "outside", strip.text = element_text(
      axis.text.y = element_markdown(), plot.tag = element_text(face = "bold", size = 16),
      plot.title = element_text(face = "bold", size = 15), plot.subtitle = element_text(face = "bold", size = 12),
      panel.border = element_rect(colour = "grey75", fill = NA, linewidth = .4), plot.margin = margin(10, 10, 10, 10))
  }
  ins <- function(p, f, l, b, r, t) p + inset_element(ggdraw() + draw_image(f), left = l, bottom = b, right = r, top = t)

# All panel icons are placed in the central gap BETWEEN the z and CTmax facets
# (centred horizontally on the facet divider, upper-middle vertically), where
# each panel has whitespace. Top expansion (texp) is reduced accordingly, since
# the icons no longer sit above the top ridge.

# a) Cereal aphids (3 species) - species illustration + italic name as each
# y-label, PLUS a general aphid icon raised into the top whitespace (clear of
# the posterior ridges). Extra right expansion makes room for the three
# right-stacked contrast brackets.
fig4_A <- add_brackets(fig4_panel("Cereal aphids", sp_lv,
  setNames(c(paste0(imgH(aphid_trim["M_dirhodum"], 40), "<br>*M. dirhodum*"),
    paste0(imgH(aphid_trim["S_avenae"], 38), "<br>*S. avenae*"),
    paste0(imgH(aphid_trim["R_padi"], 38), "<br>*R. padi*")), sp_lv),
  "Cereal aphids", "three species", "a)", texp = 0.72, xexp_r = 0.38, xexp_l = 0.30)
  "Cereal aphids", sp_lv, base = 0.30, step = 0.12, cbase = 0.55, cstep = 0.14)
fig4_A <- ins(fig4_A, P("aphid_Mdirhodum_V2.png"), .44, .56, .69, .96)

# b) Snow gum, dark vs light recovery. Icon enlarged ~30%.
fig4_B <- add_brackets(fig4_panel("Snow gum leaf PSII", rec_lv,
  c(Dark = "Dark", Light = "Light"), "Snow gum", "Eucalyptus pauciflora", "b)",
  texp = 0.5, xexp_r = 0.34), "Snow gum leaf PSII", rec_lv)
fig4_B <- ins(fig4_B, svg_png("snowgum_v4.svg"), .40, .52, .72, .93)

# c) Zebrafish x oxygen - LARVAL illustration raised and shrunk a touch; ridges
# scaled DOWN (0.7) so the tight hyperoxia peaks are not so tall.
fig4_C <- add_brackets(fig4_panel("Zebrafish (oxygen)", oxy_lv,
  c(normoxia = "Normoxia", hyperoxia = "Hyperoxia"), "Zebrafish", "Danio rerio", "c)",
  texp = 0.5, xexp_r = 0.34, ridge_scale = 0.7), "Zebrafish (oxygen)", oxy_lv)
fig4_C <- ins(fig4_C, svg_png("danio_larvae_v2.svg"), .41, .54, .68, .90)

# d) Vinegar fly by sex.
fig4_D <- add_brackets(fig4_panel("Vinegar fly (mortality)", c("Female", "Male"), c(Female = s, Male = s),
  "Vinegar fly (mortality)", c("Female", "Male"), c(Female = s, Male = s),
  texp = 0.5, xexp_r = 0.34, ridge_scale = 0.7), "Vinegar fly (mortality)", c("Female", "Male"), c(Female = s, Male = s))
fig4_D <- ins(fig4_D, svg_png("dros_male_v2.svg"), .44, .48, .66, .90)

fig4_final <- (fig4_A | fig4_B) / (fig4_C | fig4_D)

# Also write a standalone high-res PNG so the assembled figure can be imported
# directly (e.g. into the manuscript or slides) without re-running the chunk.
# Wider canvas (13.5 in) so the estimate labels are no longer clipped/overlapping.
dir.create(here("output", "figs"), recursive = TRUE, showWarnings = FALSE)

```

```
ggsave(here("output", "figs", "fig4_case_study_densities.png"), fig4_final,  
        width = 13.5, height = 9.6, dpi = 600, bg = "white")  
  
# Display the saved PNG rather than the live plot object, so the figure's tuned  
# canvas (11.5 x 9.6 in) and inset/bracket positions are not re-flowed by the  
# chunk's fig dimensions (mirrors the ms.qmd include_graphics approach).  
knitr::include_graphics(here("output", "figs", "fig4_case_study_densities.png"))
```

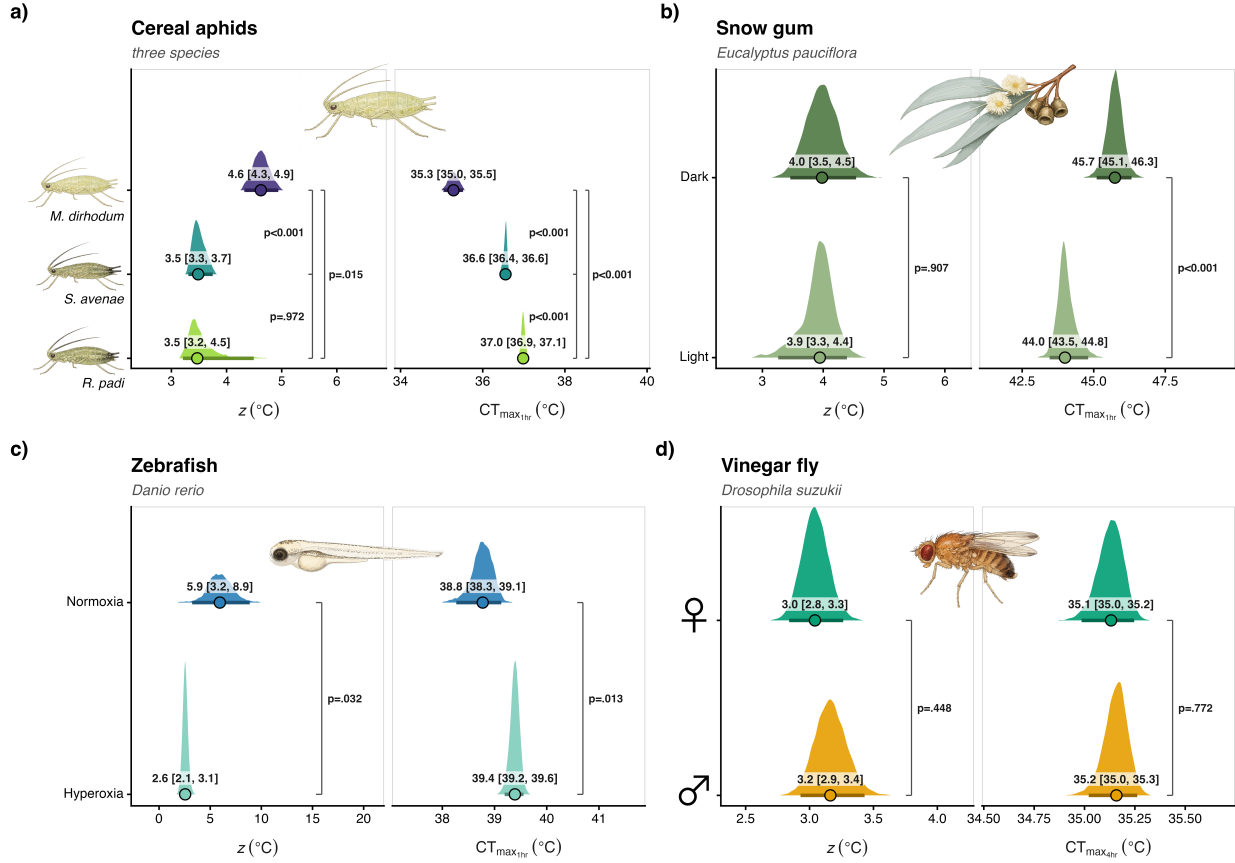


Figure S21: Manuscript Figure 5. Posterior distributions of thermal sensitivity z (left of each panel) and the critical thermal limit CT_{max} (right) from the joint Bayesian 4PL fits for four case studies: a) three cereal-aphid species (*Metopolophium dirhodum*, *Sitobion avenae*, *Rhopalosiphum padi*), b) snow gum (*Eucalyptus pauciflora*) leaf photosystem II function under dark vs 90-min-light post-heat recovery, c) zebrafish (*Danio rerio*) larvae under normoxia vs hyperoxia, and d) the vinegar fly (*Drosophila suzukii*) by sex. Filled ridges are the posterior densities, coloured by group; the black-outlined point and bar give the posterior median and 95% credible interval, also printed beside each ridge. The CT_{max} reference exposure follows each study's own analysis: 1 h for aphids, leaf and zebrafish (relative midpoint threshold) and 4 h for the vinegar fly (absolute LT50, following Ørsted *et al.* (2024)). Within each panel the groups are fit jointly, so pairwise posterior contrasts are annotated as brackets labelled with the pMCMC (twice the smaller posterior tail probability of the difference; smaller values indicate stronger evidence the groups differ). Panels carry independent x-axes.

10 Manuscript Figure 6: Heat injury and survival

Manuscript Figure 6 illustrates how a fitted joint 4PL translates a *natural* temperature trajectory into cumulative heat-injury (HI) accumulation and predicted survival, with uncertainty, for two case studies side by side: the vinegar fly (*Drosophila suzukii*, by sex) and the three cereal aphids (*Metopolophium dirhodum*, *Sitobion avenae*, *Rhopalosiphum padi*). It reuses the fly

and aphid fits built above (`wf_dros_mort_F/wf_dros_mort_M` and `wf_aphid`) and the package's `predict_heat_injury()`, so no model is refit here.

For the fly we use the shaded hourly **microclimate** trace from the Ørsted et al. (2024) NicheMapR workflow (Rennes, France; June–August 2018; Zenodo record 10821572) and predict HI for each sex with and without damage repair. For the aphids we use the **unmodified** hourly air-temperature trace for Wuhan — the hottest of Li *et al.* (2023)'s three Chinese wheat sites (May–June 2016, Open-Meteo ERA5; shipped with the package) — and project all three species at once with a single `predict_heat_injury(by = "species")` call. The aphid trace crosses each species' T_{crit} without any warming offset, so it is used as observed. We show the May–June window because, over the full May–August record, the most heat-vulnerable species (*M. dirhodum*, the lowest T_{crit}) accrues heat injury into the tens of thousands of percent of an LT_{50} dose, which swamps the axis; May–June keeps the scale readable and is the window in which the repair model still keeps the two more tolerant species (*R. padi*, *S. avenae*) alive.

Repair is modelled with an **illustrative Sharpe–Schoolfield kernel that is not empirically fitted** — it shows the *qualitative* effect of repair, not a calibrated rate. We set $r_{ref} = 0.03$ LT50-dose h^{-1} at the reference temperature, so a full LT50 dose would take roughly 33 h to clear at that temperature in the absence of further injury. Bands are ± 1 SE across posterior draws (not 95% credible intervals) to keep the exponential HI envelope readable. T_{crit} is the relative-threshold critical temperature from each fit (per species for the aphids).

```
# Prepare temperature traces and posterior summaries for the manuscript heat-injury figure.
## Fly: shaded hourly microclimate temperature from the Ørsted et al. (2024)
## Zenodo/NicheMapR workflow (Rennes, France; Jun-Aug 2018). The cached CSV is
## built by data-raw/build_orsted_microclimate_trace.R from the original
## microclimate_injury_accumulation.R script in Zenodo record 10821572.
orsted_micro <- utils::read.csv(here::here(
  "inst", "extdata", "orsted_2024",
  "orsted2024_nichemapr_rennes_2018_hourly.csv.gz"
))
fly_trace <- orsted_micro |>
  dplyr::mutate(datetime_utc = as.POSIXct(datetime_utc, tz = "UTC")) |>
  dplyr::filter(substr(datetime_utc, 1, 4) == "2018",
    substr(datetime_utc, 6, 7) %in% c("06", "07", "08"),
    is.finite(micro_temp_c)) |>
  dplyr::arrange(datetime_utc) |>
  dplyr::transmute(time = dplyr::row_number() - 1, temp = micro_temp_c,
    datetime = datetime_utc)

## Relative-threshold  $T_{crit}$  per group (matches the relative HI integration).
Tc_F <- extract_tdt(wf_dros_mort_F, target_surv = "relative", t_ref = 60,
  lethal = TRUE, ndraws = 150)$T_crit$summary$temp_median
Tc_M <- extract_tdt(wf_dros_mort_M, target_surv = "relative", t_ref = 60,
  lethal = TRUE, ndraws = 150)$T_crit$summary$temp_median

## Illustrative Sharpe-Schoolfield repair kernel (doses  $h^{-1}$  at TREF; NOT fitted).
fly_repair <- list(TA = 14065, TAL = 50000, TAH = 100000,
  TL = 15 + 273.15, TH = 30 + 273.15,
```

```

TREF = 25 + 273.15, r_ref = 0.03)

ND <- 150
set.seed(123) # reproducible posterior-draw subsample for both taxa
agg_draws <- function(hi, ...) {
  hi$draws |> dplyr::group_by(time) |>
    dplyr::summarise(hi_med = median(hi), hi_se = sd(hi),
                     surv_med = median(survival), surv_se = sd(survival),
                     .groups = "drop") |>
    dplyr::mutate(...)
}

fly_rows <- list()
for (sx in c("Female", "Male")) {
  wf <- if (sx == "Female") wf_dros_mort_F else wf_dros_mort_M
  Tc <- if (sx == "Female") Tc_F else Tc_M
  for (rep in c(FALSE, TRUE)) {
    hi <- predict_heat_injury(fly_trace[, c("time", "temp")], wf,
                             target_surv = "relative", T_c = Tc, trace_unit = "hours",
                             ndraws = ND, repair = rep,
                             repair_pars = if (rep) fly_repair else NULL, save_draws = TRUE)
    fly_rows[[length(fly_rows) + 1]] <-
      agg_draws(hi, sex = sx, repair = ifelse(rep, "With repair", "No repair"))
  }
}

fdat <- dplyr::bind_rows(fly_rows) |> dplyr::left_join(fly_trace, by = "time")
fTc <- mean(c(Tc_F, Tc_M))

## Aphids (d-f): UNMODIFIED hourly Wuhan air-temperature trace (no warming
## offset), all three species projected at once with by = "species". Per-species
## T_crit comes from the aphid fit; repair uses the same illustrative kernel.
## Trimmed to May-June: over the full May-Aug record the heat injury for the most
## vulnerable species (M. dirhodum, lowest T_crit) runs to ~28000% of an LT50 dose,
## swamping the axis; May-June keeps it readable and spans the window in which the
## repair model still protects the two more tolerant species (R. padi, S. avenae).
aphid_trace <- readr::read_csv(
  here::here("inst", "extdata", "data_temp_trace_aphid_summer2016.csv"),
  show_col_types = FALSE) |>
  dplyr::filter(city == "Wuhan",
                as.integer(format(datetime, "%m")) <= 6) |> # May-June only
  dplyr::transmute(time = time_h, temp = temp_c,
                   datetime = as.POSIXct(sub("T", " ", as.character(datetime)), tz = "UTC"))
aphid_repair <- list(TA = 14065, TAL = 50000, TAH = 100000,
                    TL = 15 + 273.15, TH = 35 + 273.15,
                    TREF = 25 + 273.15, r_ref = 0.03)
aphid_Tc <- aphid_tls$summary |> dplyr::filter(quantity == "Tcrit") |>
  dplyr::transmute(species = as.character(species), Tc = median)

```

```

aphid_rows <- list()
for (rep in c(FALSE, TRUE)) {
  hi <- predict_heat_injury(aphid_trace[, c("time", "temp")], wf_aphid,
    target_surv = "relative", trace_unit = "hours", by = "species",
    ndraws = ND, repair = rep,
    repair_pars = if (rep) aphid_repair else NULL, save_draws = TRUE)
  aphid_rows[[length(aphid_rows) + 1]] <- hi$draws |>
    dplyr::group_by(species, time) |>
    dplyr::summarise(hi_med = median(hi), hi_se = sd(hi),
      surv_med = median(survival), surv_se = sd(survival),
      .groups = "drop") |>
    dplyr::mutate(repair = ifelse(rep, "With repair", "No repair"))
}
adat <- dplyr::bind_rows(aphid_rows) |> dplyr::mutate(species = as.character(species)) |>
  dplyr::left_join(aphid_trace, by = "time")

dir.create(here::here("output", "data"), recursive = TRUE, showWarnings = FALSE)
saveRDS(list(fly = fdat, fly_trace = fly_trace, fly_Tc = fTc,
  aphid = adat, aphid_trace = aphid_trace, aphid_Tc = aphid_Tc),
  here::here("output", "data", "fig5_hi_data.rds"))

```

```

# Build the manuscript heat-injury figure from the in-session fly + aphid data
# (fdat / adat / traces / Tc) prepared in the chunk above. svg_png() and P() are
# the icon helpers defined in the Figure 5 chunk. Fly: colour = sex; aphids:
# colour = species; line type = repair. Colours match the Figure 5 ridge panels.
fdat$repair <- factor(fdat$repair, levels = c("No repair", "With repair"))
adat$repair <- factor(adat$repair, levels = c("No repair", "With repair"))
sp_pal <- c("M_dirhodum" = "#E69F00", "S_avenae" = "#D55E00", "R_padi" = "#CC79A7")
sp_lab <- c(M_dirhodum = "M. dirhodum", S_avenae = "S. avenae", R_padi = "R. padi")
sex_pal <- c(Female = "#009E73", Male = "#7B3294")
tcrit_col <- "#7B3294"

f6_thm <- theme_classic(base_size = 13) +
  theme(panel.grid.major.y = element_line(colour = "grey93"), plot.margin = margin(2, 7, 2, 3),
    legend.position = "none", axis.title = element_text(size = 13), axis.text = element_text(
f6_xsc <- function() scale_x_datetime(date_breaks = "1 month", date_labels = "%b")
f6_tg <- function(p, lab) p + labs(tag = lab) +
  theme(plot.tag = element_text(face = "bold", size = 12), plot.tag.position = c(0.015, 0.98))

# fly temperature panel (single Tcrit line)
f6_temp1 <- function(tr, Tc, title, ylab) ggplot(tr, aes(datetime, temp)) +
  geom_line(colour = "grey45", linewidth = 0.3) +
  geom_hline(yintercept = Tc, linetype = "dotted", colour = tcrit_col) +
  annotate("text", x = min(tr$datetime), y = Tc, label = "T[crit]", hjust = -0.1, vjust = -0.4,
    size = 3.2, colour = tcrit_col, parse = TRUE) +
  labs(y = ylab, x = NULL, title = title) + f6_xsc() + f6_thm +
  theme(axis.text.x = element_blank(), plot.title = element_markdown(face = "bold", size = 13,

```



```

  cowplot::get_legend(g)
}
leg_fly_col <- mk_leg(f6_hiF(fdat), "colour") # Female / Male
leg_aph_col <- mk_leg(f6_hiA(adat), "colour") # three species
leg_repair <- mk_leg(f6_hiA(adat), "linetype") # No repair / With repair -- shared
f6_body <- (col_f | col_a) + plot_layout(heights = c(0.5, 1, 1))
fig5_final <- cowplot::plot_grid(
  f6_body,
  cowplot::plot_grid(leg_fly_col, leg_aph_col, leg_repair, ncol = 3, rel_widths = c(1, 1.3, 1),
    ncol = 1, rel_heights = c(1, 0.09))

ggsave(here::here("output", "figs", "fig5_heat_injury.png"), fig5_final,
  width = 9.8, height = 8.0, dpi = 300, bg = "white")

# Display the saved PNG (not the live object) so the tuned 9.8 x 8.0 in canvas
# and inset positions are not re-flowed by the chunk's fig dimensions.
knitr::include_graphics(here::here("output", "figs", "fig5_heat_injury.png"))

```

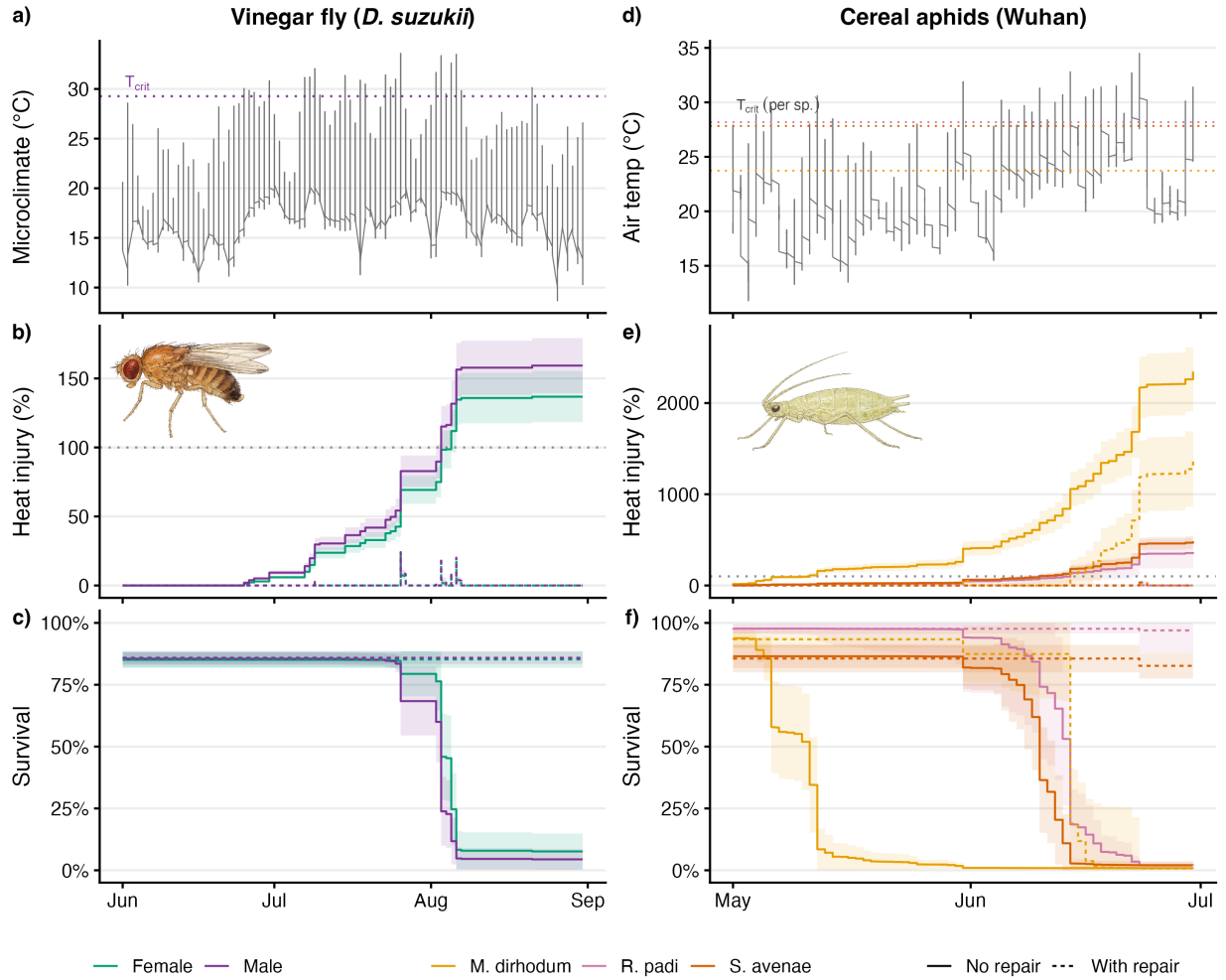


Figure S22: Manuscript Figure 6. Heat-injury accumulation and predicted survival under field temperature trajectories for the vinegar fly (*Drosophila suzukii*, by sex; panels a-c) and three cereal aphids (*Metopolophium dirhodum*, *Sitobion avenae*, *Rhopalosiphum padi*; panels d-f). (a, d) the temperature trace — shaded NicheMapR microclimate temperature for the fly (Rennes, France, Jun–Aug 2018; Ørsted et al. 2024 Zenodo record 10821572) and the unmodified hourly air-temperature trace for the aphids (Wuhan, May–June 2016; Li et al. 2023, Open-Meteo ERA5) — with the relative-threshold T_{crit} (dotted; per species for the aphids). (b, e) cumulative heat injury (100% = one LT_{50} dose, dotted grey) and (c, f) predicted survival, each with and without damage repair (line type). Fly panels are coloured by sex, aphid panels by species. Bands are ± 1 SE across posterior draws. Predicted survival begins at each group’s fitted upper asymptote rather than at 100% (the vinegar-fly 4PL estimates a maximum survival of 85%, $up < 1$). Repair uses an illustrative Sharpe–Schoolfield kernel with $r_{ref} = 0.03$ LT_{50} -dose h^{-1} at the reference temperature; it is not empirically fitted.

11 Derivation of the correction factors for threshold summaries

The manuscript states two correction factors without showing the algebra: an asymmetry correction *added* to $\text{mid}(T)$ to recover $\log_{10} t_{50}(T)$ from a 4PL whose asymptotes are not 0 and 1 (Eq. (4)), and the corresponding correction *subtracted* from the naive $CT_{max}(t_{\text{ref}})$ (Eq. (5)). Here we derive both expressions, extend them to an arbitrary survival threshold $x/100$, and check them against direct 4PL inversion.

11.1 Why a correction is needed

The 4PL midpoint is the $\log_{10}$ -time value at which survival reaches the average of the two asymptotes, $(\text{up} + \text{low})/2$. When $\text{low} \approx 0$ and $\text{up} \approx 1$, this point is also near 50% survival. With partial endpoints, however, the midpoint need not occur at $p = 0.5$: if $\text{up} = 0.85$ and $\text{low} = 0.05$, it occurs at 0.45. The absolute $\log_{10} t_{50}$ therefore comes from inverting the 4PL at $p = 0.5$. The signed offset between $\text{mid}(T)$ and this absolute crossing is the **asymmetry correction**.

11.2 Step-by-step derivation of the $\log_{10} t_{50}$ correction

Starting from Eq. (2) of the manuscript with the temperature-dependent midpoint from Eq. (3), the survival probability is

$$p(t, T) = \text{low} + \frac{\text{up} - \text{low}}{1 + \exp(k(\log_{10} t - \text{mid}(T)))}.$$

By definition $\log_{10} t_{50}(T)$ is the value of $\log_{10} t$ at which $p = 0.5$. Solving for it gives:

Step 1 — set $p = 0.5$:

$$0.5 = \text{low} + \frac{\text{up} - \text{low}}{1 + \exp(k(\log_{10} t_{50}(T) - \text{mid}(T)))}.$$

Step 2 — move low to the left:

$$0.5 - \text{low} = \frac{\text{up} - \text{low}}{1 + \exp(k(\log_{10} t_{50}(T) - \text{mid}(T)))}.$$

Step 3 — invert both sides:

$$\frac{1}{0.5 - \text{low}} = \frac{1 + \exp(k(\log_{10} t_{50}(T) - \text{mid}(T)))}{\text{up} - \text{low}}.$$

Step 4 — multiply both sides by $(\text{up} - \text{low})$:

$$\frac{\text{up} - \text{low}}{0.5 - \text{low}} = 1 + \exp(k(\log_{10} t_{50}(T) - \text{mid}(T))).$$

Step 5 — subtract 1 to isolate the exponential:

$$\frac{\text{up} - \text{low}}{0.5 - \text{low}} - 1 = \exp(k(\log_{10} t_{50}(T) - \text{mid}(T))).$$

The left-hand side simplifies:

$$\frac{\text{up} - \text{low}}{0.5 - \text{low}} - 1 = \frac{(\text{up} - \text{low}) - (0.5 - \text{low})}{0.5 - \text{low}} = \frac{\text{up} - 0.5}{0.5 - \text{low}}.$$

Hence

$$\frac{\text{up} - 0.5}{0.5 - \text{low}} = \exp(k(\log_{10} t_{50}(T) - \text{mid}(T))).$$

Step 6 — take natural logs of both sides:

$$\log \frac{\text{up} - 0.5}{0.5 - \text{low}} = k(\log_{10} t_{50}(T) - \text{mid}(T)).$$

Step 7 — divide by k and add $\text{mid}(T)$:

$$\log_{10} t_{50}(T) = \text{mid}(T) + \frac{1}{k} \log \frac{\text{up} - 0.5}{0.5 - \text{low}}. \quad (\text{S7})$$

Thus $\log_{10} t_{50}$ is the midpoint at that temperature plus a correction that depends only on (low, up) and k , not on T . This is Eq. (4) of the manuscript.

Step 8 — substitute the temperature-dependent midpoint from Eq. (3) of the manuscript:

$$\log_{10} t_{50}(T) = [\beta_0 + \beta_1(T - \bar{T})] + \frac{1}{k} \log \frac{\text{up} - 0.5}{0.5 - \text{low}}.$$

Step 9 — distribute and group all T -independent terms into an intercept α :

$$\boxed{\log_{10} t_{50}(T) = \alpha + \beta_1 T, \quad \alpha = \beta_0 - \beta_1 \bar{T} + \frac{1}{k} \log \frac{\text{up} - 0.5}{0.5 - \text{low}}.} \quad (\text{S8})$$

This recovers the classical TDT relation: $\log_{10} t_{50}$ is linear in T .

11.3 From $\log_{10} t_{50}$ to the CT_{max} correction

By definition, $CT_{max}(t_{\text{ref}})$ is the temperature at which $t_{50} = t_{\text{ref}}$, or $\log_{10} t_{50}(T) = \log_{10} t_{\text{ref}}$. From Equation S8:

$$\log_{10} t_{\text{ref}} = \alpha + \beta_1 CT_{max}(t_{\text{ref}}) \implies CT_{max}(t_{\text{ref}}) = \frac{\log_{10} t_{\text{ref}} - \alpha}{\beta_1}.$$

Substituting the full α from Equation S8 and tidying:

$$CT_{max}(t_{\text{ref}}) = \bar{T} + \frac{\log_{10} t_{\text{ref}} - \beta_0}{\beta_1} - \frac{1}{\beta_1} \cdot \frac{1}{k} \log \frac{\text{up} - 0.5}{0.5 - \text{low}}. \quad (\text{S9})$$

The first two terms are the *naive* CT_{max} : the temperature at which $\beta_0 + \beta_1(T - \bar{T})$ crosses $\log_{10} t_{\text{ref}}$. The final term is the asymmetry correction divided by β_1 , appearing as a subtraction because a vertical shift on the log-time axis becomes a horizontal temperature shift when divided by the TDT slope. This is Eq. (5) of the manuscript.

11.4 Generalisation to arbitrary survival thresholds $x/100$

Replacing $p = 0.5$ with $p = x/100$ gives

$$\log_{10} t_x(T) = \text{mid}(T) + \frac{1}{k} \log \frac{\text{up} - x/100}{x/100 - \text{low}}, \quad (\text{S10})$$

which reduces to Equation S7 when $x = 50$. The correction is still added to $\text{mid}(T)$; its sign is set by the ratio inside the log. For $x < 50$ (e.g. LT10), the correction is positive and LT_x occurs later than the midpoint. For $x > 50$ (e.g. LT90), it is negative and LT_x occurs earlier. The corresponding $CT_{max,x}(t_{\text{ref}})$ follows by substituting this LT_x correction into Equation S9.

11.5 The slope (and hence z) is robust to the correction

Because the correction in Equation S7 does not depend on T , it shifts $\log_{10} t_{50}(T)$ vertically without changing the slope:

$$z = -\frac{1}{\beta_1}$$

holds provided the 4PL is correctly specified. This is why the joint Bayesian 4PL and the classical two-stage pipeline can recover the same z even when their CT_{max} estimates differ. CT_{max} has no analogous protection: the same asymmetry term enters Equation S9 divided by β_1 , so errors in low, up, or k shift the temperature estimate directly.

11.6 When the correction vanishes

When the experimental design anchors $\text{low} \approx 0$ and $\text{up} \approx 1$:

$$\frac{\text{up} - 0.5}{0.5 - \text{low}} \approx \frac{1 - 0.5}{0.5 - 0} = 1 \implies \log \frac{\text{up} - 0.5}{0.5 - \text{low}} \approx \log 1 = 0.$$

The correction then shrinks to zero and $\text{mid}(T) \approx \log_{10} t_{50}(T)$. This is the equivalence regime in Section 3.0.2 and Section 3.0.3, and the strict-equivalence baseline of the simulation in §3.1. The correction matters when the design is partial: sublethal endpoints, sparse survival-range coverage, or asymptotes that vary with T (the last case is analysed in Section 3.0.6).

11.7 Numerical sanity check

The chunk below uses asymmetric asymptotes so the correction is non-zero. It checks that Equation S7 matches direct 4PL inversion at $p = 0.5$, and that Equation S9 matches the temperature obtained by direct inversion at $t_{\text{ref}} = 60$ min.

```
# Check the one-step correction against numerical inversion.
low_a  <- 0.05
up_a   <- 0.85          # asymmetric: (up - 0.5) != (0.5 - low), so the correction is non-zero
kk     <- 4
beta_0 <- log10(60)      # midpoint at T_bar sits exactly at log10(60 min)
beta_1 <- -0.25          # so z = -1/beta_1 = 4 °C
```

```

T_bar <- 36

correction <- (1 / kk) * log((up_a - 0.5) / (0.5 - low_a))

# Closed form (eq-supp-corr-lt50) at T = 38 °C.
T_test <- 38
mid_T <- beta_0 + beta_1 * (T_test - T_bar)
log10_LT50_closed <- mid_T + correction

# Direct numerical inversion: find log10 t such that p(t, T) = 0.5.
p_4pl <- function(log10_t, T_val) {
  midT <- beta_0 + beta_1 * (T_val - T_bar)
  low_a + (up_a - low_a) / (1 + exp(kk * (log10_t - midT)))
}
log10_LT50_direct <- stats::uniroot(
  function(x) p_4pl(x, T_test) - 0.5,
  interval = c(-5, 10)
)$root

# Closed-form CTmax_1hr from eq-supp-corr-ctmax. Because mid(T) was written
# with centred temperature, beta_0 + beta_1 * (T - T_bar), alpha converts the
# line to the uncentred form alpha + beta_1 * T before inversion.
alpha <- beta_0 - beta_1 * T_bar + correction
CT_max_1hr_closed <- (log10(60) - alpha) / beta_1
z_from_b1 <- -1 / beta_1

# Direct cross-check for CTmax_1hr: find T such that the 4PL crosses p = 0.5
# exactly at t = 60 min.
CT_max_1hr_direct <- stats::uniroot(
  function(T_val) p_4pl(log10(60), T_val) - 0.5,
  interval = c(20, 50)
)$root

data.frame(
  Quantity = c("Asymmetry correction (1/k) log[(up-0.5)/(0.5-low)]",
    "log10(LT50) at T = 38 °C, closed form (eq-supp-corr-lt50)",
    "log10(LT50) at T = 38 °C, direct 4PL inversion",
    " |closed - direct|",
    "z derived from beta_1 (expected: 4)",
    "CTmax_1hr, closed form (eq-supp-corr-ctmax)",
    "CTmax_1hr, direct 4PL inversion at t = 60 min",
    " |closed - direct|"),
  Value = c(sprintf("%+.6f", correction),
    sprintf("%.6f", log10_LT50_closed),
    sprintf("%.6f", log10_LT50_direct),
    sprintf("%.3e", abs(log10_LT50_closed - log10_LT50_direct)),
    sprintf("%.4f °C", z_from_b1),

```

Table S30: Verification that the closed-form $\log_{10} t_{50}$ correction (Equation S7) agrees with direct numerical inversion of the 4PL at $p = 0.5$, and that $CT_{max,1h}$ from Equation S9 agrees with direct inversion at $t = 60$ min.

Quantity	Value
Asymmetry correction $(1/k) \log[(up-0.5)/(0.5-low)]$	-0.062829
$\log_{10}(LT50)$ at $T = 38$ °C, closed form (eq-supp-corr-lt50)	1.215323
$\log_{10}(LT50)$ at $T = 38$ °C, direct 4PL inversion	1.215323
closed - direct	1.624e-08
z derived from β_1 (expected: 4)	4.0000 °C
CT_{max_1hr} , closed form (eq-supp-corr-ctmax)	35.7487 °C
CT_{max_1hr} , direct 4PL inversion at $t = 60$ min	35.7487 °C
closed - direct	8.151e-08

```

    sprintf("%.4f °C", CT_max_1hr_closed),
    sprintf("%.4f °C", CT_max_1hr_direct),
    sprintf("%.3e", abs(CT_max_1hr_closed - CT_max_1hr_direct)))
) |>
tinytable::tt(escape = TRUE)

```

Here the correction is **non-zero** (-0.0628), and the closed forms match direct inversion to floating-point precision. The slope still gives $z = -1/\beta_1 = 4$ °C. In contrast, $CT_{max,1h}$ is shifted from $\bar{T} = 36$ °C by about $-\text{correction}/\beta_1$, showing why the correction matters when low and up are not equidistant from 0.5.

12 References

- Arnold, P.A., Harris, R.J., Aitken, S.M., Hoek, M.M., Cook, A.M., Leigh, A., *et al.* (2026). [Towards a standard approach to investigating the Thermal Load Sensitivity of photosystem II via chlorophyll fluorescence](#). *bioRxiv : the preprint server for biology*.
- Arnold, P.A., Noble, D.W.A., Nicotra, A.B., Kearney, M.R., Rezende, E.L., Andrew, S.C., *et al.* (2025). [A framework for modelling thermal load sensitivity across life](#). *Global Change Biology*, 31, e70315.
- Faber, A.H., Møller, F.D., Ørsted, M., Ehlers, B.K. & Overgaard, J. (2026). Separating good from bad – a methodological assessment of the critical temperature that separates stressful and permissive temperatures in ectotherms.
- Jørgensen, L.B., Malte, H. & Overgaard, J. (2021). [A unifying model to estimate thermal tolerance limits in ectotherms across static, dynamic and fluctuating exposures to thermal stress](#). *Scientific Reports*, 11, 12840.
- Li, Y.-J., Chen, S.-Y., Jørgensen, L.B., Overgaard, J., Renault, D., Colinet, H., *et al.* (2023). [Interspecific differences in thermal tolerance landscape explain aphid community abundance under climate change](#). *Journal of Thermal Biology*, 114, 103583.

- Noble, D.W.A., Mayfield, M.M., Hoffmann, A.A., Chen, Z.-H., Lade, S.J., Bai, X., *et al.* (2026). [A systems modelling approach to predict biological responses to extreme heat](#). *Trends in Ecology & Evolution*, 41, 451–464.
- Ørsted, M., Jørgensen, L.B. & Overgaard, J. (2022). [Finding the right thermal limit: A framework to reconcile ecological, physiological and methodological aspects of CTmax in ectotherms](#). *Journal of Experimental Biology*, 225, jeb244514.
- Ørsted, M., Willot, Q., Olsen, A.K., Kongsgaard, V. & Overgaard, J. (2024). [Thermal limits of survival and reproduction depend on stress duration: A case study of *Drosophila suzukii*](#). *Ecology Letters*, 27, e14421.
- Rezende, E.L., Bozinovic, F., Szilágyi, A. & Santos, M. (2020). [Predicting temperature mortality and selection in natural *Drosophila* populations](#). *Science*, 369, 1242–1245.
- Rezende, E.L., Castañeda, L.E. & Santos, M. (2014). [Tolerance landscapes in thermal ecology](#). *Functional Ecology*, 28, 799–809.
- Saruhashi, S., Boerrigter, J.G.J., Hooymans, M.H.L., Sinclair, B.J. & Verberk, W.C.E.P. (2026). [Oxygen availability and oxygen delivery but not oxidative stress shape heat tolerance in diploid and triploid zebrafish larvae](#). *Journal of Experimental Biology*, 229, jeb251548.

13 Session Information

```
# Record package versions and session details for reproducibility.
pander::pander(sessionInfo())
```

R version 4.5.3 (2026-03-11)

Platform: aarch64-apple-darwin20

locale: C.UTF-8||C.UTF-8||C.UTF-8||C||C.UTF-8||C.UTF-8

attached base packages: *stats*, *graphics*, *grDevices*, *utils*, *datasets*, *methods* and *base*

other attached packages: *magick*(v.2.9.1), *cowplot*(v.1.2.0), *ggtext*(v.0.1.2), *ggridges*(v.0.5.7), *bayesTLS*(v.1.0.0), *readxl*(v.1.4.5), *pander*(v.0.6.6), *patchwork*(v.1.3.2), *tinytable*(v.0.16.0), *tidybayes*(v.3.0.7), *posterior*(v.1.7.1), *brms*(v.2.23.0), *Rcpp*(v.1.1.1-1.1), *ggplot2*(v.4.0.3), *purrr*(v.1.2.2), *tibble*(v.3.3.1), *tidyr*(v.1.3.2), *dplyr*(v.1.2.1) and *here*(v.1.0.1)

loaded via a namespace (and not attached): *RColorBrewer*(v.1.1-3), *tensorA*(v.0.36.2.1), *jsonlite*(v.2.0.0), *magrittr*(v.2.0.5), *TH.data*(v.1.1-3), *estimability*(v.1.5.1), *farver*(v.2.1.2), *nloptr*(v.2.2.1), *rmarkdown*(v.2.31), *ragg*(v.1.5.2), *vctr*(v.0.7.3), *minqa*(v.1.2.8), *tinytex*(v.0.59), *htmltools*(v.0.5.9), *distributional*(v.0.7.1), *curl*(v.7.1.0), *cellranger*(v.1.1.0), *StanHeaders*(v.2.36.0.9000), *plyr*(v.1.8.9), *sandwich*(v.3.1-1), *emmeans*(v.2.0.3), *zoo*(v.1.8-14), *TMB*(v.1.9.17), *commonmark*(v.2.0.0), *lifecycle*(v.1.0.5), *pkgconfig*(v.2.0.3), *Matrix*(v.1.7-4), *R6*(v.2.6.1), *fastmap*(v.1.2.0), *rbibutils*(v.2.3), *digest*(v.0.6.39), *numDeriv*(v.2016.8-1.1), *rsvg*(v.2.6.2), *rprojroot*(v.2.1.1), *textshaping*(v.1.0.5), *labeling*(v.0.4.3), *abind*(v.1.4-8), *mgcv*(v.1.9-4), *compiler*(v.4.5.3), *bit64*(v.4.8.0), *withr*(v.3.0.3), *S7*(v.0.2.2), *backports*(v.1.5.1), *inline*(v.0.3.21), *QuickJSR*(v.1.10.0), *pkg-build*(v.1.4.8), *MASS*(v.7.3-65), *loo*(v.2.9.0.9000), *tools*(v.4.5.3), *otel*(v.0.2.0), *glue*(v.1.8.1), *nlme*(v.3.1-168), *gridtext*(v.0.1.5), *grid*(v.4.5.3), *cmdstanr*(v.0.9.0), *checkmate*(v.2.3.4), *reshape2*(v.1.4.5), *generics*(v.0.1.4), *glmmTMB*(v.1.1.14), *gtable*(v.0.3.6), *tzdb*(v.0.5.0), *hms*(v.1.1.4),

xml2(v.1.5.2), *utf8(v.1.2.6)*, *markdown(v.2.0)*, *pillar(v.1.11.1)*, *ggdist(v.3.3.3)*, *stringr(v.1.6.0)*, *vroom(v.1.7.1)*, *splines(v.4.5.3)*, *lattice(v.0.22-9)*, *survival(v.3.8-6)*, *bit(v.4.6.0)*, *tidyselect(v.1.2.1)*, *knitr(v.1.51)*, *reformulas(v.0.4.4)*, *arrayhelpers(v.1.1-0)*, *gridExtra(v.2.3)*, *V8(v.6.0.4)*, *lite-down(v.0.7)*, *stats4(v.4.5.3)*, *xfun(v.0.57)*, *bridgesampling(v.1.2-1)*, *matrixStats(v.1.5.0)*, *rstan(v.2.36.0.9000)*, *stringi(v.1.8.7)*, *yaml(v.2.3.12)*, *pacman(v.0.5.1)*, *boot(v.1.3-32)*, *evaluate(v.1.0.5)*, *codetools(v.0.2-20)*, *cli(v.3.6.6)*, *RcppParallel(v.5.1.11-2)*, *systemfonts(v.1.3.2)*, *xtable(v.1.8-8)*, *Rdpack(v.2.6.4)*, *processx(v.3.9.0)*, *png(v.0.1-8)*, *coda(v.0.19-4.1)*, *svUnit(v.1.0.8)*, *parallel(v.4.5.3)*, *rstantools(v.2.6.0.9000)*, *readr(v.2.2.0)*, *bayesplot(v.1.15.0.9000)*, *Broddingnag(v.1.2-9)*, *lme4(v.1.1-37)*, *viridisLite(v.0.4.3)*, *mvtnorm(v.1.4-1)*, *scales(v.1.4.0)*, *crayon(v.1.5.3)*, *rlang(v.1.2.0)* and *multcomp(v.1.4-28)*